Statistics for Insurance

Theory and methods for life and non-life insurance data

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Preface

The practice of actuarial science has undergone a profound transformation in recent decades. Where the field once primarily relied on more deterministic and aggregate methods, it now commonly deals with the complexity of stochastic processes and the sheer volume of granular data that describes them. The fundamental challenge, however, remains the same: to manage the risk of processes arising in insurance and financial operations. The discipline of statistics for insurance is our primary scientific tool for this task. It offers the required methods to describe the behaviour of the underlying risks, which is the first and arguably the most fundamental step in designing any effective risk mitigation strategy.

Modern insurance data comes in a great variety of shapes and sizes, and the statistical methods required to analyse it must be equally varied and sophisticated. In this monograph, we study this complexity from different angles. We are interested in data that can give rise to a claim, which may possess both discrete and continuous components, such as in the interplay between claim frequency and severity. We consider processes that evolve through time, requiring pathwise analysis, and we address the practical challenges of varying exposure periods and reporting delays. We also face rare or catastrophic events that manifest as heavy-tails, the influence of different sources of randomness, and the common issues of truncated or censored data. The main aim of these notes is to provide a representative and central exposition of statistical tools that can tackle these characteristics. While the classical distinction between life and non-life insurance has its place, a further goal of this work is to close the gap between these two perspectives, presenting a more unified approach, striving to use the same set of mathematical tools to treat both cases.

In selecting the material, a deliberate choice was made to emphasise not primarily the application of statistical methods, but rather their mathematical foundations. This aligns with my conviction that a deep understanding of the mathematical structure is what drives intuition the most, even when practical application is the ultimate goal. The development of novel methods to solve the emerging problems in insurance often builds upon established theories. For this reason, we strive to document and understand how mathematical statistics quantifies risk, and in particular embarking on the understanding of the detailed proofs of precise statements. Practical examples, however, are also amply provided in both the main text and the exercises.

The order of the presented chapters creates a path through the core of modern statistics for insurance, though clearly only one of many. The material has grown from a shorter set of lecture notes used to teach a semester course at the masters level; in its current length, a selection of about half of the material is recommended. We begin with preliminaries on stochastic convergence, before moving to exploratory tools, the study of heavy-tailed data, and parametric models. We then explore selected topics of machine learning, with an emphasis on model selection and explainability, before turning to specialised topics like stochastic claims reserving, survival analysis, and nonparametric inference for Markov processes. The later chapters cover more advanced topics on multi-state modelling, including both parametric and nonparametric approaches. A Bayesian chapter finalises our treatment. A notable omission is a systematic treatment of parametric dependence modelling and time-series analysis. However, the reader may find that the mathematical tools presented can often be adapted to cover such cases. Numerous references have been pivotal to the development of this monograph, and they are purposefully collected in an organised manner at the end of each chapter, albeit merely representative and far from being comprehensive.

This work would not have been possible without the support and inspiration of many people, but I prominently want to thank my masters and PhD students at the University of Copenhagen, as well as my collaborators. They have been a constant source of motivation, and their intellectual curiosity, dedication to learning and teaching the material, and their help with earlier versions of the manuscript have been essential. Above all, their enthusiasm for the subject has sharpened my own understanding in many ways.

Martin Bladt, Copenhagen, 2026.

Contents

Chapter 1

Probabilistic Foundations

The mathematical foundations of statistics rest on results in probability theory, such as laws of large numbers, central limit theorems, and general principles of stochastic convergence. These results provide the framework for analysing the behaviour of statistical procedures and for establishing their validity in large samples. A broad range of methods in insurance statistics, developed in later chapters, build directly on these tools. We begin with finite-dimensional convergence and limit theorems that are used repeatedly later on, then move to empirical summaries and process-level convergence.

1.1 Stochastic convergence in finite dimensions

We now recall some of the main results used to analyse independent and identically distributed (iid) random vectors. Here, we interpret such vectors as real-valued risks, e.g. claim sizes from a non-life insurance portfolio. We start by fixing notation for deterministic and stochastic orders, and then list standard convergence tools. From real analysis we recall the following notation, which makes precise the notion of speed of convergence.

Definition 1.1 (o-notation) Let a_n, b_n be sequences of real numbers. We write $a_n = o(b_n)$ if $\lim_{n\to\infty} a_n/b_n = 0$. We write $a_n = O(b_n)$ if there exists $K < \infty$ such that $|a_n/b_n| \leq K$ for all $n \geq 1$.

When working with stochastic convergence, this notation can be ambiguous because several modes of convergence exist. Nevertheless, it can be adapted as a convenient shorthand that proves useful in practice.

Definition 1.2 (Types of convergence) Let (Z_n) and Z be $\mathbb{R}^p$-valued random variables defined on some probability space. We say that Z_n converges almost surely (a.s.) to Z, denoted by $Z_n \overset{a.s.}{\to} Z$, if for the metric d of interest

$$\mathbb{P}\big(\lim_{n\to\infty} d(Z_n, Z) = 0\big) = 1.$$

We say that Z_n converges in probability to Z, denoted by $Z_n \overset{\mathbb{P}}{\to} Z$, if for all

$\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\big(d(Z_n, Z) > \varepsilon\big) = 0.$$

Finally, we say that Z_n converges in distribution to Z, denoted by $Z_n \xrightarrow{d} Z$, if for all bounded continuous (equivalently, bounded Lipschitz) functions $f : \mathbb{R}^p \to \mathbb{R}$,

$$\lim_{n\to\infty} \mathbb{E}[f(Z_n)] = \mathbb{E}[f(Z)].$$

Convergence in distribution is also characterised by convergence of cumulative distribution functions at continuity points, or by convergence of characteristic functions.

Almost sure statements are often proved by controlling whether a sequence of exceptional events happens infinitely many times. The next definition and lemma formalise this idea and are used repeatedly in this chapter.

Definition 1.3 (Infinitely often) For events (A_n), define

$$\{A_n \text{ i.o.}\} := \bigcap_{n=1}^{\infty} \bigcup_{k\geq n} A_k.$$

This is the event that infinitely many of the events A_n occur.

Lemma 1.1 (Borel–Cantelli) *Let $(A_n)_{n\geq 1}$ be events.*

1. *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then*
$$\mathbb{P}(A_n \text{ i.o.}) = 0.$$
2. *If, instead, the events A_n are independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then*
$$\mathbb{P}(A_n \text{ i.o.}) = 1.$$

In particular, under independence,

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty \quad \Leftrightarrow \quad \mathbb{P}(A_n \text{ i.o.}) = 0.$$

Proof. If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then by monotone convergence,

$$\mathbb{E}\Big[\sum_{n=1}^{\infty} 1_{A_n}\Big] = \sum_{n=1}^{\infty} \mathbb{E}[1_{A_n}] = \sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty.$$

Hence $\sum_{n=1}^{\infty} 1_{A_n} < \infty$ almost surely, which means only finitely many A_n occur, i.e. $\mathbb{P}(A_n \text{ i.o.}) = 0$. Now assume independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$. For $m \geq n$,

$$\mathbb{P}\Big(\bigcup_{k=n}^{m} A_k\Big) = 1 - \prod_{k=n}^{m} \mathbb{P}(A_k^c) = 1 - \prod_{k=n}^{m} (1 - \mathbb{P}(A_k)) \geq 1 - \exp\Big(-\sum_{k=n}^{m} \mathbb{P}(A_k)\Big),$$

where we used $1 - x \leq e^{-x}$. Letting $m \to \infty$ and by continuity from below, $\mathbb{P}(\cup_{k\geq n} A_k) = 1$ for every n. Since these events decrease in n, we get by continuity from above $\mathbb{P}(A_n \text{ i.o.}) = 1$. □

The above theorem is often specialised to a criterion of almost sure convergence.

Corollary 1.1 (Borel–Cantelli criterion for almost sure convergence) *Let* (Z_n) *and* Z *be* $\mathbb{R}^p$*-valued random variables. If for every* $\varepsilon > 0$*,*

$$\sum_{n=1}^{\infty} \mathbb{P}(\|Z_n - Z\| > \varepsilon) < \infty,$$

then $Z_n \xrightarrow{a.s.} Z$*.*

Proof. For $\varepsilon > 0$, define $A_n^\varepsilon = \{\|Z_n - Z\| > \varepsilon\}$. By Lemma 1.1, $\mathbb{P}(A_n^\varepsilon \text{ i.o.}) = 0$ for every $\varepsilon > 0$. Hence, for each $m \geq 1$, with probability one, only finitely many of the events $A_n^{1/m}$ occur. Intersecting over $m \geq 1$, we obtain, with probability one, that for every $m \geq 1$ there exists $N_m < \infty$ such that $\|Z_n - Z\| \leq 1/m$ for all $n \geq N_m$. □

Example 1.1 (Laws of large numbers under fourth moment condition) Let (Z_n) be iid real-valued random variables with $\mathbb{E}[Z_1] = 0$, $\mathbb{E}[Z_1^4] < \infty$, and define

$$S_n = \frac{1}{n} \sum_{i=1}^{n} Z_i, \quad n \geq 1.$$

For any $\varepsilon > 0$, Markov's inequality gives $\mathbb{P}(|S_n| > \varepsilon) \leq \mathbb{E}[S_n^4]/\varepsilon^4$. Using

$$\mathbb{E}\Big[\Big(\sum_{i=1}^{n} Z_i\Big)^4\Big] = n\mathbb{E}[Z_1^4] + 3n(n-1)\big(\mathbb{E}[Z_1^2]\big)^2,$$

it follows that there exists $C < \infty$ such that $\mathbb{P}(|S_n| > \varepsilon) \leq C/n^2\varepsilon^4$, which immediately yields $S_n \xrightarrow{\mathbb{P}} 0$. Moreover, since $\sum_{n=1}^{\infty} n^{-2} = \pi^2/6 < \infty$, Corollary 1.1 strengthens the statement to $S_n \xrightarrow{a.s.} 0$. ◇

A complementary tool for weak convergence is a set-wise characterisation. One can test convergence in distribution through probabilities of open or closed sets. This is also a main tool used below.

Theorem 1.1 (Portmanteau theorem) *Let* Z *and* (Z_n) *be a random element, and sequence of random elements, respectively, in a common metric space* (S, d)*. The following are equivalent:*

1. $Z_n \xrightarrow{d} Z$*.*
2. $\liminf_{n\to\infty} \mathbb{P}(Z_n \in G) \geq \mathbb{P}(Z \in G)$ *for every open set* $G \subset S$*.*
3. $\limsup_{n\to\infty} \mathbb{P}(Z_n \in F) \leq \mathbb{P}(Z \in F)$ *for every closed set* $F \subset S$*.*

Proof. Assume convergence in distribution, and let $G \subset S$ be open. Define

$$f_m(s) = \min\{1, m\, d(s, G^c)\}, \quad s \in S,\ m \geq 1.$$

Then each f_m is continuous, $0 \le f_m \le 1_G$, and $f_m \uparrow 1_G$ pointwise. Hence

$$\mathbb{E}[f_m(Z_n)] \le \mathbb{P}(Z_n \in G).$$

Taking $\liminf_{n\to\infty}$ and using the assumption, $\mathbb{E}[f_m(Z)] \le \liminf_{n\to\infty} \mathbb{P}(Z_n \in G)$. Now let $m \to \infty$ and apply monotone convergence to obtain the second condition. The equivalence between the second and third conditions follows by complements. Finally, assume the second condition, and let $f \ge 0$ be continuous on S. Since $\{f > t\}$ is open for every $t > 0$, the assumption gives

$$\mathbb{P}(f(Z) > t) \le \liminf_{n\to\infty} \mathbb{P}(f(Z_n) > t).$$

Using Fatou's lemma,

$$\begin{aligned}\mathbb{E}[f(Z)] &= \int_0^\infty \mathbb{P}(f(Z) > t)\ \mathrm{d}t \le \int_0^\infty \liminf_{n\to\infty} \mathbb{P}(f(Z_n) > t)\ \mathrm{d}t \\ &\le \liminf_{n\to\infty} \int_0^\infty \mathbb{P}(f(Z_n) > t)\ \mathrm{d}t = \liminf_{n\to\infty} \mathbb{E}[f(Z_n)].\end{aligned}$$

Now let f be bounded continuous, with $|f| \le c < \infty$. Applying the previous inequality to $c \pm f$ yields

$$\mathbb{E}[f(Z)] \le \liminf_{n\to\infty} \mathbb{E}[f(Z_n)] \quad \text{and} \quad \limsup_{n\to\infty} \mathbb{E}[f(Z_n)] \le \mathbb{E}[f(Z)].$$

Hence $\mathbb{E}[f(Z_n)] \to \mathbb{E}[f(Z)]$ for every bounded continuous f, so convergence in distribution holds. □

The above types of convergence are the three most commonly encountered in statistics, and they are intrinsically related in several ways. The next lemma collects the standard implications:

Lemma 1.2 (Relation between types of convergence) *Let (Z_n) and Z be $\mathbb{R}^p$-valued random variables. Then*

1. $Z_n \overset{a.s.}{\to} Z$ *implies* $Z_n \overset{\mathbb{P}}{\to} Z$.
2. $Z_n \overset{\mathbb{P}}{\to} Z$ *implies* $Z_n \overset{d}{\to} Z$.
3. *Let c be a constant. Then $Z_n \overset{\mathbb{P}}{\to} c$ if and only if $Z_n \overset{d}{\to} c$.*

Proof. Assume first that $Z_n \overset{a.s.}{\to} Z$. Fix $\varepsilon > 0$ and set

$$A_n = \{\|Z_n - Z\| > \varepsilon\}.$$

Then $1_{A_n} \to 0$ almost surely and $0 \le 1_{A_n} \le 1$, so dominated convergence yields

$$\mathbb{P}(A_n) = \mathbb{E}[1_{A_n}] \to 0,$$

which is exactly $Z_n \overset{\mathbb{P}}{\to} Z$. Next assume $Z_n \overset{\mathbb{P}}{\to} Z$, and let $f : \mathbb{R}^p \to \mathbb{R}$ be bounded and continuous. Take an arbitrary subsequence (Z_{n_k}). Since $Z_{n_k} \overset{\mathbb{P}}{\to} Z$, we can choose a further subsequence $(Z_{n_{k_j}})$ such that for all $j \ge 1$

$$\mathbb{P}(\|Z_{n_{k_j}} - Z\| > 2^{-j}) < 2^{-j}$$

By Lemma 1.1 (first part), $\|Z_{n_{k_j}} - Z\| \xrightarrow{a.s.} 0$, hence $f(Z_{n_{k_j}}) \xrightarrow{a.s.} f(Z)$. Because f is bounded, the dominated convergence theorem gives

$$\mathbb{E}[f(Z_{n_{k_j}})] \to \mathbb{E}[f(Z)].$$

Thus every subsequence of $\mathbb{E}[f(Z_n)]$ has a further subsequence converging to $\mathbb{E}[f(Z)]$, so $\mathbb{E}[f(Z_n)] \to \mathbb{E}[f(Z)]$, and this proves $Z_n \xrightarrow{d} Z$. Finally, assume $Z_n \xrightarrow{d} c$ and fix $\varepsilon > 0$. The set

$$F_\varepsilon = \{x \in \mathbb{R}^p : \|x - c\| \geq \varepsilon\}$$

is closed, so Theorem 1.1 yields

$$\limsup_{n\to\infty} \mathbb{P}(\|Z_n - c\| \geq \varepsilon) = \limsup_{n\to\infty} \mathbb{P}(Z_n \in F_\varepsilon) \leq \mathbb{P}(c \in F_\varepsilon) = 0.$$

Hence $\mathbb{P}(\|Z_n - c\| > \varepsilon) \to 0$, i.e. $Z_n \xrightarrow{\mathbb{P}} c$. □

It is interesting to note that the converse implications in general fail.

Example 1.2 (Converse implications fail) The implications in Lemma 1.2 are, in general, not reversible.

1. *Convergence in distribution does not imply convergence in probability.* Let Y satisfy $\mathbb{P}(Y = 1) = \mathbb{P}(Y = -1) = 1/2$, and define

$$Z_n = (-1)^n Y, \quad n \geq 1.$$

Since Y and $-Y$ have the same law, each Z_n has the same distribution as Y, hence $Z_n \xrightarrow{d} Y$. However,

$$\|Z_{n+1} - Z_n\| = 2|Y|,$$

so $\mathbb{P}(\|Z_{n+1} - Z_n\| > 1) = 1$ for all n. If (Z_n) converged in probability to some random variable Z, then

$$\mathbb{P}\big(\|Z_{n+1} - Z_n\| > 2\varepsilon\big) \leq \mathbb{P}\big(\|Z_{n+1} - Z\| > \varepsilon\big) + \mathbb{P}\big(\|Z_n - Z\| > \varepsilon\big) \to 0$$

for each $\varepsilon > 0$, a contradiction.

2. *Convergence in probability does not imply almost sure convergence.* Let $U \sim \mathrm{Unif}(0,1)$. For $n = 2^m + j$ with $m \geq 0$ and $j = 0, \dots, 2^m - 1$, set

$$A_n = \left[j2^{-m}, (j+1)2^{-m}\right), \quad Z_n = 1_{A_n}(U).$$

Equivalently, (A_n) is the typewriter sequence

$$[0,1),\ [0,1/2),\ [1/2,1),\ [0,1/4),\ [1/4,1/2),\ [1/2,3/4),\ [3/4,1), \dots$$

Then $\mathbb{P}(Z_n = 1) = |A_n| = 2^{-m} \to 0$, hence $Z_n \xrightarrow{\mathbb{P}} 0$. On the other hand, fix $u \in (0,1)$ and define the m-th block of indices

$$B_m = \{2^m, \dots, 2^{m+1} - 1\}.$$

In each block B_m, exactly one interval among

$$\big[0, 2^{-m}\big), \big[2^{-m}, 2 \cdot 2^{-m}\big), \dots, \big[(2^m - 1)2^{-m}, 1\big)$$

contains u. Hence, for exactly one $n \in B_m$ we have $Z_n(u) = 1$, and for all other $n \in B_m$ we have $Z_n(u) = 0$. Therefore

$$\limsup_{n\to\infty} Z_n(u) = 1, \quad \liminf_{n\to\infty} Z_n(u) = 0,$$

so $(Z_n(u))$ has no limit for any $u \in (0,1)$, and in particular Z_n does not converge almost surely.

$\diamond$

Remark 1.1 (Verifying convergences) There are several useful ways of verifying convergence in probability and in distribution apart from the definition. For instance *Chebyshev's inequality* states that for any increasing function $\varphi : [0,\infty) \to [0,\infty)$ one has, for any $\varepsilon > 0$,

$$\mathbb{P}\big(\|Z_n - Z\| > \varepsilon\big) \le \frac{\mathbb{E}[\varphi(\|Z_n - Z\|)]}{\varphi(\varepsilon)}, \tag{1.1}$$

so if the numerator on the right-hand side tends to zero, convergence in probability follows. Likewise, the *Cramér–Wold device* states that

$$Z_n \xrightarrow{d} Z \quad \Leftrightarrow \quad a^\top Z_n \xrightarrow{d} a^\top Z, \quad \forall a \in \mathbb{R}^p,$$

linking one-dimensional convergence in distribution, which can be verified using standard results such as convergence of distribution functions at continuity points, and multidimensional convergence. Characteristic functions provide yet another standard route. $\circ$

Theorem 1.2 (Strong law of large numbers) *Let (Z_n) be iid random variables with $\mathbb{E}[|Z_1|] < \infty$. Then*

$$\frac{1}{n}\sum_{i=1}^{n} Z_i \xrightarrow{a.s.} \mathbb{E}[Z_1].$$

Proof. It is enough to treat $\mathbb{E}[Z_1] = 0$. Write $S_n = Z_1 + \cdots + Z_n$. For any $\varepsilon > 0$ define

$$A_\varepsilon = \big\{ \limsup_{n\to\infty} S_n/n > \varepsilon \big\}, \quad Z_i^\varepsilon = (Z_i - \varepsilon)1_{A_\varepsilon}, \quad S_n^\varepsilon = \sum_{i=1}^{n} Z_i^\varepsilon.$$

Let $M_n = S_1 \vee \cdots \vee S_n$ and $M_n' = \max_{1\le i\le n}(Z_2 + \cdots + Z_{i+1})$, we have for $i \le n$, $S_i = Z_1 + (Z_2 + \cdots + Z_i) \le Z_1 + M_n'$, hence $M_n \le Z_1 + M_n'$. By stationarity, M_n' has the same distribution as M_n, so

$$\mathbb{E}[Z_1; M_n > 0] \ge \mathbb{E}[M_n - (M_n')_+; M_n > 0] \ge \mathbb{E}[(M_n)_+ - (M_n')_+] = 0,$$

and letting $n \to \infty$ gives $\mathbb{E}\big[Z_1; \sup_n S_n > 0\big] \ge 0$, which holds for any stationary sequence.

Now, A_ε is measurable with respect to $\mathcal{T} := \cap_n \sigma(Z_i : i > n)$, and since $Z_1, \dots, Z_n$ are independent of $\mathcal{T}$, a generating-class argument yields $\sigma(Z_n : n \geq 1) \perp\!\!\!\perp \mathcal{T}$, so $\mathcal{T} \perp\!\!\!\perp \mathcal{T}$, and hence $\mathbb{P}(A_\varepsilon) = 1$ or 0. Moreover $\{\sup_n S_n^\varepsilon > 0\} = \{\sup_n S_n/n > \varepsilon\} \cap A_\varepsilon = A_\varepsilon$. Further, since A_ε is shift-invariant, (Z_n^ε) is stationary. Then, by the above inequality,

$$0 \leq \mathbb{E}[Z_1^\varepsilon; \sup_n S_n^\varepsilon > 0] = \mathbb{E}[Z_1 - \varepsilon; A_\varepsilon] = -\varepsilon\, \mathbb{P}(A_\varepsilon),$$

and hence $\mathbb{P}(A_\varepsilon) = 0$. Thus $\limsup_n S_n/n \leq 0$ a.s. Applying the same argument to $-S_n$ yields $\liminf_n S_n/n \geq 0$ a.s., so $S_n/n \overset{a.s.}{\to} 0$. $\square$

A further useful inequality, which helps bound the speed of weak convergence from above, is the following. It can also be paired with the Borel–Cantelli Lemma 1.1 when proving almost sure convergence.

Theorem 1.3 (Bernstein's inequality) *Let (Z_n) be independent random variables with values in a measurable space $\mathcal{Z}$ and let γ be a real-valued function on $\mathcal{Z}$ such that $\mathbb{E}[\gamma(Z_n)] = 0$ for all $n \geq 1$. Assume that for some constant $\kappa > 0$,*

$$\frac{1}{n}\sum_{i=1}^n \mathbb{E}\big[|\gamma(Z_i)|^m\big] \leq \frac{m!}{2}\kappa^{m-2}, \quad m = 2, 3, \dots$$

Then for any $t > 0$,

$$\mathbb{P}\Big(\frac{1}{n}\sum_{i=1}^n \gamma(Z_i) \geq \kappa t + \sqrt{2t}\Big) \leq e^{-nt}.$$

Proof. For any centred real-valued random variable X with $\mathbb{E}[X] = 0$ and $c > 0$,

$$\exp[X - c] - 1 \leq \frac{\exp[X]}{1+c} - 1 = \frac{e^X - 1 - X + X - c}{1+c} \leq \frac{e^{|X|} - 1 - |X| + X - c}{1+c}.$$

Choosing $c = \mathbb{E}[e^{|X|}] - 1 - \mathbb{E}[|X|]$ and taking expectations gives $\mathbb{E}[e^{X-c}] \leq 1$, or

$$\log \mathbb{E}[e^X] \leq c = \mathbb{E}[e^{|X|}] - 1 - \mathbb{E}[|X|].$$

Applying this to $X_i = \gamma(Z_i)/L$, expanding the exponential, and using the moment condition we obtain

$$\begin{aligned}
\log \mathbb{E}\Big[\exp\Big(\frac{S_n}{L}\Big)\Big] &\leq \sum_{i=1}^n (\mathbb{E}[e^{|\gamma(Z_i)|/L}] - 1 - \mathbb{E}[|\gamma(Z_i)|/L]) \\
&\leq \sum_{m=2}^\infty \sum_{i=1}^n \frac{\mathbb{E}[|\gamma(Z_i)|^m]}{L^m m!} \\
&\leq \sum_{m=2}^\infty \frac{1}{L^m m!}\Big(\frac{nm!}{2}\kappa^{m-2}\Big) = \frac{n}{2L^2(1-\kappa/L)}.
\end{aligned}$$

Let $S_n = \sum_{i=1}^n \gamma(Z_i)$. Fix $a > 0$ and let $L > \kappa$. By Markov's inequality

$$\mathbb{P}(S_n \geq a) = \mathbb{P}(e^{S_n/L} \geq e^{a/L}) \leq \exp\Big(-\frac{a}{L} + \frac{n}{2(L^2 - L\kappa)}\Big).$$

Optimising over L by taking $\kappa/L = 1 - (1 + 2a\kappa/n)^{-1/2}$ and then reparametrising by taking $a/n = \kappa t + \sqrt{2t}$ yields the result. $\square$

Remark 1.2 In Monte Carlo estimation, $n^{-1}\sum_{i=1}^{n}\gamma(Z_i)$ is an average of iid simulation outputs. Bernstein's inequality provides a non-asymptotic error bound for

$$\mathbb{P}\left(\left|n^{-1}\sum_{i=1}^{n}\gamma(Z_i)-\mathbb{E}[\gamma(Z_1)]\right|\geq\varepsilon\right),$$

under moment/boundedness conditions, which can be used to pick a simulation size n for a desired accuracy and confidence. ○

Numeric example 1.1 (SLLN paths and Bernstein concentration for bounded risks) Let (Z_n) be iid with $Z_i\sim\mathrm{Unif}(-1,1)$, and define $\bar{Z}_n=n^{-1}\sum_{i=1}^{n}Z_i$. Then $\mathbb{E}[Z_i]=0$ and $\mathrm{Var}(Z_i)=1/3$. Taking $\gamma(z)=z$ in Theorem 1.3, the moment condition is satisfied (for instance) with $\kappa=1/3$, so that for $t>0$,

$$\mathbb{P}(\bar{Z}_n\geq\kappa t+\sqrt{2t})\leq e^{-nt}.$$

By symmetry, for any fixed $\varepsilon>0$ and t_ε defined by $\varepsilon=\kappa t_\varepsilon+\sqrt{2t_\varepsilon}$, we obtain

$$\mathbb{P}(|\bar{Z}_n|\geq\varepsilon)\leq 2e^{-nt_\varepsilon}.$$

For comparison, Chebyshev gives

$$\mathbb{P}(|\bar{Z}_n|\geq\varepsilon)\leq\frac{\mathrm{Var}(Z_1)}{n\varepsilon^2}=\frac{1}{3n\varepsilon^2}.$$

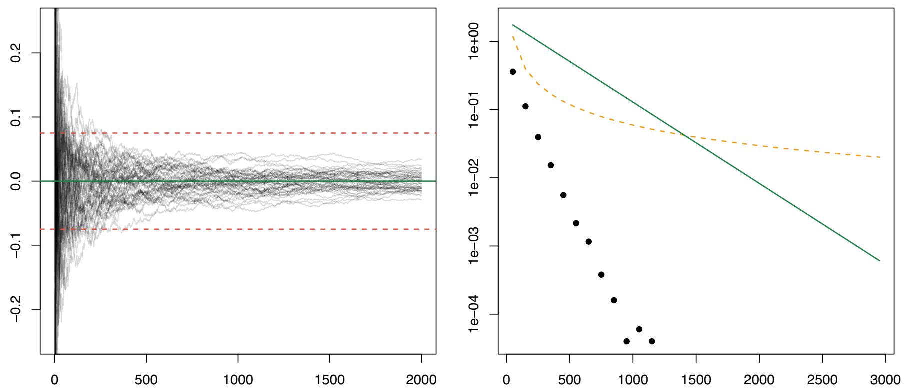

Figure 1.1: Left: 50 simulated trajectories of $\bar{Z}_n$ (black) with horizontal reference lines at 0 (green) and $\pm\varepsilon$ (red dashed), with $\varepsilon=0.075$. Right (log scale): Monte Carlo empirical probabilities $\mathbb{P}(|\bar{Z}_n|\geq\varepsilon)$ (black points), Chebyshev bound $1/(3n\varepsilon^2)$ (orange dashed), and Bernstein bound $2e^{-nt_\varepsilon}$ (green solid), where t_ε solves $\varepsilon=\kappa t+\sqrt{2t}$ with $\kappa=1/3$.

The left panel of Figure 1.1 visualises the pathwise statement behind Theorem 1.2, with visible early fluctuations, and trajectories stabilising and eventually remaining inside the $\pm\varepsilon$ region. The right panel highlights the rate contrast on a logarithmic scale, and although both inequalities are not tight, Chebyshev's inequality decays polynomially in n, while Bernstein's inequality decays exponentially in n. ◇

$X \sim \text{Unif}(-1,1)$

so $f(x) = \frac{1}{2} \cdot 1_{[-1,1]}(x)$ and $\mathbb{E}[X] = 0$ (Symmetry)

and

$*$ $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

$= \mathbb{E}[X^2] = \int_{-1}^{1} \frac{x^2}{2} dx = \left[\frac{x^3}{6}\right]_{-1}^{1} = \frac{1}{6} - \left(\frac{-1}{6}\right) = \frac{1}{3}$

~~$**$~~ Note also

$m \geq 2$ $\quad \mathbb{E}[|X|^m] = \int_{-1}^{1} \frac{|x|^m}{2} = 2 \cdot \underbrace{\left[\frac{x^{m+1}}{2(m+1)}\right]_0^1}_{\int_0^1 \frac{x^m}{2} dx} = \frac{1}{m+1}$

So:

$$\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[|X|^m] = \frac{1}{m+1} \overset{*!}{\leq} \frac{m!}{2}\left(\frac{1}{3}\right)^{m-2}$$

$*!$ Lad $m=2$

$$\frac{1}{3} \leq \frac{2!}{2} \cdot 1 = 1 \quad \checkmark$$

Lad $m>2$ og lad det holde for $m-1$

Så vil

$$\frac{1}{m} \leq \frac{(m-1)!}{2}\left(\frac{1}{3}\right)^{m-3}$$

Men så vil også: (gang med $\frac{m}{3}$)

$$\frac{m!}{2}\left(\frac{1}{3}\right)^{m-2} \geq \frac{1}{3} \geq \frac{1}{m+1} \quad \text{for } m>2$$

Lowest bound must be

$K = \frac{1}{4}$!

så vil $\quad K = \frac{1}{3}$

$\Rightarrow$

$$\mathbb{P}\left(\bar{Z}_n \geq Kt + \sqrt{2t}\right) \leq e^{-nt}$$

Symmetry (Z_n Symmetric $\rightarrow \bar{Z}_n$ Symmetric)

$$\mathbb{P}\left(|\bar{Z}_n| \geq Kt + \sqrt{2t}\right) \leq 2 e^{-nt}$$

Example 1.3 (Law of large numbers for non-iid risks) Let (Z_n) be independent (not necessarily identically distributed) random variables. Write $\mu_i = \mathbb{E}[Z_i]$ and $Y_i = Z_i - \mu_i$. Assume there exist constants $v > 0$ and $\kappa > 0$ such that for all i and all $m \geq 2$,

$$\mathbb{E}\big[|Y_i|^m\big] \leq \frac{m!}{2}\, v^2 \kappa^{m-2}.$$

Applying Theorem 1.3 to the centred variables $Y_1, \dots, Y_n$ with the scaling function $\gamma(y) = y/v$, the moment condition holds with parameter κ/v. Hence,

$$\mathbb{P}\Big(\frac{1}{n}\sum_{i=1}^{n} Y_i \geq \kappa t + v\sqrt{2t}\Big) \leq e^{-nt}, \quad t > 0,$$

and the same bound applies to $-Y_i$. For any $\varepsilon > 0$, set $t = \kappa^{-2}(\sqrt{v^2 + 2\kappa\varepsilon} - v)^2/2$, so that $\kappa t + v\sqrt{2t} = \varepsilon$. Therefore

$$\mathbb{P}\Big(\Big|\frac{1}{n}\sum_{i=1}^{n}(Z_i - \mu_i)\Big| > \varepsilon\Big) \leq 2e^{-nt},$$

which is clearly summable in n. Therefore, Lemma 1.1 yields

$$\frac{1}{n}\sum_{i=1}^{n}(Z_i - \mu_i) \overset{a.s.}{\to} 0.$$

◇

Two results that are often combined to analyse the asymptotic behaviour of complex expressions by reducing to simpler components are the following.

Theorem 1.4 (Continuous mapping theorem) *Let Z and (Z_n) be an $\mathbb{R}^p$-valued random variables and sequence, respectively, and let $g : \mathbb{R}^p \to \mathbb{R}^q$ be continuous. Then*

1. *$Z_n \overset{a.s.}{\to} Z$ implies $g(Z_n) \overset{a.s.}{\to} g(Z)$.*
2. *$Z_n \overset{\mathbb{P}}{\to} Z$ implies $g(Z_n) \overset{\mathbb{P}}{\to} g(Z)$.*
3. *$Z_n \overset{d}{\to} Z$ implies $g(Z_n) \overset{d}{\to} g(Z)$.*

Proof. For the first statement, let $\Omega_0 = \{\omega : \ Z_n(\omega) \to Z(\omega)\}$. Then $\mathbb{P}(\Omega_0) = 1$. For $\omega \in \Omega_0$, continuity of g gives $g(Z_n(\omega)) \to g(Z(\omega))$. Hence $g(Z_n) \overset{a.s.}{\to} g(Z)$. For the second statement, fix $\varepsilon, \eta > 0$. Choose M such that $\mathbb{P}(\|Z\| > M) \leq \eta$ and define

$$K = \{x \in \mathbb{R}^p : \|x\| \leq M + 1\}.$$

Since g is continuous on the compact set K, it is uniformly continuous on K, so there exists $\delta \in (0, 1)$ such that

$$\|x - y\| \leq \delta, \ x, y \in K \quad \Rightarrow \quad \|g(x) - g(y)\| \leq \varepsilon.$$

On the event $\{\|Z\| \leq M,\ \|Z_n - Z\| \leq \delta\}$ we have $\|Z_n\| \leq M+1$, so $Z, Z_n \in K$, hence $\|g(Z_n) - g(Z)\| \leq \varepsilon$. Therefore

$$\mathbb{P}\big(\|g(Z_n) - g(Z)\| > \varepsilon\big) \leq \mathbb{P}\big(\|Z_n - Z\| > \delta\big) + \mathbb{P}(\|Z\| > M).$$

Using $Z_n \xrightarrow{\mathbb{P}} Z$ now gives

$$\limsup_{n\to\infty} \mathbb{P}\big(\|g(Z_n) - g(Z)\| > \varepsilon\big) \leq \eta.$$

Since $\eta > 0$ is arbitrary, $g(Z_n) \xrightarrow{\mathbb{P}} g(Z)$. Finally, for the third statement, let $F \subset \mathbb{R}^q$ be closed. Then $g^{-1}(F) \subset \mathbb{R}^p$ is closed, and by Theorem 1.1 (iii),

$$\limsup_{n\to\infty} \mathbb{P}\big(g(Z_n) \in F\big) = \limsup_{n\to\infty} \mathbb{P}\big(Z_n \in g^{-1}(F)\big) \leq \mathbb{P}\big(Z \in g^{-1}(F)\big) = \mathbb{P}\big(g(Z) \in F\big).$$

Applying Theorem 1.1 again, with the reverse implication, yields $g(Z_n) \xrightarrow{d} g(Z)$. □

Lemma 1.3 (Slutsky's lemma for random variables) *Let (Z_n) and Z be $\mathbb{R}^p$-valued random variables. If $Z_n \xrightarrow{d} Z$ and $W_n \xrightarrow{d} c$ with c a constant, then*

$$(Z_n, W_n) \xrightarrow{d} (Z, c).$$

Proof. We first compare (Z_n, W_n) to (Z_n, c) in probability, and then use convergence in distribution for the latter. We have that $\|(Z_n, W_n) - (Z_n, c)\| = \|W_n - c\| \xrightarrow{\mathbb{P}} 0$ by the third point of Lemma 1.2. We also have $(Z_n, c) \xrightarrow{d} (Z, c)$ by the definition of convergence in distribution. Indeed, for any bounded continuous function f, the map $g(x) = f(x, c)$ is again bounded and continuous, so

$$\mathbb{E}[f(Z_n, c)] = \mathbb{E}[g(Z_n)] \to \mathbb{E}[g(Z)] = \mathbb{E}[f(Z, c)].$$

The two pieces together imply the result, as per Exercise 6. □

We now state a useful approximation theorem which helps to prove $Z_n \xrightarrow{d} Z$ by choosing approximations $Y_n^{(k)}$ of Z_n and $Y^{(k)}$ of Z such that $Y_n^{(k)} \xrightarrow{d} Y^{(k)}$ for fixed k, when the approximation errors are uniformly small.

Theorem 1.5 (Weak convergence by approximation) *Let (S, d) be a metric space and let $Z, Z_n, Y^{(k)}, Y_n^{(k)}$ be S-valued random elements. Assume that for each fixed k, $Y_n^{(k)} \xrightarrow{d} Y^{(k)}$ as $n \to \infty$, and that $Y^{(k)} \xrightarrow{d} Z$ as $k \to \infty$. If in addition, for any given $\varepsilon > 0$,*

$$\lim_{k\to\infty} \limsup_{n\to\infty} \mathbb{P}\Big(d\big(Y_n^{(k)}, Z_n\big) > \varepsilon\Big) = 0, \tag{1.2}$$

then $Z_n \xrightarrow{d} Z$.

Proof. Let $F \subset S$ be closed and fix $\varepsilon > 0$. Write $F^\varepsilon = \{s \in S :\ d(s, F) \leq \varepsilon\}$. Then

$$\mathbb{P}(Z_n \in F) \leq \mathbb{P}(Y_n^{(k)} \in F^\varepsilon) + \mathbb{P}\big(d\big(Y_n^{(k)}, Z_n\big) > \varepsilon\big).$$

For fixed k, Theorem 1.1 (iii) yields

$$\limsup_{n\to\infty} \mathbb{P}(Z_n \in F) \le \mathbb{P}(Y^{(k)} \in F^\varepsilon) + \limsup_{n\to\infty} \mathbb{P}\big(d\big(Y_n^{(k)}, Z_n\big) > \varepsilon\big).$$

By (1.2) the second term vanishes as $k \to \infty$. Since F^ε is closed and $Y^{(k)} \xrightarrow{d} Z$, another application of Theorem 1.1 gives

$$\limsup_{k\to\infty} \mathbb{P}(Y^{(k)} \in F^\varepsilon) \le \mathbb{P}(Z \in F^\varepsilon).$$

Hence

$$\limsup_{n\to\infty} \mathbb{P}(Z_n \in F) \le \mathbb{P}(Z \in F^\varepsilon).$$

Letting $\varepsilon \downarrow 0$ and using continuity from above for the closed sets $F^\varepsilon \downarrow F$ yields $\limsup_{n\to\infty} \mathbb{P}(Z_n \in F) \le \mathbb{P}(Z \in F)$. Since F was arbitrary, Theorem 1.1 implies $Z_n \xrightarrow{d} Z$. $\square$

Then we may make the following definitions, which are the stochastic analogues of Definition 1.1.

Definition 1.4 (Stochastic o-notation) Let (Z_n) be $\mathbb{R}^p$-valued random variables and r_n be positive random variables. We say that Z_n is bounded in probability, denoted by $Z_n = O_{\mathbb{P}}(1)$ if

$$\lim_{M\to\infty} \limsup_{n\to\infty} \mathbb{P}(\|Z_n\| > M) = 0.$$

Likewise, we say that Z_n converges to zero in probability, denoted by $Z_n = o_{\mathbb{P}}(1)$ if $Z_n \xrightarrow{\mathbb{P}} 0$. We also define $Z_n = O_{\mathbb{P}}(r_n)$ if $Z_n/r_n = O_{\mathbb{P}}(1)$ and similarly $Z_n = o_{\mathbb{P}}(r_n)$ if $Z_n/r_n = o_{\mathbb{P}}(1)$.

An immediate consequence is that bounded in probability follows from distributional convergence, a useful fact which is repeatedly used in the remaining chapters.

Example 1.4 (Convergence in distribution implies bounded in probability) Let (Z_n) be a real-valued random sequence converging in distribution to Z with distribution function F. Then at continuity points $M, -M$ we have

$$\mathbb{P}(Z_n > M) \to 1 - F(M), \quad \mathbb{P}(Z_n \le -M) \to F(-M).$$

Since both of the expressions tend to zero as $M \to \infty$ it follows that

$$Z_n \xrightarrow{d} Z \quad \Rightarrow \quad Z_n = O_{\mathbb{P}}(1),$$

which is the desired implication. $\diamond$

A companion transfer rule lets us propagate stochastic bounds across approximations, simplifying notation.

Example 1.5 (Convergence in distribution of an approximating sequence) We show that if $Z_n = O_{\mathbb{P}}(a_n)$ and $W_n = o_{\mathbb{P}}(\|Z_n\|)$, then $W_n = o_{\mathbb{P}}(a_n)$.

The assumptions that $Z_n = O_{\mathbb{P}}(a_n)$ and $W_n = o_{\mathbb{P}}(\|Z_n\|)$ mean that

$$\lim_{M\to\infty} \limsup_{n\to\infty} \mathbb{P}(\|Z_n\|/a_n > M) = 0, \quad \lim_{n\to\infty} \mathbb{P}(\|W_n\|/\|Z_n\| > \varepsilon) = 0$$

for all $\varepsilon > 0$. We want to show that for all $\varepsilon, \delta > 0$ there exists L such that for $n \geq L$, we have $\mathbb{P}(\|W_n/a_n\| \leq \varepsilon) > 1 - \delta$. For $\varepsilon_1, \varepsilon > 0$ we get

$$\begin{aligned}\mathbb{P}(\|W_n\|/a_n \leq \varepsilon) &\geq \mathbb{P}(\{\|W_n\|/\|Z_n\| < \varepsilon_1\} \cap \{\|Z_n\|/a_n < \varepsilon/\varepsilon_1\}) \\ &\geq \mathbb{P}(\|W_n\|/\|Z_n\| < \varepsilon_1) + \mathbb{P}(\|Z_n\|/a_n < \varepsilon/\varepsilon_1) - 1.\end{aligned}$$

By assumption we can find L such that for $n \geq L$ we have $\mathbb{P}(\|Z_n\|/a_n < \varepsilon/\varepsilon_1) > 1 - \delta/2$ and $\mathbb{P}(\|W_n\|/\|Z_n\| < \varepsilon_1) > 1 - \delta/2$. The result follows. ◇

1.2 Fundamental theorems in finite dimensions

In insurance, we need not only point estimates of risks but also uncertainty quantification. The main results we encounter in asymptotic statistics are limit theorems describing the distribution of suitably normalised estimators. These results fall under the umbrella term "central limit theorems", or CLTs, and play a central role in describing the *large sample behaviour* of estimators.

Theorem 1.6 (Laplace's multivariate CLT) *Let (Z_n) be a sequence of iid random vectors in $\mathbb{R}^p$. Assume that $\mathbb{E}\big[\|Z_1\|^2\big] < \infty$. It holds that*

$$\sqrt{n}\Big(\frac{1}{n}\sum_{i=1}^n Z_i - \mathbb{E}\left[Z_1\right]\Big) \xrightarrow{d} N\left(0, \mathrm{Var}(Z_1)\right),$$

where N denotes the p-dimensional normal distribution.

Proof. Let $\mu = \mathbb{E}[Z_1]$ and fix $a \in \mathbb{R}^p$. Put $\xi_k = a^\top(Z_k - \mu)$ so that $\mathbb{E}[\xi_1] = 0$ and $\mathbb{E}[\xi_1^2] = a^\top \mathrm{Var}(Z_1)a$. Let $\varphi(t) = \mathbb{E}[e^{it\xi_1}]$ be the characteristic function of ξ_1. Since $\mathbb{E}[\xi_1^2] < \infty$ and $|e^{it} - 1| \leq |t|$, dominated convergence yields

$$\varphi'(t) = \mathbb{E}[(i\xi_1)e^{it\xi_1}], \quad \varphi''(t) = \mathbb{E}[(i\xi_1)^2 e^{it\xi_1}], \quad t \in \mathbb{R}.$$

Hence $\varphi'(0) = i\mathbb{E}[\xi_1] = 0$ and $\varphi''(0) = -\mathbb{E}[\xi_1^2]$, and Taylor's formula gives

$$\varphi(t) = 1 - \frac{t^2}{2}\mathbb{E}[\xi_1^2] + o(t^2), \quad t \to 0.$$

Hence the characteristic function of $n^{-1/2}\sum_{k=1}^n \xi_k$ is

$$\varphi\Big(\frac{t}{\sqrt{n}}\Big)^n = \Big(1 - \frac{t^2}{2n}\mathbb{E}[\xi_1^2] + o(n^{-1})\Big)^n \to \exp\Big(-\frac{t^2}{2}\mathbb{E}[\xi_1^2]\Big),$$

which is the characteristic function of $N\big(0,\, a^\top \mathrm{Var}(Z_1)a\big)$. By the Cramér–Wold device this implies

$$\sqrt{n}\Big(\frac{1}{n}\sum_{i=1}^n Z_i - \mu\Big) \xrightarrow{d} N\left(0, \mathrm{Var}(Z_1)\right),$$

as claimed. □

Sometimes the distribution of the random variables may depend on the sample size, or the risks may be transformed using an n-dependent function. In that case, it results convenient to work with the following notion.

Definition 1.5 (Triangular array) A triangular array of random variables (or in general of stochastic elements) is a sequence with a representation on the form

$$Z_{nm}, \quad n = 1, 2, \ldots, \quad m = 1, \ldots, n.$$

We require mutual independence for elements within the same row but do not make any assumptions on the distribution, nor on the dependence between rows. A triangular array can be visualised as

$$\begin{array}{llll} Z_{11} & & & \\ Z_{21} & Z_{22} & & \\ Z_{31} & Z_{32} & Z_{33} & \\ \vdots & \vdots & \vdots & \ddots \end{array}$$

Theorem 1.7 (Lindeberg's multivariate CLT) *Let (Z_{nm}) be a triangular array of random variables with values in $\mathbb{R}^p$ with $\mathbb{E}\left[Z_{nm}\right] = 0$ and $\mathbb{E}\big[\|Z_{nm}\|^2\big] < \infty$ for all $m \leq n$. Assume that*

$$\sum_{m=1}^{n} \operatorname{Var}(Z_{nm}) \to \Sigma,$$

for a fixed $p \times p$ positive semi-definite matrix Σ, and that the associated real-valued array $(\|Z_{nm}\|)$ satisfies Lindeberg's condition, i.e. for all $c > 0$:

$$\sum_{m=1}^{n} \mathbb{E}\big[\|Z_{nm}\|^2; \|Z_{nm}\| > c\big] \to 0.$$

Then

$$\sum_{m=1}^{n} Z_{nm} \xrightarrow{d} N(0, \Sigma),$$

where N denotes the p-dimensional normal distribution.

Proof. Fix $a \in \mathbb{R}^p$ and set $X_{nm} = a^\top Z_{nm}$. If $a = 0$ there is nothing to prove, so assume $a \neq 0$. Then $\mathbb{E}[X_{nm}] = 0$ and

$$c_{nm} = \mathbb{E}[X_{nm}^2] = a^\top \operatorname{Var}(Z_{nm})a, \quad c_n = \sum_{m=1}^{n} c_{nm} \to a^\top \Sigma a.$$

Moreover, for any $\varepsilon > 0$,

$$\sum_{m=1}^{n} \mathbb{E}\big[X_{nm}^2; |X_{nm}| > \varepsilon\big] \leq \|a\|^2 \sum_{m=1}^{n} \mathbb{E}\big[\|Z_{nm}\|^2; \|Z_{nm}\| > \varepsilon/\|a\|\big] \to 0,$$

so the (one-dimensional) Lindeberg condition holds for (X_{nm}). It therefore suffices, by the Cramér–Wold device, to prove that $\sum_{m=1}^n X_{nm} \xrightarrow{d} N(0,\, a^\top \Sigma a)$.

We first note that for any $\varepsilon > 0$,

$$\sup_m c_{nm} \le \varepsilon^2 + \sup_m \mathbb{E}\big[X_{nm}^2; |X_{nm}| > \varepsilon\big] \le \varepsilon^2 + \sum_{m=1}^n \mathbb{E}\big[X_{nm}^2; |X_{nm}| > \varepsilon\big],$$

so $\sup_m c_{nm} \to 0$ by the Lindeberg condition. Let $\varphi_{nm}(t) = \mathbb{E}[e^{itX_{nm}}]$. Introduce independent $\zeta_{nm} \sim N(0, c_{nm})$ with characteristic functions $\psi_{nm}(t) = \exp(-t^2 c_{nm}/2)$, so $\sum_{m=1}^n \zeta_{nm} \sim N(0, c_n)$. Using the elementary bound (proved by induction)

$$\Big|\prod_{m=1}^n z_m - \prod_{m=1}^n z'_m\Big| \le \sum_{m=1}^n |z_m - z'_m|, \quad |z_m| \le 1,\ |z'_m| \le 1,$$

To control the remainders, set $h_m(x) = e^{ix} - \sum_{k=0}^m (ix)^k/k!$. Then $h_m(x) = i\int_0^x h_{m-1}(s)\, \mathrm{d}s$, $|h_{-1}| \equiv 1$, and $|h_0| \le 2$, so by induction

$$\Big|e^{ix} - \sum_{k=0}^m \frac{(ix)^k}{k!}\Big| \le \frac{2|x|^m}{m!} \wedge \frac{|x|^{m+1}}{(m+1)!}.$$

With $m = 2$ this gives $\big|e^{itx} - 1 - itx + \frac{1}{2}t^2x^2\big| \le |tx|^2(1 \wedge |tx|)$, so

$$\begin{aligned}
|\varphi_{nm}(t) - 1 + \tfrac{1}{2}t^2 c_{nm}| &\le t^2 \mathbb{E}\big[X_{nm}^2(1 \wedge |tX_{nm}|)\big],\\
|\psi_{nm}(t) - 1 + \tfrac{1}{2}t^2 c_{nm}| &\le t^2 \mathbb{E}\big[\zeta_{nm}^2(1 \wedge |t\zeta_{nm}|)\big].
\end{aligned}$$

Combining this with the product bound and the triangle inequality yields

$$\begin{aligned}
&\Big|\prod_{m=1}^n \varphi_{nm}(t) - \prod_{m=1}^n \psi_{nm}(t)\Big|\\
&\le \sum_{m=1}^n |\varphi_{nm}(t) - 1 + \tfrac{1}{2}t^2 c_{nm}| + \sum_{m=1}^n |\psi_{nm}(t) - 1 + \tfrac{1}{2}t^2 c_{nm}|\\
&\le |t|^2 \sum_{m=1}^n \mathbb{E}\big[X_{nm}^2(1 \wedge |tX_{nm}|)\big] + |t|^2 \sum_{m=1}^n \mathbb{E}\big[\zeta_{nm}^2(1 \wedge |t\zeta_{nm}|)\big].
\end{aligned}$$

It remains to note that

$$\sum_{m=1}^n \mathbb{E}\big[X_{nm}^2(1 \wedge |tX_{nm}|)\big] \le \varepsilon|t| \sum_{m=1}^n c_{nm} + \sum_{m=1}^n \mathbb{E}\big[X_{nm}^2; |X_{nm}| > \varepsilon\big],$$

vanishes as $n \to \infty$ and $\varepsilon \downarrow 0$, by the Lindeberg condition and $c_n \to a^\top \Sigma a$. Likewise,

$$\sum_{m=1}^n \mathbb{E}\big[\zeta_{nm}^2(1 \wedge |t\zeta_{nm}|)\big] \le |t|^3 \mathbb{E}\big[|\xi|^3\big] \sum_{m=1}^n c_{nm}^{3/2} \le |t|^3 \mathbb{E}\big[|\xi|^3\big]\, c_n \sup_m c_{nm}^{1/2} \to 0,$$

where $\xi \sim N(0,1)$. □

A simple and effective way of checking Lindeberg's condition is through a moment condition, as given by the following result.

Lemma 1.4 (Lyapounov's condition) *If a triangular array (Z_{nm}) satisfies Lyapounov's condition of order $\alpha > 2$, i.e.*

$$\sum_{m=1}^{n} \mathbb{E}\left[\|Z_{nm}\|^{\alpha}\right] \to 0,$$

then Lindeberg's condition is fulfilled.

Proof. Fix $c > 0$. On the event $\{\|Z_{nm}\| > c\}$ we have $\|Z_{nm}\|^2 \le c^{2-\alpha}\|Z_{nm}\|^{\alpha}$ since $\alpha > 2$. Hence

$$\sum_{m=1}^{n} \mathbb{E}\big[\|Z_{nm}\|^2; \|Z_{nm}\| > c\big] \le c^{2-\alpha} \sum_{m=1}^{n} \mathbb{E}\big[\|Z_{nm}\|^{\alpha}\big] \to 0,$$

which is exactly Lindeberg's condition. □

To see the Lindeberg condition in action, we consider a portfolio with non-identical risks.

Example 1.6 (Portfolio with heterogeneous risks) Let (Z_n) represent $\mathbb{R}$-valued independent but not identically distributed risks. By subtracting their means we can assume w.l.o.g. that they have mean zero. Let

$$S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Z_i, \quad \sigma_i = \mathrm{Var}(Z_i)/n, \quad s_n = \Big(\sum_{i=1}^{n} \sigma_i\Big)^{1/2}.$$

We are interested in conditions under which $S_n/s_n \xrightarrow{d} N(0,1)$. One can achieve this by applying Lindeberg's condition to the array defined by $Z_{nm} = n^{-1/2} Z_m / s_n$, assuming s_n has a positive finite limit $\bar{\sigma}$. For instance, if there exists a constant M such that $|Z_m| \le M$ for all m, then since $n^{1/2} s_n \to \infty$, there exists N such that for $n \ge N$ we have $n^{1/2} s_n > M/c$, and so

$$\sum_{m=1}^{n} \int_{(|Z_{nm}|>c)} |Z_{nm}|^2 \, \mathrm{d}\mathbb{P} = \frac{1}{n s_n^2} \sum_{m=1}^{n} \mathbb{E}[|Z_m|^2 1_{(|Z_m| > c n^{1/2} s_n)}] = 0.$$

for every $n \ge N$, i.e. Lindeberg's condition is fulfilled. ◇

We turn our attention to uniform consistency. By the law of large numbers, we know that the empirical distribution function

$$\mathbb{F}^{(n)}(t) = \frac{1}{n} \sum_{i=1}^{n} 1_{(Z_i \le t)}$$

converges almost surely to $F(t) = \mathbb{P}(Z \le t)$ for any fixed t. Sometimes, we may be interested in providing a pathwise result of the convergence, for instance guaranteeing that the latter convergence is uniform in t. That is exactly what the following celebrated theorem states.

Theorem 1.8 (Glivenko–Cantelli) *Let (Z_n) be a sequence of iid real-valued random variables with distribution function F. It holds that*

$$\sup_{t\in\mathbb{R}} |\mathbb{F}^{(n)}(t) - F(t)| = \sup_{t\in\mathbb{R}} \left|\frac{1}{n}\sum_{i=1}^{n} 1_{(Z_i\le t)} - F(t)\right| \xrightarrow{a.s.} 0.$$

Using the same proof technique, we prove a slightly more general result which is useful when generalising to other estimators that are not bounded but still non-decreasing and càdlàg (right-continuous with left limits); then the above result follows as a corollary. Crucially, the limit must be bounded, but the requirement does not extend to the approximating sequence.

Lemma 1.5 (Abstract Glivenko–Cantelli) *Assume that $T\colon \mathbb{R}\to\mathbb{R}$ is a non-decreasing, bounded and càdlàg function. Assume that $(T^{(n)})$ is a sequence of random non-decreasing càdlàg maps $T^{(n)}\colon \mathbb{R}\to\mathbb{R}$ such that $T^{(n)}(x) \xrightarrow{a.s.} T(x)$ and $T^{(n)}(x-) \xrightarrow{a.s.} T(x-)$ for all $x\in\mathbb{R}$. It then holds that*

$$\sup_{x\in\mathbb{R}} |T^{(n)}(x) - T(x)| \xrightarrow{a.s.} 0.$$

Proof. The idea is to control oscillations between jumps by a finite partition and use pointwise convergence on the grid. Given $\varepsilon > 0$ choose some $k\in\mathbb{N}$ and a sequence $-\infty = x_0 < \cdots < x_k = \infty$ satisfying $T(x_j-) - T(x_{j-1}) < \varepsilon$. We can do this by including possible jumps in the sequence. By assumption it holds for each finite point in the partition x_j that

$$|T^{(n)}(x_j) - T(x_j)| \xrightarrow{a.s.} 0, \quad |T^{(n)}(x_j-) - T(x_j-)| \xrightarrow{a.s.} 0.$$

Let

$$\Delta^{(n)} = \max_{j=1,\ldots,k-1} \left\{|T^{(n)}(x_j) - T(x_j)|, |T^{(n)}(x_j-) - T(x_j-)|\right\}$$

Let $x\in\mathbb{R}$ be arbitrary. By construction there is a j such that $x\in[x_{j-1}, x_j)$ and we have

$$T^{(n)}(x_{j-1}) - T(x_{j-1}) - \varepsilon \le T^{(n)}(x_{j-1}) - T(x_j-) \le T^{(n)}(x) - T(x)$$

and also

$$T^{(n)}(x) - T(x) \le T^{(n)}(x_j-) - T(x_{j-1}) \le T^{(n)}(x_j-) - T(x_j-) + \varepsilon.$$

Then

$$\begin{aligned} &T^{(n)}(x_{j-1}) - T(x_{j-1}) - \varepsilon \le T^{(n)}(x) - T(x) \le T^{(n)}(x_j-) - T(x_j-) + \varepsilon \\ &\Rightarrow |T^{(n)}(x) - T(x)| \le \Delta^{(n)} + \varepsilon. \end{aligned}$$

Since x was arbitrary, it holds that

$$\sup_{x\in\mathbb{R}} |T^{(n)}(x) - T(x)| \le \Delta^{(n)} + \varepsilon.$$

Note that $\Delta^{(n)}$ is the maximum of finitely many terms converging to zero almost surely, so $\Delta^{(n)} \xrightarrow{a.s.} 0$, and in turn

$$\lim_{n\to\infty} \sup_{x\in\mathbb{R}} |T^{(n)}(x) - T(x)| \le \varepsilon, \quad \text{a.s.}$$

Since $\varepsilon > 0$ was arbitrary, the result follows. □

A similar result can be established for arrays, though the proof is not available through standard techniques, see Theorem 3.2.1 in Shorack and Wellner [2009].

Theorem 1.9 (Extended Glivenko–Cantelli) *Let (Z_{nm}) be a triangular array of real-valued random variables with arbitrary distribution functions F_{nm} and empirical distribution function*

$$\mathbb{F}^{(n)}(t) = \frac{1}{n}\sum_{m=1}^{n} 1_{(Z_{nm} \leq t)}$$

for every n. Let $F_n = \frac{1}{n}\sum_{m=1}^{n} F_{nm}$. Then it holds that

$$\sup_{t \in \mathbb{R}} |\mathbb{F}^{(n)}(t) - F_n(t)| \overset{a.s.}{\to} 0.$$

We now introduce an important tool for transferring asymptotic distributions through transformations of estimators. It is to be distinguished from the continuous mapping theorem in that the transformation happens before normalisation.

Theorem 1.10 (δ-method) *Let (T_n) and Z be $\mathbb{R}^p$-valued random variables and a_n be a positive deterministic sequence. Let $h : \mathbb{R}^p \to \mathbb{R}$ be a differentiable function at c, with derivative $\nabla h(c)$. If $a_n(T_n - c) \overset{d}{\to} Z$ then*

$$a_n(h(T_n) - h(c)) \overset{d}{\to} \nabla h(c)^\top Z.$$

Proof. We linearise h around c and then show the remainder is negligible at the scale a_n. By Slutsky's Lemma 1.3 (with the constant sequence) and the continuous mapping Theorem 1.4 (for the inner product operation) we have that

$$\nabla h(c)^\top a_n(T_n - c) \overset{d}{\to} \nabla h(c)^\top Z.$$

Now, from Example 1.4 we know that $a_n(T_n - c) = O_{\mathbb{P}}(1)$, or in other words $(T_n - c) = O_{\mathbb{P}}(a_n^{-1})$. By Taylor expansion and Example 1.5 we get

$$h(T_n) - h(c) = \nabla h(c)^\top (T_n - c) + o(\|T_n - c\|) = \nabla h(c)^\top (T_n - c) + o_{\mathbb{P}}(a_n^{-1}),$$

so that by multiplication with a_n we may piece the two results together to obtain

$$a_n(h(T_n) - h(c)) = \nabla h(c)^\top a_n(T_n - c) + o_{\mathbb{P}}(1) \overset{d}{\to} \nabla h(c)^\top Z$$

by Exercise 6. □

1.3 Basic exploratory tools

In anticipation of specialised tools in the subsequent chapters, we now introduce some basic functionals for exploring the distribution of a dataset and indicate how the above results can be used to quantify their uncertainty. The focus is on functionals that later appear in tail modelling and dependence analysis, and we emphasise their large-sample behaviour.

1.3.1 Some concrete estimators

We start with variance as a baseline measure of dispersion and then turn to tail-oriented diagnostics. For measuring the spread of a distribution, it is common to use the empirical variance, which is defined as

$$\hat{\sigma}_n^2 = \frac{1}{n}\sum_{i=1}^n (Z_i - \bar{Z}_n)^2, \quad \bar{Z}_n = \frac{1}{n}\sum_{i=1}^n Z_i.$$

Proposition 1.1 (Asymptotics of the empirical variance) *Let (Z_n) be an iid sequence with finite fourth moment, and central moments $\mu^{(k)} = \mathbb{E}[(Z_1 - 1_{(k>1)}\mathbb{E}[Z_1])^k]$, $k = 1, \dots, 4$. Then we have*

$$\begin{aligned} \hat{\sigma}_n^2 &\overset{a.s.}{\to} \mu^{(2)}, \\ \sqrt{n}(\hat{\sigma}_n^2 - \mu^{(2)}) &\overset{d}{\to} N\big(0, \mu^{(4)} - (\mu^{(2)})^2\big). \end{aligned}$$

Proof. First note that

$$\hat{\sigma}_n^2 = h(T_n), \quad T_n = \Big(\frac{1}{n}\sum_{i=1}^n Z_i, \frac{1}{n}\sum_{i=1}^n Z_i^2\Big)^\top, \quad h(x,y) = y - x^2.$$

From this decomposition, the first part follows from the law of large numbers and the continuous mapping Theorem 1.4. Let the raw moments be given by μ_k, $k = 1, \dots, 4$. By Laplace's CLT, Theorem 1.6, $\sqrt{n}(T_n - (\mu_1, \mu_2)^\top) \overset{d}{\to} N(0, \Sigma)$ where

$$\Sigma = \begin{pmatrix} \mu_2 - \mu_1^2 & \mu_3 - \mu_1\mu_2 \\ \mu_3 - \mu_1\mu_2 & \mu_4 - \mu_2^2. \end{pmatrix}$$

We then apply the δ-method, Theorem 1.10, to the function h, with gradient $\nabla h(x,y) = (-2x, 1)^\top$. The resulting variance is calculated by

$$\nabla h(\mu_1, \mu_2)^\top \, \Sigma \, \nabla h(\mu_1, \mu_2) = \mu^{(4)} - (\mu^{(2)})^2.$$

□

As an illustration of the multivariate delta method, consider the sample correlation.

Example 1.7 (Sample correlation and Fisher's z-transform) Let (X_n, Y_n) be iid bivariate random variables with $\mathbb{E}[X_i] = \mathbb{E}[Y_i] = 0$, $\mathbb{E}[X_i^2] = \mathbb{E}[Y_i^2] = 1$, and $\mathrm{Corr}(X_i, Y_i) = \rho \in (-1, 1)$. Define

$$S_{XX} = \frac{1}{n}\sum_{i=1}^n X_i^2, \quad S_{YY} = \frac{1}{n}\sum_{i=1}^n Y_i^2, \quad S_{XY} = \frac{1}{n}\sum_{i=1}^n X_i Y_i,$$

and the sample correlation

$$r_n = \frac{S_{XY}}{\sqrt{S_{XX} S_{YY}}}.$$

By Laplace's CLT, Theorem 1.6,

$$\sqrt{n}\big((S_{XY}, S_{XX}, S_{YY}) - (\rho, 1, 1)\big) \xrightarrow{d} N(0, \Sigma),$$

where direct moment calculations for the bivariate normal yield

$$\Sigma = \begin{pmatrix} 1+\rho^2 & 2\rho & 2\rho \\ 2\rho & 2 & 2\rho^2 \\ 2\rho & 2\rho^2 & 2 \end{pmatrix}.$$

Let $g(a, b, c) = a/(bc)^{1/2}$. Then $\nabla g(\rho, 1, 1) = (1, -\rho/2, -\rho/2)^\top$, and the δ-method, Theorem 1.10, gives

$$\sqrt{n}(r_n - \rho) \xrightarrow{d} N\big(0, (1 - \rho^2)^2\big).$$

Define the Fisher z-transform $z_n = \frac{1}{2}\log\big((1 + r_n)/(1 - r_n)\big)$. Since $h(r) = \frac{1}{2}\log((1 + r)/(1 - r))$ has derivative $h'(\rho) = 1/(1 - \rho^2)$, another application of the δ-method yields

$$\sqrt{n}\big(z_n - \tfrac{1}{2}\log\big(\tfrac{1+\rho}{1-\rho}\big)\big) \xrightarrow{d} N(0, 1).$$

$\diamond$

The tail of a distribution refers to its survival behaviour at very large values. Such values are sometimes deemed "catastrophic" from a non-life insurance perspective, and their accurate estimation is crucial for risk management in both insurance and reinsurance. For determining or classifying tail behaviour, one can look at the following function.

Definition 1.6 (Empirical mean excess function) Let u be a threshold. Then

$$\mathbb{e}_n(u) = \frac{\sum_{i=1}^n Z_i 1(Z_i > u)}{\sum_{i=1}^n 1(Z_i > u)} - u,$$

is called the empirical mean excess function.

Clearly, the analysis of $\mathbb{e}_n(u)$ is equivalent to that of $\mathbb{e}_n(u) + u$. The following result establishes convergence to the true mean excess function, defined as $e_F(u) = \mathbb{E}[Z_1 - u \mid Z_1 > u]$. Define also $e_F^{(2)}(u) = \mathbb{E}[(Z_1 - u)^2 \mid Z_1 > u]$. A list of mean excess functions of several parametric families may be found in [Embrechts et al., 1997, p. 161].

Theorem 1.11 (Asymptotics of the empirical mean excess function) *Let (Z_n) be an iid sequence with finite second moment and with distribution function F. For any u such that $F(u) < 1$ we have*

$$\mathbb{e}_n(u) \xrightarrow{a.s.} e_F(u),$$
$$\sqrt{n}(\mathbb{e}_n(u) - e_F(u)) \xrightarrow{d} N\Big(0, \frac{e_F^{(2)}(u) - e_F(u)^2}{1 - F(u)}\Big).$$

Proof. We express $\mathbb{e}_n(u)$ as a smooth function of two sample averages and apply a bivariate CLT. The first part follows directly from the law of large numbers and the first part of the continuous mapping Theorem 1.4. By Laplace's CLT, Theorem 1.6,

$$T_n = \Big(\frac{1}{n}\sum_{i=1}^{n} Z_i 1_{(Z_i>u)}, \frac{1}{n}\sum_{i=1}^{n} 1_{(Z_i>u)}\Big)^\top$$

converges, when normalised, to a centred Gaussian vector with covariance matrix Σ with entries

$$\begin{aligned}\sigma_1^2 &= \mathbb{E}[Z_1^2 1_{(Z_1>u)}] - \mathbb{E}[Z_1 1_{(Z_1>u)}]^2\\ \sigma_2^2 &= \mathbb{P}(Z_1 > u)\mathbb{P}(Z_1 \le u)\\ \sigma_{12} &= \mathbb{E}[Z_1 1_{(Z_1>u)}]\mathbb{P}(Z_1 \le u).\end{aligned}$$

We then apply the δ-method, Theorem 1.10, to the function $h(x, y) = x/y$, with gradient $\nabla h(x, y) = (1/y, -x/y^2)^\top$. The resulting variance is calculated straightforwardly through

$$\nabla h(\mathbb{E}[Z_1 1_{(Z_1>u)}], \mathbb{P}(Z_1 > u))^\top \, \Sigma \, \nabla h(\mathbb{E}[Z_1 1_{(Z_1>u)}], \mathbb{P}(Z_1 > u)).$$

□

Notice, however, that the above result is pointwise in u and does not describe the behaviour of the process $u \mapsto \mathbb{e}_n(u)$ across thresholds.

Definition 1.7 (Empirical cumulative hazard without product integral) Let (Z_n) be an iid sequence. We define its empirical cumulative hazard by

$$\mathbb{A}^{(n)}(t) = \log(n) - \log\Big(\sum_{i=1}^{n} 1_{(Z_i>t)}\Big).$$

The following result establishes convergence to the cumulative hazard function

$$\Lambda(t) = -\log(1 - F(t)),$$

which for absolutely continuous F coincides with the usual integral representation

$$\int_{-\infty}^{t} \frac{\mathrm{d}F(s)}{\mathrm{d}s} \frac{1}{1-F(s)} \, \mathrm{d}s = \int_{-\infty}^{t} \frac{1}{1-F(s)} \, \mathrm{d}F(s).$$

Theorem 1.12 (Asymptotics of the cumulative hazard) *Let (Z_n) be an iid sequence with distribution function F. For any t such that $F(t) < 1$ we have*

$$\begin{aligned}\mathbb{A}^{(n)}(t) &\xrightarrow{a.s.} \Lambda(t),\\ \sqrt{n}(\mathbb{A}^{(n)}(t) - \Lambda(t)) &\xrightarrow{d} N\Big(0, \frac{F(t)}{1-F(t)}\Big).\end{aligned}$$

Proof. This is another direct δ-method application, now for the log transform of the empirical CDF. The first part follows directly from the law of large numbers and the first part of the continuous mapping Theorem 1.4. By Laplace's CLT, Theorem 1.6,

$$\sqrt{n}(\mathbb{F}^{(n)}(t) - F(t)) \xrightarrow{d} N\big(0, F(t)(1-F(t))\big).$$

Notice that

$$\mathbb{A}^{(n)}(t) = -\log(1-\mathbb{F}^{(n)}(t)).$$

Thus, we apply the δ-method, Theorem 1.10, to the function $h(x) = -\log(1-x)$, with gradient $h'(x) = 1/(1-x)$. The resulting variance is given by

$$h'(F(t))^2 F(t)(1-F(t)) = \frac{F(t)}{1-F(t)}.$$

□

1.3.2 Finite sample behaviour

Recall that given any estimator T_n with $\sqrt{n}(T_n - \eta) \xrightarrow{d} N(0,\sigma^2)$, one can approximate its distribution by considering

$$T_n \overset{d}{\approx} N(\eta, \sigma^2/n) \tag{1.3}$$

and obtain quantiles on the right-hand side expression to match the ones on the left-hand side. Of course, the variance is in general unknown, so that a common strategy is to approximate it by a suitable estimator $\hat{\sigma}_n^2$ and plug it into the relevant quantile calculation.

It should be noted that the quality of the approximation (1.3) can vary substantially. For instance, Figure 1.2 shows that for the given sample sizes, the cumulative hazard performs better visually than the mean excess function. The behaviour of an estimator before its limit is well approximated is often referred to as *finite sample behaviour*.

We see from the above theorems that we are able to capture the behaviour of several of the exploratory tools for a fixed time (or threshold). Similar results hold when considering finitely many time points. For instance,

$$\sqrt{n}\big(\mathbb{F}^{(n)}(t) - F(t), \mathbb{F}^{(n)}(s) - F(s)\big) \xrightarrow{d} N\big(0, \Sigma(s,t)\big). \tag{1.4}$$

where $\sigma_{12}(s,t) = \mathbb{E}[1_{(Z_1\le t)}1_{(Z_1\le s)}] - \mathbb{E}[1_{(Z_1\le t)}]\mathbb{E}[1_{(Z_1\le s)}] = F(s\wedge t) - F(s)F(t)$, and the variances are unchanged from the unidimensional case. This raises the question of whether one-dimensional central limit theorems suffice to describe the full behaviour of $\mathbb{F}^{(n)}$, especially for hypotheses that depend on the entire curve. The answer is negative. For instance, in Figure 1.3 we plot empirical versus true cumulative distribution functions for two sample sizes, where we wish to test whether F belongs to one of three given distributions. The probability that an empirical curve jumps at any fixed time $t_0 < \cdots < t_m$ is zero, so the maximal distance between curves requires a genuinely continuous inspection of the paths.

In fact, although finite-dimensional distributions are enough to characterise the distribution of a stochastic process, their convergence is not enough to establish *pathwise* (see next section) convergence between e.g. $t \mapsto \mathbb{F}^{(n)}(t)$ and a

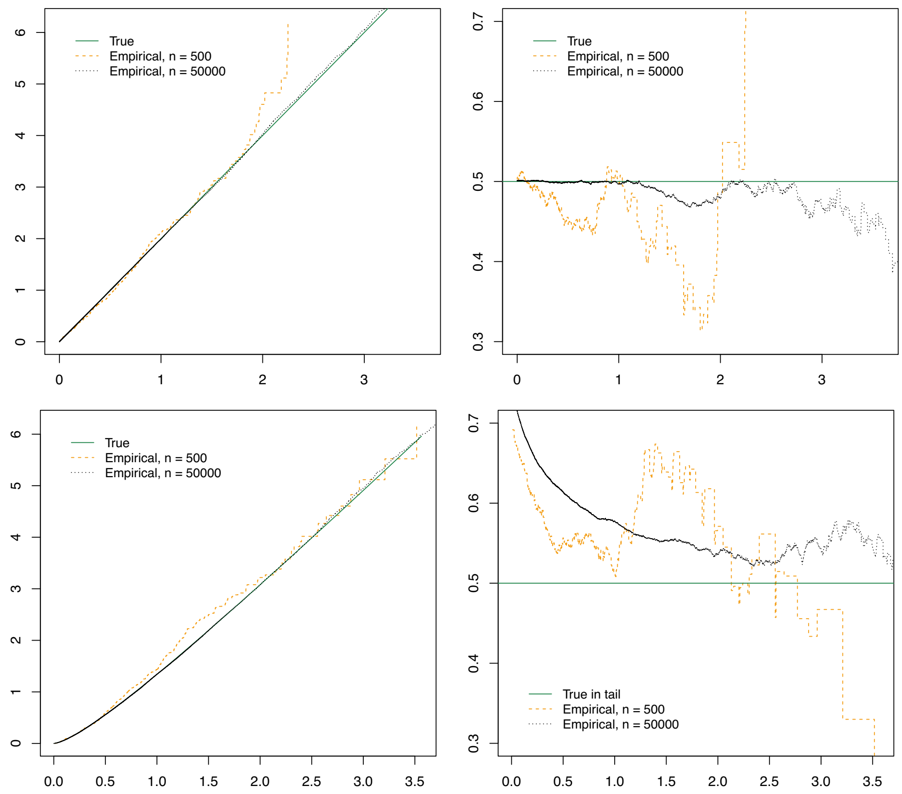

Figure 1.2: Empirical versus theoretical cumulative hazard (left, as a function of t) and mean excess functions (right, as a function of threshold u) for Exponential (top) and Gamma (bottom) distributed random variables.

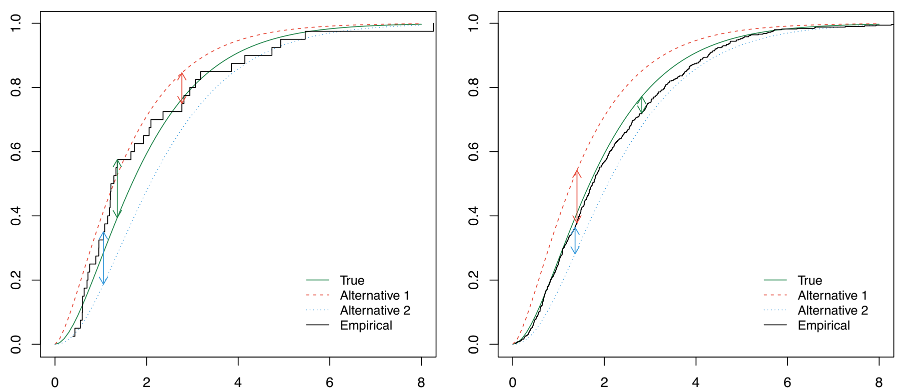

Figure 1.3: Empirical versus theoretical cumulative distribution functions $F(t)$ for Gamma distributed samples of size $n = 40$ (left) and $n = 500$ (right). Vertical arrows provide the maximal distance between curves.

suitable limit process. Therefore, our next task is to define pathwise convergence and provide additional conditions under which we may formulate *functional central limit theorems*.

1.4 Convergence of stochastic processes

In this section, we provide an overview of the relevant theory on stochastic processes and stochastic convergence, along with a couple of preliminary examples. The main application of these methods lies in nonparametric statistical procedures. The notation introduced here is used extensively in the sequel, particularly when establishing limit results for models in parametric models, extremes, survival analysis, and multi-state models. Additional results on the convergence of stochastic processes and the functional δ-method may be found in references such as van der Vaart and Wellner [1996] and van der Vaart [1998].

1.4.1 Introduction to stochastic processes

A stochastic process in continuous time is a collection of random variables indexed by some real subset, that is $X = (X_t)$, where typically $t \in I \subseteq \mathbb{R}$. Equivalently, from a pathwise point of view, given an underlying probability space $(\Omega, \mathcal{A}, \mathbb{P})$, the maps

$$\omega \mapsto X(\omega) = (X_t(\omega))_{t \in I}$$

define a random element in a suitable metric space of functions $(\mathbb{D}, d)$, where d is a suitable metric. Let $\mathcal{D}$ be the Borel σ-field generated by d. We usually refer to the space as $(\mathbb{D}, \mathcal{D})$ when no ambiguity arises. A schematic representation is provided in Figure 1.4.

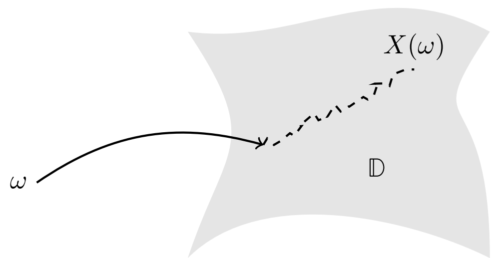

Figure 1.4: A schematic representation of a trajectory of a stochastic process X in the space $\mathbb{D}$.

Common choices for this metric space of functions are:

1. A finite-dimensional Euclidean space $\mathbb{D} = \mathbb{R}^p$, equipped with any norm.
2. $\mathbb{D} = C[a, b]$, the set of all continuous functions $z\colon [a, b] \to \mathbb{R}$ for $a, b \in \bar{\mathbb{R}}$, equipped with the uniform norm $\|z\|_{[a,b]} = \sup_{s \in [a,b]} |z(s)|$.

3. $\mathbb{D} = D[a,b]$, the set of all càdlàg functions $z\colon [a,b] \to \mathbb{R}$ for $a,b \in \bar{\mathbb{R}}$, equipped with the uniform norm $\|z\|_{[a,b]} = \sup_{s\in[a,b]} |z(s)|$. Another common choice is the Skorokhod norm.

4. $\mathbb{D} = \ell^\infty(\mathcal{F})$, the set of all uniformly bounded real functions on an arbitrary set $\mathcal{F}$: all functions $z\colon \mathcal{F} \to \mathbb{R}$ such that $\|z\|_{\mathcal{F}} = \sup_{f\in\mathcal{F}} |z(f)| < \infty$, equipped with the uniform norm $\|z\|_{\mathcal{F}}$. For instance, $\mathcal{F}$ could be a Euclidean subset.

Note that the convergence theory sketched below reduces to the classical one in finite dimensions for the first case above. The key observation in that case is that all norms in finite-dimensional spaces are equivalent.

In any of the four cases, to speak about the distribution of X, and hence about convergence of distributions, we define $C_b(\mathbb{D})$, the set of all bounded continuous functions on $\mathbb{D}$, and consider expectations of the kind

$$\mathbb{E}f(X), \quad f \in C_b(\mathbb{D}). \tag{1.5}$$

The spaces in items 1 and 2 are usually measurable and thus the above expectation is well-defined. Unfortunately, when dealing with processes in 3 or 4, subtle but important measurability issues arise when using the supremum norm. For instance, for $\mathbb{F}^{(n)}(t) = n^{-1}\sum_{i=1}^n 1_{(Z_i\le t)}$ the process

$$t \mapsto X_n(t) = \sqrt{n}\big(\mathbb{F}^{(n)}(t) - F(t)\big), \quad t \in [a,b],$$

has paths in $\mathbb{D} = D[a,b]$ but the Borel σ-field $\mathcal{D}$ is too large for the pre-image to satisfy $X_n^{-1}(\mathcal{D}) \subseteq \mathcal{B}^n$, where $\mathcal{A} = \mathcal{B}^n$ is the underlying product space supporting $Z_1,\ldots,Z_n$.

Equipping the target space with the sup-norm is natural, since it allows us to study uniform convergence of paths. However, as mentioned, it also causes measurability issues. We address these by working with outer expectations, which coincide with the usual expectation when measurability holds, and allow us to extend standard convergence results.

1.4.2 Convergence of processes

We introduce the concept of *weak convergence* for stochastic processes. Note, however, that thanks to the wealth of theorems available for this extended notion, we seldom have to deal with the foundational definitions directly. First, we introduce the so-called *outer integrals* and *outer probabilities*. They play a fundamental role in the rigorous construction of the theory, though their practical role is minor in terms of applications to statistics for insurance.

Definition 1.8 (Inner and outer integral and probability) Let $(\Omega,\mathcal{A},\mathbb{P})$ be a probability space and let $T\colon \Omega \to \bar{\mathbb{R}}$ be an arbitrary map. The outer integral of T with respect to $\mathbb{P}$ is defined as

$$\mathbb{E}^*[T] = \inf\big\{\mathbb{E}[U] \text{ such that } U \ge T, U\colon \Omega \to \bar{\mathbb{R}} \text{ measurable with } \mathbb{E}[U] < \infty\big\}.$$

The outer probability of a subset $B \subseteq \Omega$ is

$$\mathbb{P}^*(B) = \inf\{\mathbb{P}(A) : B \subseteq A, A \in \mathcal{A}\}.$$

The inner integral and inner probability are respectively defined as

$$\mathbb{E}_* [T] = -\mathbb{E}^* [-T] , \quad \mathbb{P}_*(B) = 1 - \mathbb{P}^* (\Omega \setminus B) .$$

Outer probabilities enter the picture because sup-norm functionals over uncountable index sets need not be measurable. In most applications we work with separable or càdlàg versions, in which case the outer and ordinary notions coincide for sets of interest (more precisely on sets generated by balls in $\mathbb{D}$; see [van der Vaart and Wellner, 1996, Section 1.7]), so the machinery is primarily technical rather than conceptual. Nonetheless, it is important to remember that there is a subtle distinction.

Convergence in distribution of a stochastic process evaluated at fixed indices is referred to as convergence in finite dimensions. The latter notion is implied by the following stronger condition.

Definition 1.9 (Weak convergence) Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. Let $(\mathbb{D}, d)$ be a metric space and let $\mathcal{D}$ be the Borel σ-field on $\mathbb{D}$. Let (X_n) be a sequence of arbitrary maps $X_n \colon \Omega \to \mathbb{D}$. The sequence (X_n) converges weakly to a Borel probability measure[1] L if

$$\mathbb{E}^* [f(X_n)] \to \int f \, \mathrm{d}L \quad \text{for every } f \in C_b(\mathbb{D}).$$

This is denoted $X_n \xrightarrow{d} L$ or $X_n \xrightarrow{d} X$ if $X \sim L$.

Notice that the definition subsumes the distributional convergence in finite dimensions, and therefore the use of the same symbol is justified.

Remark 1.3 The limit X is usually a continuous stochastic process, which is measurable with the uniform metric, and so we can write $X_n \xrightarrow{d} X$ whenever, equivalently,

$$\mathbb{E}^* [f(X_n)] \to \mathbb{E} [f(X)] \quad \text{for every } f \in C_b(\mathbb{D}).$$

Also note that the metric on $\mathbb{D}$ does not play a role in the existence within this definition, since the outer expectation always exists (we have $\inf \emptyset = \infty$). ○

The following lemma shows that establishing weak convergence in a larger space than required is not a problem, as it is equivalent to showing it in the possibly smaller space where the processes take values.

Lemma 1.6 (Subspaces) *Let $\mathbb{D}_0 \subseteq \mathbb{D}$ be arbitrary metric spaces equipped with the same metric. If X and every X_n take their values in $\mathbb{D}_0$, then $X_n \xrightarrow{d} X$ as maps in $\mathbb{D}_0$ if and only if $X_n \xrightarrow{d} X$ as maps in $\mathbb{D}$.*

For stochastic processes we may also define convergence in probability and almost surely using the outer probability notion. Again, the definitions coincide in finite dimensions and therefore the same notation is justified.

Definition 1.10 (Convergence in outer probability and outer almost surely) Let (X_n) be a sequence of arbitrary maps $X_n \colon \Omega \to \mathbb{D}$ and let $X \colon \Omega \to \mathbb{D}$.

[1] A Borel probability measure L is a probability measure defined on the Borel σ-field $\mathcal{D}$.

(i) (X_n) converges in *outer probability* to X if $\mathbb{P}^*(d(X_n, X) > \varepsilon) \to 0$ for every $\varepsilon > 0$. This is denoted $X_n \overset{\mathbb{P}}{\to} X$.

(ii) (X_n) converges *outer almost surely* to X if $\mathbb{P}^*(\lim_{n\to\infty} d(X_n, X) = 0) = 1$ for d the metric of interest. This is denoted $X_n \overset{a.s.}{\to} X$.

The following result generalises Lemma 1.2.

Lemma 1.7 (Relation between types of convergence) *Let (X_n) be a sequence of arbitrary maps and X Borel measurable. Then*

1. $X_n \overset{a.s.}{\to} X$ *implies* $X_n \overset{\mathbb{P}}{\to} X$.
2. $X_n \overset{\mathbb{P}}{\to} X$ *implies* $X_n \overset{d}{\to} X$.
3. *Let* $c \in \mathbb{D}$. $X_n \overset{\mathbb{P}}{\to} c$ *if and only if* $X_n \overset{d}{\to} c$.

Sometimes it can be convenient to deal with stochastic processes and their distributional representations using results from real analysis directly; in that connection the following result allows for such treatment in an appropriate probability space, sometimes referred to as a Skorokhod construction.

Theorem 1.13 (A.s. representation) *Let $X_n : \Omega \to \mathbb{D}$ be an arbitrary sequence of maps, and let X be Borel measurable and separable. If $X_n \overset{d}{\to} X$, then there exists a probability space $\left(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{\mathbb{P}}\right)$ and maps $\tilde{X}_n : \tilde{\Omega} \to \mathbb{D}$ with*

1. $\tilde{X}_n \overset{a.s.}{\to} \tilde{X}$;
2. $\mathbb{E}^*\left[f(\tilde{X}_n)\right] = \mathbb{E}^*\left[f(X_n)\right]$, *for every bounded* $f : \mathbb{D} \to \mathbb{R}$ *and every* n.

The following definition is quite technical, but required for full understanding of weak convergence of stochastic processes. All limit processes considered in the sequel are tight, as a consequence of some general-purpose central limit theorems given below.

Definition 1.11 (Tightness) A Borel probability measure L is *tight* if for every $\varepsilon > 0$ there exists a compact set K with $L(K) \geq 1 - \varepsilon$. Likewise we say the process $X \sim L$ is tight if L is tight.

The following lemma allows us to establish the joint weak convergence of two stochastic processes from that of the individual processes and the cross-finite-dimensional distributions. This is useful since the latter condition usually follows easily by elementary means.

Lemma 1.8 (From marginal to joint weak convergence) *Let $X_n, X\colon \Omega \to \ell^\infty(T)$ and $Y_n, Y\colon \Omega \to \ell^\infty(S)$, satisfy $X_n \overset{d}{\to} X$ and $Y_n \overset{d}{\to} Y$ with tight X and Y. If additionally*

$$(X_n(t_1), \ldots, X_n(t_k), Y_n(s_1), \ldots, Y_n(s_l)) \overset{d}{\to} (X(t_1), \ldots, X(t_k), Y(s_1), \ldots, Y(s_l)),$$

for any $t_1, \ldots, t_k \in T$ and $s_1, \ldots, s_l \in S$, then there exist versions of X and Y with bounded sample paths and $(X_n, Y_n) \overset{d}{\to} (X, Y)$ in $\ell^\infty(T) \times \ell^\infty(S)$.

We state a general version of the continuous mapping theorem. Slutsky's lemma in combination with the continuous mapping theorem is a strong tool used to conclude weak convergence for different kinds of processes in what follows.

Theorem 1.14 (Continuous mapping) *Let $\mathbb{D}_n \subseteq \mathbb{D}$ and $g_n \colon \mathbb{D}_n \to \mathbb{E}$ satisfy the following: if $x_n \to x$ with $x_n \in \mathbb{D}_n$ for every n and $x \in \mathbb{D}_0$, then $g_n(x_n) \to g(x)$, where $\mathbb{D}_0 \subseteq \mathbb{D}$ and $g \colon \mathbb{D}_0 \to \mathbb{E}$. Let (X_n) be a sequence of maps with values in $\mathbb{D}_n$, let X be Borel measurable and tight, and take values in $\mathbb{D}_0$. Then*

1. $X_n \overset{a.s.}{\to} X$ *implies* $g_n(X_n) \overset{a.s.}{\to} g(X)$.
2. $X_n \overset{\mathbb{P}}{\to} X$ *implies* $g_n(X_n) \overset{\mathbb{P}}{\to} g(X)$.
3. $X_n \overset{d}{\to} X$ *implies* $g_n(X_n) \overset{d}{\to} g(X)$.

Lemma 1.9 (Slutsky's lemma for general random elements) *If $X_n \overset{d}{\to} X$ and $Y_n \overset{d}{\to} c$ with X tight and c a constant then $(X_n, Y_n) \overset{d}{\to} (X, c)$.*

Notice that the following theorem is akin to the Glivenko–Cantelli result, and the proof is also very similar.

Theorem 1.15 (Dini's theorem, for monotone processes) *Let (X_n) be a sequence of monotone stochastic processes indexed by (some subset of) $\mathbb{R}$ converging marginally in probability to a continuous deterministic function f. Then the convergence is uniform in probability on compact subsets of the index set.*

Proof. Monotonicity lets us reduce uniform deviations to a finite grid. Assume w.l.o.g. that the processes are non-decreasing. Let $\eta > 0$ and consider a partition $a = s_0 < s_1 < \cdots < s_K = b$ such that $f(s_i) - f(s_{i-1}) \leq \eta/2$. For $s \in [s_i, s_{i+1})$ we have

$$\begin{aligned} &X_n(s) - f(s) \leq X_n(s_{i+1}) - f(s_{i+1}) + f(s_{i+1}) - f(s) \\ &\leq X_n(s_{i+1}) - f(s_{i+1}) + \eta/2 \leq \max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| + \eta/2. \end{aligned}$$

Similarly $X_n(s) - f(s) \geq -\max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| - \eta/2$, which gives

$$\sup_{s \in [a,b]} |X_n(s) - f(s)| \leq \max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| + \eta/2.$$

Using the convergence in probability of the K finite terms on the right-hand side, we obtain

$$\begin{aligned} &\limsup_{n \to \infty} \mathbb{P}\Big(\sup_{s \in [a,b]} |X_n(s) - f(s)| > \eta \Big) \\ &\leq \limsup_{n \to \infty} \mathbb{P}\Big(\max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| > \eta/2 \Big) = 0. \end{aligned}$$

This finishes the proof. $\square$

1.4.3 The functional δ-method

When dealing with stochastic processes, the nature of transformations that we can apply to their paths can be rather complex, since they may vary in a pathwise manner. The *functional δ-method* is a tool that preserves weak convergence of stochastic processes when transformed under pathwise maps that satisfy the natural but important requirement of *Hadamard differentiablility.*

Definition 1.12 (Hadamard differentiability) Let $\mathbb{E}$ and $\mathbb{D}$ be two normed spaces. A map $\phi\colon \mathbb{D}_\phi \subseteq \mathbb{D} \to \mathbb{E}$ is called Hadamard differentiable at $\theta \in \mathbb{D}_\phi$ if there is a continuous linear map $\phi'_\theta : \mathbb{D}_\phi \to \mathbb{E}$ such that

$$\frac{\phi\left(\theta + t_n h_n\right) - \phi\left(\theta\right)}{t_n} \to \phi'_\theta(h)$$

for all converging scalar sequences $t_n \to 0$ and functions $h_n \to h$ such that $\theta + t_n h_n \in \mathbb{D}_\phi$ for every n. We say that ϕ is *Hadamard differentiable tangentially* to a set $\mathbb{D}_0 \subseteq \mathbb{D}$ by requiring that every $h_n \to h$ in the definition has $h \in \mathbb{D}_0$ and the derivative need only be defined on $\mathbb{D}_0$. For a specific h, we speak of Hadamard differentiability in the direction of h.

Remark 1.4 (Hadamard differentiability implies continuity) Hadamard differentiability implies continuity at θ in the norm of $\mathbb{E}$ (e.g. the supremum norm). Indeed, let $x_n \to \theta$ in $\mathbb{D}$ and set $r_n = x_n - \theta$. Define $t_n = \|r_n\|^{1/2}$ and $h_n = r_n/t_n$, so that $t_n \to 0$, $h_n \to 0$, and $\theta + t_n h_n = x_n$. The definition of Hadamard differentiability yields

$$\frac{\phi(\theta + r_n) - \phi(\theta)}{t_n} \to \phi'_\theta(0),$$

hence $\|\phi(x_n) - \phi(\theta)\| = o(t_n) \to 0$. Thus ϕ is continuous at θ for whichever norm is placed on $\mathbb{E}$, as long as Hadamard differentiability holds in that norm. ○

Notice that the above defines a pathwise derivative as a map of functions, so one needs to know how it transforms functions to be able to evaluate it. An effective tool is the following.

Theorem 1.16 (Chain rule) *Let the following maps be Hadamard differentiable:*

1. $\phi : \mathbb{D}_\phi \subseteq \mathbb{D} \to \mathbb{E}_\phi$ *at* $\theta \in \mathbb{D}_\phi$ *tangentially to* $\mathbb{D}_0$*, and*
2. $\psi : \mathbb{E}_\psi \supseteq \mathbb{E}_\phi \to \mathbb{F}$ *at* $\phi(\theta)$ *tangentially to* $\phi'_\theta(\mathbb{D}_0)$*.*

Then $\psi \circ \phi$ *is Hadamard differentiable at* θ *tangentially to* $\mathbb{D}_0$ *with derivative* $\psi'_{\phi(\theta)} \circ \phi'_\theta$*.*

Proof. We rewrite the difference quotient for $\psi \circ \phi$ and pass to the limit in two stages. We may simply rewrite

$$\frac{\psi \circ \phi(\theta + t_n h_n) - \psi \circ \phi(\theta)}{t_n} = \frac{\psi(\phi(\theta) + t_n\{\frac{\phi(\theta + t_n h_n) - \phi(\theta)}{t_n}\}) - \psi(\phi(\theta))}{t_n},$$

and obtain the limit by the composition rule for limits in normed spaces, using the Hadamard differentiability of ϕ with derivative ϕ'_θ for the expression in brackets, and then the Hadamard differentiability of ψ. □

Before providing more specialised Hadamard differentiability results of intricate functions, we provide a few elementary examples, all with respect to the uniform norm.

Proposition 1.2 (Elementary Hadamard differentiability) *Fix $\tau > 0$ and equip $\mathbb{D} = \ell^\infty([0,\tau])$ with the sup norm $\|\cdot\|_\infty$.*

1. *Weighted integral functional. Let w be bounded and measurable on $[0,\tau]$, and define*
$$\phi(F) = \int_0^\tau F(t)w(t)\ \mathrm{d}t, \quad F \in \mathbb{D}.$$
Then $\phi : \mathbb{D} \to \mathbb{R}$ is Hadamard differentiable at every F tangentially to $\mathbb{D}$ with derivative $\phi'_F(h) = \int_0^\tau h(t)w(t)\ \mathrm{d}t$.

2. *Cumulative integral map. Define*
$$\Phi(F)(t) = \int_0^t F(s)\ \mathrm{d}s, \quad 0 \le t \le \tau.$$
Then $\Phi : \mathbb{D} \to \mathbb{D}$ is Hadamard differentiable at every F tangentially to $\mathbb{D}$ with derivative $\Phi'_F(h)(t) = \int_0^t h(s)\ \mathrm{d}s$.

3. *Ratio of integrals. Let w be bounded and measurable on $[0,\tau]$, and set*
$$\psi(F) = \frac{\int_0^\tau F(t)\ \mathrm{d}t}{\int_0^\tau F(t)w(t)\ \mathrm{d}t}, \quad F \in \mathbb{D}_\psi := \Big\{F \in \mathbb{D} : \int_0^\tau F(t)w(t)\ \mathrm{d}t \ne 0\Big\}.$$
Then ψ is Hadamard differentiable at any $F \in \mathbb{D}_\psi$, with
$$\psi'_F(h) = \frac{\int_0^\tau h(t)\ \mathrm{d}t}{\int_0^\tau F(t)w(t)\ \mathrm{d}t} - \frac{\int_0^\tau F(t)\ \mathrm{d}t}{\big(\int_0^\tau F(t)w(t)\ \mathrm{d}t\big)^2}\int_0^\tau h(t)w(t)\ \mathrm{d}t.$$

Proof. For the first statement, let $t_n \to 0$ and $h_n \to h$ in $\|\cdot\|_\infty$, and simply note that
$$\frac{\phi(F + t_n h_n) - \phi(F)}{t_n} - \phi'_F(h) = \int_0^\tau (h_n(t) - h(t))w(t)\ \mathrm{d}t.$$
Hence
$$\left|\int_0^\tau (h_n - h)w\ \mathrm{d}t\right| \le \|h_n - h\|_\infty \int_0^\tau |w(t)|\ \mathrm{d}t \le \tau\|w\|_\infty\|h_n - h\|_\infty \to 0.$$
For the second statement, for each t, $\{\Phi(F+t_nh_n)-\Phi(F)\}(t)/t_n = \int_0^t h_n(s)\ \mathrm{d}s$. Thus
$$\sup_{0\le t\le\tau}\left|\int_0^t (h_n - h)(s)\ \mathrm{d}s\right| \le \tau\|h_n - h\|_\infty \to 0,$$
so the convergence holds in $\|\cdot\|_\infty$. For the third statement, let $L_1(F) = \int_0^\tau F(t)\ \mathrm{d}t$ and $L_2(F) = \int_0^\tau F(t)w(t)\ \mathrm{d}t$. By the previous statements and Remark 1.4 both

are continuous linear functionals on $\mathbb{D}$. For $F \in \mathbb{D}_\psi$, we have $L_2(F) \neq 0$. Hence, by continuity, there exists N such that for $n \geq N$,

$$\big|L_2(F + t_n h_n) - L_2(F)\big| < \tfrac{1}{2}|L_2(F)|,$$

so $|L_2(F + t_n h_n)| \geq \frac{1}{2}|L_2(F)|$ and $F + t_n h_n \in \mathbb{D}_\psi$. Using linearity,

$$\frac{\psi(F + t_n h_n) - \psi(F)}{t_n} = \frac{L_1(h_n)L_2(F) - L_1(F)L_2(h_n)}{L_2(F + t_n h_n)L_2(F)}.$$

Since $L_1(h_n) \to L_1(h)$ and $L_2(h_n) \to L_2(h)$, and $L_2(F + t_n h_n) \to L_2(F)$, the limit is the stated derivative. □

We now provide some more complicated maps, which are used sporadically in the remaining chapters.

Lemma 1.10 (Hadamard differentiability of inverses and compositions) *The following hold.*

1. *We consider the inverse map with different support assumptions.*

   (i) Let $0 < u_1 < u_2 < 1$ and let F be continuously differentiable on $[a,b] = [F^{-1}(u_1) - \varepsilon, F^{-1}(u_2) + \varepsilon]$ with strictly positive derivative f. Then the inverse $\phi(h) = h^{-1}$ as a map from the set of restrictions of distribution functions to $[a,b]$, to $\ell^\infty([u_1,u_2])$ is Hadamard differentiable at F tangentially to $C[a,b]$. The derivative is

   $$\phi'_F(h)(\cdot) = -(h/f) \circ F^{-1}(\cdot).$$

   (ii) Let F have compact support $[a,b]$ and be continuously differentiable on $[a,b]$ with strictly positive derivative f. Then the inverse $\phi(h) = h^{-1}$ as a map from the set of distribution functions supported on $[a,b]$, to $\ell^\infty(0,1)$ is Hadamard differentiable at F tangentially to $C[a,b]$. The derivative is again

   $$\phi'_F(h)(\cdot) = -(h/f) \circ F^{-1}(\cdot).$$

2. *Let $\varphi : (a,b) \to \mathbb{R}$ be a function with uniformly continuous and bounded derivative. Let $\phi(F) = \varphi(F)$ be a map from the subset of functions in $\ell^\infty(\mathcal{F})$ with image in (a,b), to $\ell^\infty(\mathcal{F})$. The derivative of ϕ for F in such subset is*

   $$\phi'_F(h)(\cdot) = \varphi'(F(\cdot))h(\cdot).$$

3. *Suppose $F_1 : \mathcal{X} \to \mathcal{Y}$ is a map, and $F_2 : \mathcal{Y} \to \mathcal{Z}$ is Fréchet differentiable*[2] *uniformly in y in the range of F_1, with derivatives $F'_{2,y}$ such that $y \mapsto F'_{2,y}$ is uniformly norm-bounded. Then the composition $\phi(F_1,F_2)(x) = F_2 \circ F_1(x)$ is Hadamard differentiable at (F_1,F_2) tangentially to the set $\ell^\infty(\mathcal{X},\mathcal{Y}) \times UC(\mathcal{Y},\mathcal{Z})$*[3] *with derivative*

   $$\phi'_{(F_1,F_2)}(h_1,h_2)(x) = h_2 \circ F_1(x) + F'_{2,F_1(x)}(h_1(x)).$$

[2] A technical strengthening of Hadamard differentiability, where the limit should be continuous and linear, and the convergence uniform in h on bounded subsets of $\mathbb{D}_\phi$.

[3] The space of uniformly continuous functions from $\mathcal{Y} \to \mathcal{Z}$.

Proof. We treat the three items in turn.

1. Let $t_n \to 0$ and let $h_n \to h$ uniformly on $[a,b]$, with $h \in C[a,b]$, such that $F + t_n h_n$ lies in the stated domain for all n. For $u \in (0,1)$ set $q_u = F^{-1}(u)$ and $q_{u,n} = (F + t_n h_n)^{-1}(u)$.

   We first prove a generic pointwise expansion for fixed $u \in (0,1)$. Choose $\varepsilon_n > 0$ with $\varepsilon_n = o(t_n)$. By the definition of the inverse,

   $$(F + t_n h_n)(q_{u,n} - \varepsilon_n) \leq u \leq (F + t_n h_n)(q_{u,n}).$$

   Since (h_n) is uniformly bounded, this yields

   $$F(q_{u,n} - \varepsilon_n) + O(t_n) \leq u \leq F(q_{u,n}) + O(t_n).$$

   Since F is monotone and continuous with $F(q_u) = u$ and strictly positive derivative at q_u, the function F is bounded away from u outside any neighbourhood of q_u. Hence the preceding inequalities imply $q_{u,n} \to q_u$. Moreover, uniform convergence of h_n and continuity of h give $h_n(q_{u,n} - \varepsilon_n) \to h(q_u)$. By Taylor's theorem at q_u we obtain

   $$\begin{aligned} &u + (q_{u,n} - q_u) f(q_u) - o(|q_{u,n} - q_u|) + O(\varepsilon_n) + t_n h(q_u) - o(t_n) \leq u \\ &\leq u + (q_{u,n} - q_u) f(q_u) + o(|q_{u,n} - q_u|) + O(\varepsilon_n) + t_n h(q_u) + o(t_n), \end{aligned}$$

   and hence $q_{u,n} - q_u = O(t_n)$ and

   $$\frac{q_{u,n} - q_u}{t_n} \to -\frac{h(q_u)}{f(q_u)}.$$

   (i) The continuity and strict positivity of f on $[a,b]$ imply uniform continuity and a positive lower bound. The Taylor remainder above is therefore uniform in $u \in [u_1, u_2]$, and the same argument yields

   $$\sup_{u \in [u_1, u_2]} \left| \frac{q_{u,n} - q_u}{t_n} + \frac{h(q_u)}{f(q_u)} \right| \to 0,$$

   which proves Hadamard differentiability as a map into $\ell^\infty([u_1, u_2])$ with the stated derivative.

   (ii) Set $\varepsilon_{u,n} = t_n^2 \wedge (q_{u,n} - a) > 0$ for each $u \in (0,1)$. Because $F + t_n h_n$ lies in the stated domain, it is supported on $[a,b]$, so $a < q_{u,n} \leq b$ and thus

   $$(F + t_n h_n)(q_{u,n} - \varepsilon_{u,n}) \leq u \leq (F + t_n h_n)(q_{u,n}).$$

   The smoothness of F on $[a,b]$ implies that

   $$F(q_{u,n} - \varepsilon_{u,n}) = F(q_{u,n}) + O(\varepsilon_{u,n}),$$

   uniformly in $u \in (0,1)$. Since $h_n \to h$ uniformly, we may replace h_n by h at the cost of an $o(t_n)$ term. Hence

   $$-t_n h(q_{u,n}) + o(t_n) \leq F(q_{u,n}) - F(q_u) \leq -t_n h(q_{u,n} - \varepsilon_{u,n}) + o(t_n),$$

with $o(t_n)$ uniform in u. Since f is continuous and strictly positive on $[a,b]$, the middle term is bounded above and below by a constant times $|q_{u,n} - q_u|$. This yields $|q_{u,n} - q_u| = O(t_n)$ uniformly in u. Using uniform convergence of h_n and uniform continuity of h, the preceding Taylor expansion is uniform in $u \in (0,1)$, and we obtain

$$\sup_{u\in(0,1)} \left| \frac{q_{u,n} - q_u}{t_n} + \frac{h(q_u)}{f(q_u)} \right| \to 0.$$

This proves Hadamard differentiability as a map into $\ell^\infty(0,1)$ with the stated derivative.

2. Let $t_n \to 0$ and let $h_n \to h$ in $\ell^\infty(\mathcal{F})$ such that $F + t_n h_n$ remains in the stated domain for all n. For each $x \in \mathcal{F}$, the mean value theorem yields a $\xi_{n,x} \in (0,1)$ with

$$\varphi(F(x) + t_n h_n(x)) = \varphi(F(x)) + \varphi'(F(x) + \xi_{n,x} t_n h_n(x))\, t_n h_n(x).$$

Therefore

$$\begin{aligned} &\| \frac{\phi(F + t_n h_n) - \phi(F)}{t_n} - \varphi'(F) h \|_\infty \\ &\le \| \big(\varphi'(F + \xi_n t_n h_n) - \varphi'(F)\big) h_n \|_\infty + \| \varphi'(F)\, (h_n - h) \|_\infty, \end{aligned}$$

where ξ_n denotes the pointwise selection $x \mapsto \xi_{n,x}$. Since $h_n \to h$ in the sup norm, the sequence (h_n) is bounded and $t_n \|h_n\|_\infty \to 0$. By uniform continuity of φ' on (a,b) the first term vanishes, and the second term tends to 0 because φ' is bounded. Hence ϕ is Hadamard differentiable at F with derivative $\phi'_F(h)(\cdot) = \varphi'(F(\cdot)) h(\cdot)$.

3. Let $t_n \to 0$ and let $h_{1n} \to h_1$ and $h_{2n} \to h_2$ in $\ell^\infty(\mathcal{X}, \mathcal{Y})$ and $\ell^\infty(\mathcal{Y}, \mathcal{Z})$, respectively, with $h_2 \in UC(\mathcal{Y}, \mathcal{Z})$, and such that $F_1 + t_n h_{1n}$ and $F_2 + t_n h_{2n}$ remain in the stated domain for all n. Write

$$F'_{2,F_1}(h_1)(x) = F'_{2,F_1(x)}(h_1(x)), \quad x \in \mathcal{X},$$

and similarly for $F'_{2,F_1}(h_{1n})$. Then

$$\begin{aligned} &\frac{(F_2 + t_n h_{2n}) \circ (F_1 + t_n h_{1n}) - F_2 \circ F_1}{t_n} - h_2 \circ F_1 - F'_{2,F_1}(h_1) \\ &\quad = (h_{2n} - h_2) \circ (F_1 + t_n h_{1n}) + h_2(F_1 + t_n h_{1n}) - h_2(F_1) \\ &\qquad + \frac{F_2(F_1 + t_n h_{1n}) - F_2(F_1)}{t_n} - F'_{2,F_1}(h_{1n}) + F'_{2,F_1}(h_{1n} - h_1). \end{aligned}$$

The first term converges to zero in $\ell^\infty(\mathcal{X}, \mathcal{Z})$ because

$$\|(h_{2n} - h_2) \circ (F_1 + t_n h_{1n})\|_\infty \le \|h_{2n} - h_2\|_\infty \to 0.$$

The second term converges to zero since h_2 is uniformly continuous and $\|t_n h_{1n}\|_\infty \to 0$. For the third term, uniform Fréchet differentiability of F_2 in the range of F_1 yields

$$\sup_{x\in\mathcal{X}} \| \frac{F_2(F_1(x) + t_n h_{1n}(x)) - F_2(F_1(x))}{t_n} - F'_{2,F_1(x)}(h_{1n}(x)) \|_\infty \to 0.$$

Finally, by uniform boundedness of $y \mapsto F'_{2,y}$,

$$\|F'_{2,F_1}(h_{1n} - h_1)\|_\infty \le \sup_y \|F'_{2,y}\| \, \|h_{1n} - h_1\|_\infty \to 0.$$

Thus the difference quotient converges to $h_2 \circ F_1 + F'_{2,F_1}(h_1)$, and the stated derivative follows.

□

The two maps presented below are useful because of their link to the cumulative hazard function and to the distribution function and because they are Hadamard differentiable. They are used extensively in the final three chapters, where censoring plays an important role.

Lemma 1.11 (Hadamard differentiability of integrals) *Let $\psi\colon [0, M] \to \mathbb{R}$ be a C^2-function. Then the function*

$$g(F_1, F_2)(\cdot) = \int_{(-\infty,\cdot]} \psi(F_1(s)) \, \mathrm{d}F_2(s)$$

is Hadamard differentiable as a map from the domain[4]

$$\mathbb{D}_g = \mathrm{BV}_M[-\infty, \infty] \times \mathrm{BV}_M[-\infty, \infty] \subseteq D[-\infty, \infty] \times D[-\infty, \infty] = \mathbb{D}$$

to $\mathbb{E} = D[-\infty, \infty]$ where we equip the spaces with the supremum norm. The derivative is given by

$$\begin{aligned} g'_{(F_1,F_2)}(h_1, h_2)(\cdot) = h_2(s)\psi(F_1(s))\Big|_{s=-\infty}^{s=\cdot} - \int_{(-\infty,\cdot]} h_2(s-) \, \mathrm{d}\psi(F_1(s)) \\ + \int_{(-\infty,\cdot]} \psi'(F_1(s))h_1(s) \, \mathrm{d}F_2(s). \end{aligned}$$

Proof. Let $t_n \downarrow 0$ and let $h_{1n} \to h_1$, $h_{2n} \to h_2$ in $D[-\infty, \infty]$ be such that $F_{2n} = F_2 + t_n h_{2n} \in \mathrm{BV}_M$ for all n. Then each h_{2n} has bounded variation. Set $F_{1n} = F_1 + t_n h_{1n}$ and fix $x \in [-\infty, \infty]$. By adding and subtracting $\psi'(F_1)h_1$ we obtain

$$\begin{aligned} &\frac{g(F_{1n}, F_{2n})(x) - g(F_1, F_2)(x)}{t_n} \\ &= \int_{(-\infty,x]} \Big(\frac{\psi(F_{1n}) - \psi(F_1)}{t_n} - \psi'(F_1)h_1 \Big) \, \mathrm{d}F_{2n} \\ &\quad + \int_{(-\infty,x]} \psi(F_1) \, \mathrm{d}h_{2n} + \int_{(-\infty,x]} \psi'(F_1)h_1 \, \mathrm{d}F_{2n}. \end{aligned}$$

For the first term, Taylor's theorem and boundedness of (h_{1n}) in the supremum norm yield

$$\|(\psi(F_{1n}) - \psi(F_1))/t_n - \psi'(F_1)h_1\|_\infty \le \|\psi'\|_\infty \|h_{1n} - h_1\|_\infty + \tfrac{1}{2}\|\psi''\|_\infty t_n \|h_{1n}\|_\infty^2,$$

[4] Here, we use BV_M to denote the space of càdlàg functions of bounded variation with total variation at most M.

and therefore

$$\begin{aligned}\sup_x\Big|\int_{(-\infty,x]}\Big(\frac{\psi(F_{1n})-\psi(F_1)}{t_n}-\psi'(F_1)h_1\Big)\,\mathrm{d}F_{2n}\Big|\\ \leq\big(\|\psi'\|_\infty\|h_{1n}-h_1\|_\infty+\tfrac{1}{2}\|\psi''\|_\infty t_n\|h_{1n}\|_\infty^2\big)\,M,\end{aligned}$$

since $F_{2n}\in\mathrm{BV}_M$ and as $t_n\to 0$, this term converges to zero uniformly in x.

By integration by parts,

$$\int_{(-\infty,x]}\psi(F_1)\,\mathrm{d}h_{2n}=\big(\psi(F_1(s))\,h_{2n}(s)\big)\big|_{s=-\infty}^{s=x}-\int_{(-\infty,x]}h_{2n}(s-)\,\mathrm{d}\psi(F_1(s)),$$

and the right-hand side converges uniformly in x to the corresponding expression with h_2, because $h_{2n}\to h_2$ in supremum norm and $\psi(F_1)$ is of bounded variation.

Finally, write

$$\int_{(-\infty,x]}\psi'(F_1)h_1\,\mathrm{d}F_{2n}=\int_{(-\infty,x]}\psi'(F_1)h_1\,\mathrm{d}F_2+\int_{(-\infty,x]}\psi'(F_1)h_1\,\mathrm{d}(F_{2n}-F_2).$$

Set $r=\psi'(F_1)h_1$, which is bounded and càdlàg on $[-\infty,\infty]$. Given $\varepsilon>0$, choose a finite partition $-\infty=u_0<\cdots<u_m=\infty$ such that the oscillation of r on each $[u_{i-1},u_i)$ is at most ε. Then, uniformly in x,

$$\Big|\int_{(-\infty,x]}r\,\mathrm{d}(F_{2n}-F_2)\Big|\leq 2M\varepsilon+\sum_{i=1}^m\big|r(u_{i-1})\big|\,\big|(F_{2n}-F_2)[u_{i-1},u_i)\big|.$$

For fixed partition the sum tends to zero because $F_{2n}\to F_2$ uniformly, and the first term is bounded by $2M\varepsilon$. Hence the second integral vanishes uniformly in x as $n\to\infty$.

Collecting terms shows that the difference quotient converges uniformly in x to $g'_{(F_1,F_2)}(h_1,h_2)$, proving the required Hadamard differentiability. □

Definition 1.13 (Product integral) Let $F\in D(0,b]$ and let $\Delta F(t)=F(t)-F(t-)$ and $F^c(t)=F(t)-\sum_{0<s\leq t}\Delta F(s)$. The product integral is defined as

$$\phi(F)(t)=\prod_{0<s\leq t}(1+\mathrm{d}F(s))=\exp\left(F^c(t)\right)\prod_{0<s\leq t}(1+\Delta F(s)).$$

As additional notation we have

$$\phi(F)(s,t]=\prod_{s<u\leq t}(1+\mathrm{d}F(u))=\frac{\phi(F)(t)}{\phi(F)(s)},\quad\text{for } s<t.$$

Let X be a finite signed $(k\times k)$ matrix-valued measure defined on the Borel sets $((0,b],\mathcal{B}(0,b])$. By this we mean that each entry $X(i,j)$ is a finite signed measure on $((0,b],\mathcal{B}(0,b])$. The following definition does not require that X is commutative:

$$\prod_{0<s\leq t}\big(\mathrm{Id}+\mathrm{d}X(s)\big)=\lim_{\max|t_j-t_{j-1}|\to 0}\prod_j\left(\mathrm{Id}+X(t_{j-1},t_j]\right),$$

for partitions $0=t_0<\cdots<t_n=t$ of $(0,t]$.

Remark 1.5 The product integral can be defined in several ways, and more generally than Definition 1.13. For some equivalent definitions, see Gill and Johansen [1990]. ○

Lemma 1.12 (Hadamard differentiability of the product integral) *For fixed, finite, positive constants b and M, the product integral map $\phi\colon \mathrm{BV}_M(0,b] \subseteq D(0,b] \to D(0,b]$ is Hadamard differentiable with derivative*

$$\phi'_F\,(h)\,(\cdot) = \int_{(0,\cdot]} \phi(F)(0,u-)\phi(F)(u,\cdot]\;\mathrm{d}h(u).$$

The scalar result above is the template for product-integral arguments throughout the book. In multi-state settings we need the matrix analogue, so we provide it next before invoking the general functional δ-method.

Lemma 1.13 (Hadamard differentiability of the product integral - matrix version) *Let $\mathbb{D}$ be a space of matrices with function entries each of bounded variation, indexed by $[0,\theta]$, and equip it with the uniform norm. For $F \in \mathbb{D}$, define*

$$\phi(F)(t,s) = \prod_{(t,s]} (\mathrm{Id} + \mathrm{d}F)\,,\;\; 0 \le t < s \le \theta.$$

The map ϕ is Hadamard differentiable at each $F \in \mathbb{D}$ w.r.t. the uniform norm on $[0,\theta]$ and has derivative

$$\phi'_F(h)(t,s) = \int_{(t,s]} \prod_{t}^{u-} (\mathrm{Id} + \mathrm{d}F)\;\;\mathrm{d}h(u) \prod_{u}^{s} (\mathrm{Id} + \mathrm{d}F),\;\; 0 \le t < s \le \theta,$$

where the integral w.r.t. h is defined by the integration by parts formula.

Theorem 1.17 (Functional δ-method) *Let $\mathbb{D}$ and $\mathbb{E}$ be normed spaces. Let $\phi\colon \mathbb{D}_\phi \subseteq \mathbb{D} \to \mathbb{E}$ be Hadamard differentiable at θ tangentially to $\mathbb{D}_0$. Let $X_n\colon \Omega \to \mathbb{D}_\phi$ be maps with $r_n(X_n - \theta) \xrightarrow{d} X$ for some sequence of constants $r_n \to \infty$, where X is tight and takes its values in $\mathbb{D}_0$. Then $r_n\big(\phi(X_n) - \phi(\theta)\big) \xrightarrow{d} \phi'_\theta(X)$.*

Proof. We simply rescale via a clever function g_n and then apply the continuous mapping theorem to the weak limit. Indeed, define the function $g_n(h) = r_n\big(\phi(\theta + r_n^{-1}h) - \phi(\theta)\big)$ on $\mathbb{D}_n = \{h : \theta + r_n^{-1}h \in \mathbb{D}_\phi\}$. By definition of Hadamard differentiability $g_n(h_n) \to \phi'_\theta(h)$ for any $h_n \to h \in \mathbb{D}_0$. Then the continuous mapping Theorem 1.14 gives

$$r_n\big(\phi(X_n) - \phi(\theta)\big) = g_n\big(r_n(X_n - \theta)\big) \xrightarrow{d} \phi'_\theta(X).$$

□

Remark 1.6 The phrase *tangentially to* $\mathbb{D}_0$ in Theorem 1.17 highlights that ϕ is defined on the full domain $\mathbb{D}_\phi$, but differentiability is only required along directions in $\mathbb{D}_0$. In the theorem, the perturbations are precisely $h = r_n(X_n - \theta)$, and the assumption $X \in \mathbb{D}_0$ means these directions live in $\mathbb{D}_0$. For example, if X_n are càdlàg and $\sqrt{n}(X_n - \theta) \xrightarrow{d} X$ with X a continuous Gaussian process on $[0,\tau]$, then $\theta \in \mathbb{D}[0,\tau]$ while $h = X \in C[0,\tau]$, so it suffices that ϕ be differentiable along continuous directions even if the derivative may or may not be defined on the whole of $\mathbb{D}[0,\tau]$. ○

1.4.4 Entropy theorems

There exist similar central limit theorems for stochastic processes that turn out very useful in the sequel. Many times, the estimators we deal with are sums of iid terms. Therefore, before introducing these results, it is useful to define the empirical distribution function indexed by a function class. Thus, we introduce the so-called *empirical process*.

The empirical measure $\mathbb{P}^{(n)}$ of a sample of iid random variables $Z_1, \ldots, Z_n \sim L$ on a measurable space $(\mathcal{Z}, \mathcal{B})$ is the discrete measure given by

$$\mathbb{P}^{(n)}(C) = \frac{1}{n}\sum_{i=1}^{n} \delta_{Z_i}(C),$$

for a measurable set $C \in \mathcal{B}$. Let $\mathcal{F}$ be a collection of measurable functions $f\colon \mathcal{Z} \to \mathbb{R}$. Then the empirical measure induces the following map from $\mathcal{F}$ to $\mathbb{R}$

$$f \mapsto \mathbb{P}^{(n)} f = \int f \ \mathrm{d}\mathbb{P}^{(n)} = \frac{1}{n}\sum_{i=1}^{n} f(Z_i).$$

We scale and centre the above map to obtain the $\mathcal{F}$-indexed empirical process $\mathbb{G}^{(n)} = \sqrt{n}\left(\mathbb{P}^{(n)} - L\right)$ given by

$$f \mapsto \mathbb{G}^{(n)} f = \int f \ \mathrm{d}\mathbb{G}^{(n)} = \sqrt{n}\Big(\frac{1}{n}\sum_{i=1}^{n} f(Z_i) - \int f \ \mathrm{d}L\Big).$$

If $\sup_{f\in\mathcal{F}} \left|f(z) - \int f \ \mathrm{d}L\right| < \infty$, for almost every $z \in \mathcal{Z}$, then we view the empirical process $(\mathbb{G}^{(n)} f)_{f\in\mathcal{F}}$ as a well-defined stochastic process with paths in $\ell^\infty(\mathcal{F})$.

The complexity of $\mathcal{F}$ is later quantified via covering or bracketing numbers. For instance, the class of indicator half-lines $\{1_{(-\infty,t]} : t \in \mathbb{R}\}$ has small bracketing numbers because each function can be bracketed by neighbouring indicators with close thresholds, which is why this class has favourable asymptotic properties, as shown below.

Definition 1.14 (Glivenko–Cantelli class) Given an empirical measure $\mathbb{P}^{(n)}$ of a random sample (Z_n) defined on a measurable space $(\mathcal{Z}, \mathcal{B})$ where every Z_i has law L, a class $\mathcal{F}$ of measurable functions $f : \mathcal{Z} \to \mathbb{R}$, is called a $\mathbb{P}$-Glivenko–Cantelli class if

$$\|\mathbb{P}^{(n)} - L\|_{\mathcal{F}} \to 0,$$

where the convergence is in outer probability or outer almost surely.

Definition 1.15 (Donsker class) A collection $\mathcal{F}$ of measurable functions $f\colon \mathcal{Z} \to \mathbb{R}$ is a $\mathbb{P}$-Donsker class if

$$\mathbb{G}^{(n)} = \sqrt{n}\big(\mathbb{P}^{(n)} - L\big) \xrightarrow{d} \mathbb{G} \quad \text{in} \quad \ell^\infty(\mathcal{F}),$$

for a tight Gaussian limit process $\mathbb{G} \in \ell^\infty(\mathcal{F})$.

Remark 1.7 Sometimes we may treat empirical processes $\mathbb{G}^{(n)}$ as stochastic processes indexed exchangeably by either: a) classes of functions $\mathcal{F}$; or b) intervals. For instance, if we consider sets on the form

$$\mathcal{F} = \{f\colon [0,\infty) \to \{0,1\} \text{ s.t. } f(z) = 1_{[0,t]}(z) \text{ for } t \in [0,T]\},$$

implying that $\mathbb{G}^{(n)}f = \sqrt{n}(\frac{1}{n}\sum_{i=1}^{n} 1_{(Z_i \le t)} - F(t))$, for $f = 1_{[0,t]}$, and we may rewrite weak convergence as

$$\mathbb{G}^{(n)} = \sqrt{n}\big(\mathbb{P}^{(n)} - L\big) \xrightarrow{d} \mathbb{G} \quad \text{in} \quad \ell^{\infty}([0,T]),$$

meaning that the t-indexed stochastic process $\big(\mathbb{G}^{(n)}(t)\big)_{t\in[0,T]}$ converges weakly to a pathwise bounded process. ○

Definition 1.16 (Bracketing number) Given two functions l and u the bracket $[l,u]$ is the set of all functions f with $l \le f \le u$. An ε-bracket is a bracket with $\|u-l\| < \varepsilon$ where $\|\cdot\|$ is some norm. The bracketing number $N_{[\,]}(\varepsilon,\mathcal{F},\|\cdot\|)$ is the minimum number of ε-brackets needed to cover $\mathcal{F}$. The entropy (with bracketing) is defined as $\log N_{[\,]}(\varepsilon,\mathcal{F},\|\cdot\|)$.

The bracketing number increases as ε decreases. Intuitively speaking, the larger the entropy, the more the class of functions wiggles, and the more difficult it becomes to establish limit theorems. The precise size is quantified in the theorems below, collectively referred to as entropy theorems.

The next three results form a progression. Finite L^1-bracketing controls uniform laws, finite L^2 entropy integral yields Donsker limits, and Theorem 1.20 extends the same logic to changing classes, where n-dependent functions allow us to extend from the iid case to arrays.

Theorem 1.18 (Glivenko–Cantelli) *A class $\mathcal{F}$ is $\mathbb{P}$-Glivenko–Cantelli if*

$$N_{[\,]}(\varepsilon,\mathcal{F},L^1(\mathbb{P})) < \infty$$

for every $\varepsilon > 0$.

Theorem 1.19 (Bracketing CLT, Donsker's theorem) *A class $\mathcal{F}$ is $\mathbb{P}$-Donsker if*

$$J_{[\,]}(1,\mathcal{F},L^2(\mathbb{P})) = \int_{(0,1)} \sqrt{\log N_{[\,]}(\varepsilon,\mathcal{F},L^2(\mathbb{P}))}\,\mathrm{d}\varepsilon < \infty.$$

The theorem below is stated for wide classes of separable processes. For practical purposes, we only apply this theorem in the following chapters for càdlàg processes indexed by euclidean subsets, which are separable. It is also worth remarking that even separability can be relaxed.

Theorem 1.20 (Bracketing CLT for changing functions) *Let (X_n) be a sequence of iid separable stochastic processes. Define the classes $\mathcal{F}_n = \{f_{n,t} : t \in I\}$ indexed by a Euclidean subspace I. We require a specific norm, so redefine the bracketing number $N_{[\,]}(\varepsilon\|F_n(X_1)\|_{L^2},\mathcal{F}_n,L^2(\mathbb{P}))$ as the minimal number of* *sets N_ε in a partition $\mathcal{F}_n = \bigcup_{j=1}^{N_\varepsilon} \mathcal{F}^n_{\varepsilon_j}$ of the index set into sets $\mathcal{F}^n_{\varepsilon_j}$ such that for every one of these sets it holds that*

$$\mathbb{E}\Big[\sup_{f_{n,t}, f_{n,s} \in \mathcal{F}^n_{\varepsilon_j}} |f_{n,t}(X_1) - f_{n,s}(X_1)|^2\Big] \le \varepsilon^2 \|F_n(X_1)\|^2_{L^2},$$

where $F_n = \sup_t |f_{n,t}|$ is an envelope function of $\mathcal{F}^n$. Suppose

$$\mathbb{E}\left[F_n(X_1)^2\right] = O(1) \tag{1.6}$$

$$\mathbb{E}\left[F_n(X_1)^2 1_{(F_n(X_1) > \eta\sqrt{n})}\right] \to 0 \quad \textit{for every } \eta > 0 \tag{1.7}$$

$$\sup_{\|s-t\| \le \delta_n} \mathbb{E}\big[(f_{n,t}(X_1) - f_{n,s}(X_1))^2\big] \to 0 \quad \textit{for every } \delta_n \to 0 \tag{1.8}$$

$$\int_{[0,\delta_n]} \sqrt{\log\left(N_{[\,]}(\varepsilon \|F_n(X_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P}))\right)}\, \mathrm{d}\varepsilon \to 0 \quad \textit{for every } \delta_n \to 0. \tag{1.9}$$

Then the sequence $(\mathbb{G}^{(n)} f_{n,t})_{f_{n,t} \in \mathcal{F}_n}$ converges in distribution in $\ell^\infty(I)$ to a tight Gaussian process (with a continuous version) provided the sequence of covariance functions $\mathbb{E}\left[f_{n,t}(X_1) f_{n,s}(X_1)\right] - \mathbb{E}[f_{n,s}(X_1)]\mathbb{E}[f_{n,t}(X_1)]$ converge pointwise for each $(s,t) \in I^2$. If for each n the partition points (in t) can be chosen independently of n, then the condition in (1.8) is unnecessary.

The following examples illustrate the basic idea behind the entropy theorems. For low-dimensional parametric classes, a simple grid in the parameter space often yields a polynomial bound on the bracketing number.

Example 1.8 (A one-parameter linear class) Let (Z_n) be iid real-valued random variables with law L and assume $\mathbb{E}[Z_1^2] < \infty$. Consider

$$\mathcal{F}_1 = \{f_\theta : \mathbb{R} \to \mathbb{R} \,:\, f_\theta(z) = \theta z,\ \theta \in [-1,1]\}.$$

Fix $\varepsilon > 0$ and choose a grid $-1 = \theta_0 < \theta_1 < \cdots < \theta_m = 1$ with mesh size at most $\varepsilon / \|Z_1\|_{L^2}$. For $j = 0, \ldots, m-1$, define

$$l_j(z) = \min\{f_{\theta_j}(z), f_{\theta_{j+1}}(z)\}, \quad u_j(z) = \max\{f_{\theta_j}(z), f_{\theta_{j+1}}(z)\}.$$

Then every f_θ with $\theta \in [\theta_j, \theta_{j+1}]$ belongs to $[l_j, u_j]$, and

$$\|u_j - l_j\|_{L^2} = (\theta_{j+1} - \theta_j)\, \|Z_1\|_{L^2} \le \varepsilon.$$

Hence

$$N_{[\,]}(\varepsilon, \mathcal{F}_1, L^2(\mathbb{P})) \le \left\lceil \frac{2\|Z_1\|_{L^2}}{\varepsilon} \right\rceil.$$

Since $\|g\|_{L^1} \le \|g\|_{L^2}$, we also have $N_{[\,]}\left(\varepsilon, \mathcal{F}_1, L^1(\mathbb{P})\right) < \infty$ for every $\varepsilon > 0$, so $\mathcal{F}_1$ is $\mathbb{P}$-Glivenko–Cantelli by Theorem 1.18. Moreover, a simple calculation shows $J_{[\,]}(1, \mathcal{F}_1, L^2(\mathbb{P})) < \infty$ and thus $\mathcal{F}_1$ is even $\mathbb{P}$-Donsker by Theorem 1.19. ◇

Example 1.9 (A two-parameter affine class) Under the same moment assumption as the previous example, consider

$$\mathcal{F}_2 = \{f_{a,b} : \mathbb{R} \to \mathbb{R} \,:\, f_{a,b}(z) = a + bz,\ (a,b) \in [-1,1]^2\}.$$

Fix $\varepsilon > 0$ and choose grids in $[-1,1]$ with mesh sizes $\Delta_a = \varepsilon/2$ and $\Delta_b = \varepsilon/(2\|Z_1\|_{L^2})$. This partitions $[-1,1]^2$ into rectangles $R_{jk} = [a_j, a_{j+1}] \times [b_k, b_{k+1}]$. For each rectangle, define

$$l_{jk}(z) = \inf_{(a,b)\in R_{jk}} (a+bz), \quad u_{jk}(z) = \sup_{(a,b)\in R_{jk}} (a+bz).$$

Then every $f_{a,b}$ with $(a,b) \in R_{jk}$ belongs to $[l_{jk}, u_{jk}]$. Since $a+bz$ is linear in (a,b), the extrema occur at the corners, and therefore

$$u_{jk}(z) - l_{jk}(z) \le \Delta_a + \Delta_b |z|.$$

Consequently,

$$\|u_{jk} - l_{jk}\|_{L^2} \le \Delta_a + \Delta_b \|Z_1\|_{L^2} \le \varepsilon.$$

The number of rectangles is at most $\lceil 2/\Delta_a \rceil \lceil 2/\Delta_b \rceil$, and hence

$$N_{[\,]}\left(\varepsilon, \mathcal{F}_2, L^2(\mathbb{P})\right) \le \frac{C}{\varepsilon^2}$$

for some constant $C < \infty$. As in the previous example, this implies finite $L^1(\mathbb{P})$-bracketing numbers for every $\varepsilon > 0$, so $\mathcal{F}_2$ is $\mathbb{P}$-Glivenko–Cantelli by Theorem 1.18, and the entropy integral in Theorem 1.19 is again finite, hence $\mathcal{F}_2$ is $\mathbb{P}$-Donsker. ◇

1.4.5 k-nearest neighbours empirical processes

The tools above allow the study of fixed-bandwidth localisation, as is conveniently exploited in the following chapter. However, one can sometimes instead use a k-nearest-neighbour (k-NN) approach, where the neighbourhood is selected by spatial search and therefore adapts automatically to the covariate density. The corresponding local empirical-process CLT can be viewed as a corollary of the bracketing CLT for changing classes, Theorem 1.20, but the proof is technically involved; we therefore refer to the detailed arguments in Portier [2025]. Equivalently, the k-NN rule may be interpreted as a kernel method with a data-dependent bandwidth, when choosing a uniform kernel on the random ball centred at x whose radius is the k-NN distance.

Definition 1.17 (Local k-nearest-neighbour empirical measure) Let $(S, \mathcal{S})$ be a measurable space and let (X_n, Y_n) be iid random elements in $(\mathbb{R}^d \times S, \mathcal{B}(\mathbb{R}^d) \otimes \mathcal{S})$, with $x \in \mathbb{R}^d$ fixed. Let $\mathcal{K}_{n,\kappa}(x)$ be the index set of the κ nearest covariates X_i to x (ties are broken lexicographically). For $A \in \mathcal{S}$, define

$$\hat{\mu}_{n,\kappa,x}(A) = \frac{1}{\kappa} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} 1_{(Y_i \in A)}.$$

For measurable $g \colon S \to \mathbb{R}$, write $\hat{\mu}_{n,\kappa,x}(g) = \int g \, \mathrm{d}\hat{\mu}_{n,\kappa,x}$.

The next theorem is the local k-NN counterpart of Theorem 1.20, adapted from Portier [2025].

Theorem 1.21 (Bracketing CLT for local k-NN empirical processes) *Let $(S, \mathcal{S})$ be a measurable space. Let (X_n, Y_n) be iid random elements in $(\mathbb{R}^d \times S, \mathcal{B}(\mathbb{R}^d) \otimes \mathcal{S})$, let μ_z denote the regular conditional law of Y given $X = z$, and fix $x \in \mathbb{R}^d$. Assume X has density f_X with $f_X(x) > 0$ and f_X continuous at x. Let $\mathcal{F}$ be a class of measurable functions on S with envelope F, and for $u > 0$ define*

$$\mu_{x,u}(A) = \frac{\mathbb{E}[\mu_X(A)1_{(X \in B(x,u^{1/d}))}]}{\mathbb{E}[1_{(X \in B(x,u^{1/d}))}]}, \quad A \in \mathcal{S}.$$

Here $B(x, r) = \{z \in \mathbb{R}^d : \|z - x\| \le r\}$. Suppose there exists $\delta > 0$ such that with

$$\rho_\delta(f_1, f_2) = \sup_{|u| \le \delta} \|f_1 - f_2\|_{L^2(\mu_{x,u})}$$

the following holds:

$$\sup_{|u|\le\delta} \int_{[0,\delta_n]} \sqrt{\log N_{[\,]}\big(\varepsilon \|F\|_{L^2(\mu_{x,u})}, \mathcal{F}, L^2(\mu_{x,u})\big)}\, \mathrm{d}\varepsilon \to 0 \quad \textit{for every } \delta_n \to 0, \tag{1.10}$$

$$\sup_{|u|\le\delta} \mu_{x,u}(F^2 1_{(F>M)}) \to 0 \quad \textit{as} \quad M \to \infty, \tag{1.11}$$

$$\sup_{f\in\mathcal{F}} \min_{1\le j\le m_\eta} \rho_\delta(f, f_j^\eta) \le \eta \quad \textit{for every } \eta > 0 \textit{ and some } f_1^\eta, \dots, f_{m_\eta}^\eta \in \mathcal{F}, \tag{1.12}$$

$$z \mapsto \big(\mu_z(f_1 f_2), \mu_z(f_1), \mu_z(F^2)\big) \textit{ is continuous at } x, \textit{ for every } f_1, f_2 \in \mathcal{F}. \tag{1.13}$$

If $\kappa = \kappa_n$ satisfies $\kappa \to \infty$ and $\kappa/n \to 0$, then

$$\mathbb{U}_{\kappa,x}^{(n)}(f) := \kappa^{-1/2} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} \left(f(Y_i) - \mu_{X_i}(f)\right), \quad f \in \mathcal{F},$$

converges weakly in $\ell^\infty(\mathcal{F})$ to a tight centred Gaussian process $\mathbb{U}_x$ with covariance

$$\operatorname{Cov}\big(\mathbb{U}_x(f_1), \mathbb{U}_x(f_2)\big) = \mu_x(f_1 f_2) - \mu_x(f_1)\mu_x(f_2).$$

The next corollary adds a deterministic localisation-bias bound and an explicit rate condition to pass from the centred process to the full local empirical measure.

Corollary 1.2 (From centred to full k-NN empirical-process CLT) *Assume the conditions of Theorem 1.21. Suppose additionally that there are constants $\tau_x, L_x > 0$ such that*

$$\sup_{f\in\mathcal{F}} |(\mu_z - \mu_x)(f)| \le L_x \|z - x\|, \quad z \in B(x, \tau_x), \tag{1.14}$$

$$\kappa^{(d+2)/2}/n \to 0. \tag{1.15}$$

Then

$$\left\{\sqrt{\kappa}\left(\hat{\mu}_{n,\kappa,x}(f) - \mu_x(f)\right)\right\}_{f\in\mathcal{F}} \xrightarrow{d} \mathbb{U}_x \quad \textit{in } \ell^\infty(\mathcal{F}).$$

As a first elementary example we show how the previous results recover asymptotics for a k-NN regression from a singleton class. It also highlights that, when the response is bounded, the local Lindeberg requirement is automatic.

Example 1.10 (k-NN local regression at a fixed covariate) Let (X_n, Y_n) be iid with values in $\mathbb{R}^d \times \mathbb{R}$, and fix $x \in \mathbb{R}^d$. Write

$$m(z) = \mathbb{E}[Y_1 \mid X_1 = z], \quad \hat{m}_{n,\kappa}(x) = \frac{1}{\kappa} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} Y_i.$$

Assume first that $|Y_1| \le B$ a.s. for some $B < \infty$, that $f_X(x) > 0$ with f_X continuous at x, and that $z \mapsto \mathbb{E}[Y_1 \mid X_1 = z]$ and $z \mapsto \mathbb{E}[Y_1^2 \mid X_1 = z]$ are continuous at x. Assume also that there exist $\tau_x, L_x > 0$ such that

$$|m(z) - m(x)| \le L_x \|z - x\|, \quad z \in B(x, \tau_x),$$

and let $\kappa = \kappa_n$ satisfy $\kappa \to \infty$, and $\kappa^{(d+2)/2}/n \to 0$. Set $\mathcal{F} = \{g\}$ with $g(y) = y$. Then $\hat{m}_{n,\kappa}(x) = \hat{\mu}_{n,\kappa,x}(g)$, and:

$$\begin{aligned} &N_{[\,]}\big(\varepsilon \|F\|_{L^2(\mu_{x,u})}, \mathcal{F}, L^2(\mu_{x,u})\big) = 1, \quad \varepsilon > 0, \\ &\sup_{|u| \le \delta} \mu_{x,u}(F^2 1_{(F>M)}) = 0 \quad \text{for } M > B, \end{aligned}$$

with envelope $F \equiv B$. Hence (1.10)–(1.12) are immediate; condition (1.13) is exactly continuity of the conditional first and second moments at x. The boundedness assumption can be weakened; it is enough that for some $\delta_0 > 0$ and $\tau_x' > 0$,

$$\sup_{\|z - x\| \le \tau_x'} \mathbb{E}\left[|Y_1|^{2+\delta_0} \mid X_1 = z\right] < \infty.$$

Indeed, taking $F(y) = |y|$ and $\delta < (\tau_x')^d$ gives

$$\sup_{|u| \le \delta} \mu_{x,u}(F^2 1_{(F>M)}) \le M^{-\delta_0} \sup_{|u| \le \delta} \mu_{x,u}(F^{2+\delta_0}) \to 0,$$

so (1.11) still holds. Therefore Theorem 1.21 gives

$$\kappa^{-1/2} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} \big(Y_i - m(X_i)\big) \xrightarrow{d} N\big(0, \sigma^2(x)\big),$$

where $\sigma^2(x) = \mathrm{Var}(Y_1 \mid X_1 = x)$. By (1.14) and (1.15), Corollary 1.2 further yields

$$\sqrt{\kappa}\big(\hat{m}_{n,\kappa}(x) - m(x)\big) \xrightarrow{d} N\big(0, \sigma^2(x)\big).$$

This is the k-NN analogue of the fixed-point Nadaraya–Watson asymptotic normality, to be treated in the following chapter, with normalisation $\sqrt{\kappa}$ and without explicit bandwidth. ◇

The next example treats discrete responses through a finite class of indicator functions. This gives, in one step, binary, categorical and multinomial asymptotics together with the familiar conditional multinomial covariance structure.

Example 1.11 (Binary, categorical and multinomial k-NN local probabilities) Let (X_n, Y_n) be iid and fix $x \in \mathbb{R}^d$. Assume $f_X(x) > 0$, that f_X is continuous at x, and let $\kappa = \kappa_n$ satisfy $\kappa \to \infty$ and $\kappa/n \to 0$.

For the categorical case, suppose $Y_1 \in \{1, \dots, J\}$. Define

$$p_j(z) = \mathbb{P}(Y_1 = j \mid X_1 = z), \quad \hat{p}_{j,\kappa}(x) = \frac{1}{\kappa} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} 1_{(Y_i = j)}, \quad j = 1, \dots, J.$$

Take $\mathcal{F} = \{g_1, \dots, g_J\}$ with $g_j(y) = 1_{(y=j)}$ and envelope $F \equiv 1$. Then, for every $\varepsilon \in (0, 1)$ and every probability measure Q,

$$N_{[\,]}\big(\varepsilon \|F\|_{L^2(Q)}, \mathcal{F}, L^2(Q)\big) \le J,$$

so (1.10) holds; condition (1.11) is immediate since $F \equiv 1$, and condition (1.12) holds because $\mathcal{F}$ is finite. If $z \mapsto p_j(z)$ is continuous at x for each j, then (1.13) holds as well, since

$$\mu_z(g_j) = p_j(z), \quad \mu_z(g_j g_r) = \mu_z(1_{(Y=j)} 1_{(Y=r)}).$$

Hence, if $\kappa \to \infty$ and $\kappa/n \to 0$,

$$\Big(\kappa^{-1/2} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} \big(g_j(Y_i) - \mu_{X_i}(g_j)\big)\Big)_{j=1,\dots,J} \xrightarrow{d} N_J(0, \Sigma_x),$$

with

$$[\Sigma_x]_{jr} = p_j(x) 1_{(j=r)} - p_j(x) p_r(x), \quad 1 \le j, r \le J.$$

If, in addition, there exist $\tau_x, L_x > 0$ such that

$$\max_{1 \le j \le J} |p_j(z) - p_j(x)| \le L_x \|z - x\|, \quad z \in B(x, \tau_x),$$

and $\kappa^{(d+2)/2}/n \to 0$, then Corollary 1.2 gives

$$\sqrt{\kappa}\big(\hat{p}_{\cdot,\kappa}(x) - p(x)\big) \xrightarrow{d} N_J(0, \Sigma_x),$$

where $\hat{p}_{\cdot,\kappa}(x) = (\hat{p}_{1,\kappa}(x), \dots, \hat{p}_{J,\kappa}(x))^\top$ and $p(x) = (p_1(x), \dots, p_J(x))^\top$.

The binary case is $J = 2$, yielding

$$\sqrt{\kappa}\big(\hat{p}_{1,\kappa}(x) - p_1(x)\big) \xrightarrow{d} N\big(0, p_1(x)\big(1 - p_1(x)\big)\big).$$

For multinomial responses with a fixed number m of trials, write $Y = (Y_1, \dots, Y_J)$ with $\sum_{j=1}^J Y_j = m$, and apply the same argument to $g_j(y) = y_j/m \in [0, 1]$, which is valid because $g_j(Y_1)$ is the within-observation category-j proportion and satisfies $\mathbb{E}[g_j(Y_1) \mid X_1 = x] = p_j(x)$. The statistical similarity with the categorical case means they estimate the same conditional probability vector, with multinomial observations carrying within-observation replication so the covariance is divided by m (the categorical case corresponding to $m = 1$). One obtains the same limit covariance form, scaled by $1/m$:

$$[\Sigma_x]_{jr} = \frac{1}{m}\big(p_j(x) 1_{(j=r)} - p_j(x) p_r(x)\big), \quad 1 \le j, r \le J.$$

◇

1.5 Empirical distributions, quantiles and copulas

With the functional theorems in place, we can now transfer them to quantile- and dependence-based summaries that are central to risk measurement. We consider once again the process $t \mapsto \mathbb{F}^{(n)}(t) = n^{-1} \sum_{i=1}^{n} 1_{(Z_i \le t)}$. We may extend the proof of Theorem 1.12 to recognise that its finite-dimensional distributions converge to a centred Gaussian process $\mathbb{G}$ with covariance function

$$\operatorname{Cov}(\mathbb{G}(t), \mathbb{G}(s)) = F(s \wedge t) - F(s)F(t),$$

or in other words, $\mathbb{G} = \mathbb{B} \circ F$ a.s. where $\mathbb{B}$ is a standard Brownian bridge[5]. We can now assert the full pathwise convergence.

Theorem 1.22 (Pathwise convergence of the empirical distribution function) *Let (Z_n) be an iid sequence of positive random variables with distribution function F. Then*

$$\sqrt{n}(\mathbb{F}^{(n)} - F) \xrightarrow{d} \mathbb{B} \circ F \quad \text{in} \quad \ell^\infty([0,\infty)).$$

Moreover, consider another sample (W_n) and denote the joint empirical and true distribution functions for the pair (Z_n, W_n) respectively by $\mathbb{H}^{(n)}$ and H. Then

$$\sqrt{n}(\mathbb{H}^{(n)} - H) \xrightarrow{d} \mathbb{G}_H \quad \text{in} \quad \ell^\infty([0,\infty)^2)$$

with $\mathbb{G}_H$ a tight Gaussian process satisfying

$$\operatorname{Cov}(\mathbb{G}_H(s,t), \mathbb{G}_H(u,v)) = H(s \wedge u, t \wedge v) - H(s,t)H(u,v).$$

Proof. Given that we already have recognised the limit process, we simply apply the Bracketing CLT, Theorem 1.19. Define $\mathcal{F} = \{f : [0,\infty) \to \{0,1\} \text{ s.t. } f(z) = 1_{[0,t]}(z) \text{ for } t \in [0,\infty)\}$. We calculate the bracketing number. Consider for $j = 1, \dots, k$ brackets on the form $l_j(x) = 1_{[0,t_{j-1})}(x)$, $u_j(x) = 1_{[0,t_j)}(x)$ for sequences on the form $0 = t_0 < t_1 < \cdots < t_k = \infty$. Let $\mathcal{F}_j = \{f \in \mathcal{F} : l_j \le f \le u_j\}$. Then clearly we have the partition $\mathcal{F} = \bigcup_{j=1}^{k} \mathcal{F}_j$. Let $\varepsilon > 0$ be given, and pick a sequence for $k \le \lceil 2/\varepsilon^2 \rceil$ such that

$$F(t_j-) - F(t_{j-1}) < \varepsilon^2 \quad \text{for} \quad 1 \le j \le k,$$

which is possible because F is a distribution function, in particular, it is càdlàg, non-decreasing and bounded by 1. Then this sequence covers all of $\mathcal{F}$ with brackets satisfying

$$\|u_j(Z_1) - l_j(Z_1)\|_{L^2} = \big(F(t_j-) - F(t_{j-1})\big)^{1/2} < \varepsilon.$$

Consequently $N_{[\,]}\left(\varepsilon, \mathcal{F}, L^2(\mathbb{P})\right) \le k \le \lceil 2/\varepsilon^2 \rceil$. We then obtain

$$\int_{(0,1)} \sqrt{\log N_{[\,]}\left(\varepsilon, \mathcal{F}, L^2(\mathbb{P})\right)}\, \mathrm{d}\varepsilon \le \int_{(0,1]} \sqrt{\log\left(2/\varepsilon^2 + 1\right)}\, \mathrm{d}\varepsilon < \infty,$$

which shows that $\mathcal{F}$ is $\mathbb{P}$-Donsker. The second part is similar and left for the reader. □

[5] A Brownian motion and Brownian bridge are centred stochastic processes on $\mathbb{R}$ and $[0,1]$, respectively. Their covariance functions are given by $(s,t) \mapsto t \wedge s$ and $(s,t) \mapsto t \wedge s - ts$, respectively.

Quantiles are crucial in insurance for assessing risk, and they are related to the concept of Value at Risk (VaR). Indeed the VaR at a confidence level $\alpha \in (0,1)$ is defined simply as the α-quantile of the loss distribution. Formally, for a loss random variable Z, the VaR at level α is given by:

$$\mathrm{VaR}_\alpha(Z) = \inf\{z \in \mathbb{R} \,|\, \mathbb{P}(Z \leq z) \geq \alpha\} = F^{-1}(\alpha).$$

In other words, $\mathrm{VaR}_\alpha(Z)$ represents the maximum loss not exceeded with a confidence level of α. As a risk measure VaR possesses several important properties:

1. *Monotonicity*: if $Z_1 \leq Z_2$ almost surely, then $\mathrm{VaR}_\alpha(Z_1) \leq \mathrm{VaR}_\alpha(Z_2)$.
2. *Translation invariance*: for any constant $c \in \mathbb{R}$, $\mathrm{VaR}_\alpha(Z+c) = \mathrm{VaR}_\alpha(Z)+c$.
3. *Homogeneity*: for any positive scalar $\lambda > 0$, $\mathrm{VaR}_\alpha(\lambda Z) = \lambda \mathrm{VaR}_\alpha(Z)$.
4. *Subadditivity for elliptical distributions*[6]: for loss variables Z_1 and Z_2, and $\alpha \geq 1/2$ we have $\mathrm{VaR}_\alpha(Z_1 + Z_2) \leq \mathrm{VaR}_\alpha(Z_1) + \mathrm{VaR}_\alpha(Z_2)$.

For instance, Solvency II requires that pensions fulfil the Solvency Capital Requirement that enough capital is held to withstand a “200-year event". The loss of such 200-year event can be estimated as the 99.5% upper quantile of the total loss.

Given the widespread use of VaR in the insurance industry, in particular as one of the gold standard risk measures, finding limit results for nonparametric estimators for the quantile process, which considers all quantiles simultaneously, is of considerable interest. Generally speaking, nonparametric methods provide a flexible and robust approach to estimating risk without relying on specific distributional assumptions. This is true for medium and large quantiles using the method below. However, it should also be noted that for estimating *very high quantiles*, other nonparametric tools from Extreme Value Theory are better suited.

Theorem 1.23 (Pathwise convergence of the quantile process) *Let $0 < p < q < 1$. Let (Z_n) be an iid sequence of real-valued random variables such that their distribution function F is continuously differentiable on $[a,b] = [F^{-1}(p) - \varepsilon, F^{-1}(q) + \varepsilon]$ for some $\varepsilon > 0$ (which may span the support of F if compact) with strictly positive derivative f. Then*

$$\sqrt{n}\big([\mathbb{F}^{(n)}]^{-1} - F^{-1}\big) \xrightarrow{d} -\frac{\mathbb{B}}{f \circ F^{-1}} \quad \text{in} \quad \ell^\infty([p,q]).$$

Proof. By the first part of Lemma 1.10 we know that the inverse map is Hadamard differentiable tangentially to the space of functions that are continuous on $[a,b]$,

[6] Distributions with $\mathbb{E}[e^{i\langle t,(Z_1,\ldots,Z_r)-\mu\rangle}] = \psi(t\Sigma t^\top)$ for every $t \in \mathbb{R}^r$, for some location parameter μ, some nonnegative-definite matrix Σ and some scalar function ψ. For instance, Gaussian random vectors.

and hence we may apply the functional δ-method, Theorem 1.17, to the already established convergence from Theorem 1.22. This yields that

$$\sqrt{n}\big([\mathbb{F}^{(n)}]^{-1} - F^{-1}\big) \xrightarrow{d} -\frac{(\mathbb{B}\circ F)}{f} \circ F^{-1} \quad \text{in} \quad \ell^\infty([p,q]),$$

as desired. □

Quantile limits immediately deliver distribution theory for robust scale; the interquartile range is the prototypical case.

Example 1.12 (Asymptotics of the interquartile range) Let $q_p = F^{-1}(p)$ and $\hat{q}_p = [\mathbb{F}^{(n)}]^{-1}(p)$. Under the assumptions of Theorem 1.23 with $p = 1/4$ and $q = 3/4$, we have

$$\sqrt{n}\big(\hat{q}_p - q_p\big) \xrightarrow{d} -\frac{\mathbb{B}(p)}{f(q_p)},$$

so the vector $\sqrt{n}\big(\hat{q}_{1/4}-q_{1/4}, \hat{q}_{3/4}-q_{3/4}\big)$ is asymptotically normal with covariance matrix

$$\Sigma = \begin{pmatrix} \frac{(1/4)(3/4)}{f(q_{1/4})^2} & \frac{(1/4)(1-3/4)}{f(q_{1/4})f(q_{3/4})} \\ \frac{(1/4)(1-3/4)}{f(q_{1/4})f(q_{3/4})} & \frac{(1/4)(3/4)}{f(q_{3/4})^2} \end{pmatrix},$$

since $\mathrm{Cov}(\mathbb{B}(p_1), \mathbb{B}(p_2)) = p_1 \wedge p_2 - p_1 p_2$. The interquartile range is $\mathrm{IQR} = q_{3/4} - q_{1/4}$ and $\widehat{\mathrm{IQR}} = \hat{q}_{3/4} - \hat{q}_{1/4}$, so by the continuous mapping theorem,

$$\sqrt{n}\big(\widehat{\mathrm{IQR}} - \mathrm{IQR}\big) \xrightarrow{d} N(0, \sigma^2_{\mathrm{IQR}}),$$

with

$$\sigma^2_{\mathrm{IQR}} = \frac{3}{16}\left(\frac{1}{f(q_{3/4})^2} + \frac{1}{f(q_{1/4})^2}\right) - \frac{1}{8}\frac{1}{f(q_{1/4})f(q_{3/4})}.$$

◇

We now examine the Lorenz curve, Gini coefficient, and expected shortfall, which are defined as integral functionals of the quantile process.

Example 1.13 (Lorenz curve and Gini coefficient) Let $Z \geq 0$ with mean $\mu = \mathbb{E}[Z] \in (0, \infty)$ and distribution function F with strictly positive density f. Define the quantile function $q_p = F^{-1}(p)$. Fix $0 < \delta < 1/2$ and assume that f is continuous and strictly positive on $[q_\delta, q_{1-\delta}]$. The *trimmed* Lorenz curve is

$$L_\delta(p) = \frac{1}{\mu_\delta}\int_\delta^p q_u \,\mathrm{d}u, \quad p \in [\delta, 1-\delta], \quad \mu_\delta = \int_\delta^{1-\delta} q_u \,\mathrm{d}u.$$

Let $\hat{q}_p$ be the empirical quantile and $\hat{\mu}_\delta = \int_\delta^{1-\delta} \hat{q}_u \,\mathrm{d}u$. By Theorem 1.23,

$$\sqrt{n}\big(\hat{q} - q\big) \xrightarrow{d} G \quad \text{in} \quad \ell^\infty([\delta, 1-\delta]), \quad G(p) = -\frac{\mathbb{B}(p)}{f(q_p)}.$$

The map $q \mapsto \int_\delta^{(\cdot)} q_u \, \mathrm{d}u$ is Hadamard differentiable by Lemma 1.11, and the quotient map is Hadamard differentiable on $(0,\infty)^2$. Hence, by the functional δ-method,

$$\sqrt{n}\big(\widehat{L}_\delta(p) - L_\delta(p)\big) \xrightarrow{d} \frac{1}{\mu_\delta}\int_\delta^p G(u)\,\mathrm{d}u - \frac{L_\delta(p)}{\mu_\delta}\int_\delta^{1-\delta} G(u)\,\mathrm{d}u, \quad p \in [\delta, 1-\delta].$$

Define the trimmed Gini coefficient by

$$G_\delta = 1 - \frac{2}{1-2\delta}\int_\delta^{1-\delta} L_\delta(p)\,\mathrm{d}p, \quad \widehat{G}_\delta = 1 - \frac{2}{1-2\delta}\int_\delta^{1-\delta} \widehat{L}_\delta(p)\,\mathrm{d}p.$$

Another application of Lemma 1.11 yields

$$\sqrt{n}\big(\widehat{G}_\delta - G_\delta\big) \xrightarrow{d} -\frac{2}{1-2\delta}\int_\delta^{1-\delta}\Big(\frac{1}{\mu_\delta}\int_\delta^p G(u)\,\mathrm{d}u - \frac{L_\delta(p)}{\mu_\delta}\int_\delta^{1-\delta} G(u)\,\mathrm{d}u\Big)\,\mathrm{d}p,$$

which is mean-zero normal. If additional regularity holds near $p = 0$ and $p = 1$ (for instance f is bounded away from 0 on a compact support), one may let $\delta \downarrow 0$ and recover limit results for the untruncated Lorenz and Gini definitions. ◇

Remark 1.8 The quantile process limit in Theorem 1.23 is only guaranteed on compact subintervals of $(0,1)$. Near the endpoints, the factor $1/f(q_p)$ may blow up when $f(q_p)$ is small or vanishes (as often happens for distributions on $[0,\infty)$). The trimmed versions avoid this problem; as mentioned above, extending to $\delta = 0$ requires extra endpoint regularity. ○

Example 1.14 (Expected shortfall) Let $0 < \alpha < 1$ and define the *trimmed* expected shortfall

$$\mathrm{ES}_{\alpha,\delta}(Z) = \frac{1}{1-\alpha-\delta}\int_\alpha^{1-\delta} q_u\,\mathrm{d}u, \quad 0 < \delta < 1-\alpha,$$

with estimator $\widehat{\mathrm{ES}}_{\alpha,\delta} = (1-\alpha-\delta)^{-1}\int_\alpha^{1-\delta} \hat{q}_u\,\mathrm{d}u$. Under the assumptions of Theorem 1.23 on $[\alpha, 1-\delta]$, the map $q \mapsto \int_\alpha^{(\cdot)} q_u\,\mathrm{d}u$ is Hadamard differentiable by Lemma 1.11, and thus

$$\sqrt{n}\big(\widehat{\mathrm{ES}}_{\alpha,\delta} - \mathrm{ES}_{\alpha,\delta}\big) \xrightarrow{d} \frac{1}{1-\alpha-\delta}\int_\alpha^{1-\delta} G(u)\,\mathrm{d}u, \quad G(u) = -\frac{\mathbb{B}(u)}{f(q_u)}.$$

If additional endpoint regularity holds so that the quantile process extends to $[\alpha, 1)$ and $f(q_u)^{-1}$ is integrable near $u = 1$, one may let $\delta \downarrow 0$ and recover the usual expected shortfall ES_α. ◇

A *copula* is a mathematical function that allows us to study and model the dependence structure between multiple random variables independently of their marginal distributions. Formally, a copula is a multivariate distribution function with uniform marginals on the interval $[0,1]$. According to Sklar's Theorem, for any multivariate distribution function H with continuous marginals $F_1, F_2, \ldots, F_r$, there exists a copula $C : [0,1]^r \to [0,1]$ such that

$$H(x_1, x_2, \ldots, x_r) = C(F_1(x_1), F_2(x_2), \ldots, F_r(x_r)).$$

Conversely, if C is a copula and $F_1, F_2, \ldots, F_r$ are distribution functions, then H defined as above is a multivariate distribution function with marginals $F_1, \ldots, F_r$.

In risk management and insurance, copulas are commonly used for their ability to model dependencies between different risk factors. Some fundamental properties of copulas include:

1. *Uniform marginals:* for a copula $C(u_1, u_2, \ldots, u_r)$, the marginal distribution of each variable is uniform on $[0, 1]$.

2. *Grounded and monotone:* a copula is grounded, meaning $C(u_1, \ldots, u_r) = 0$ if any $u_i = 0$. It is also r-increasing, that is C is non-decreasing in each argument.

3. *Invariant under strictly increasing transformations:* the copula remains unchanged under strictly increasing transformations of the marginal distributions.

4. *Bounds:* copulas are bounded by the Fréchet–Hoeffding bounds, where the lower bound is $\max\left(\sum_{i=1}^{r} u_i - r + 1, 0\right)$ and the upper bound is given by $\min(u_1, u_2, \ldots, u_r)$.

Many parametric copula functions exist, and we refer the reader to Nelsen [2006] for general methods on how to construct parametric copulas. We now define a nonparametric estimator of a bivariate copula, i.e. for $r = 2$ and with marginals F_1 and F_2, referred to as the *empirical copula* function:

$$\mathbb{C}^{(n)}(u, v) = \mathbb{H}^{(n)}([\mathbb{F}_1^{(n)}]^{-1}(u), [\mathbb{F}_2^{(n)}]^{-1}(v)).$$

Theorem 1.24 (Convergence of the empirical copula process) *Let $0 < p < q < 1$. Let (Z_n, W_n) be iid vectors with joint distribution function H and marginal distribution functions F_1 and F_2. Assume the latter are continuously differentiable on $[a, b] = [F_1^{-1}(p) - \varepsilon, F_1^{-1}(q) + \varepsilon]$ and $[c, d] = [F_2^{-1}(p) - \varepsilon, F_2^{-1}(q) + \varepsilon]$ with positive derivatives f_1 and f_2, respectively, for some $\varepsilon > 0$. Additionally, assume that $\partial H/\partial x, \partial H/\partial y$ are continuous on the product of the intervals. Then*

$$\sqrt{n}\big(\mathbb{C}^{(n)} - C\big) \xrightarrow{d} \mathbb{G}_C \quad \text{in} \quad \ell^\infty([a, b] \times [c, d]).$$

where with $\mathbb{G}_H$ as in Theorem 1.22 we have

$$\begin{aligned}\mathbb{G}_C &= \mathbb{G}_H(F_1^{-1}, F_2^{-1}) \\ &\quad - \frac{\partial H}{\partial x}(F_1^{-1}, F_2^{-1})\frac{\mathbb{G}_H(F_1^{-1}, \infty)}{f_1(F_1^{-1})} - \frac{\partial H}{\partial y}(F_1^{-1}, F_2^{-1})\frac{\mathbb{G}_H(F_2^{-1}, \infty)}{f_2(F_2^{-1})}.\end{aligned}$$

Proof. Notice that we may construct the copula map as a composition of Hadamard differentiable maps. Indeed, define the maps

$$\begin{aligned}\phi(H)(x, y) &= (H(x, y), H(x, \infty), H(\infty, y)) \\ \psi(H, F_1, F_2)(x, y) &= (H(x, y), F_1^{-1}(x), F_2^{-1}(y)) \\ \nu(H, F_1^{-1}, F_2^{-1})(x, y) &= H \circ (F_1^{-1}(x), F_2^{-1}(y)),\end{aligned}$$

with Hadamard derivatives given by

$$\begin{aligned}
\phi'(a)(x, y) &= (a(x, y), a(x, \infty), a(\infty, y)) \\
\psi'(a, b, c)(x, y) &= (a(x, y), -(b/f_1) \circ F_1^{-1}(x), -(c/f_2) \circ F_2^{-1}(y)) \\
\nu'(a, b, c)(x, y) &= a \circ (F_1^{-1}(x), F_2^{-1}(y)) \\
&\qquad + (\partial H/\partial x, \partial H/\partial y) \circ (F_1^{-1}, F_2^{-1})(x, y)(b, c)^\top,
\end{aligned}$$

where we have used the second, first, third, and second statements in Lemma 1.10 for the first, second, and third maps, respectively (one can also apply the definition directly to see that these are indeed the derivatives). Thus, by the chain rule, Theorem 1.16, we obtain that $\nu \circ \psi \circ \phi$ is Hadamard differentiable at H with derivative

$$\nu'_{\psi\circ\phi(H)} \circ \psi'_{\phi(H)} \circ \phi'_H(h)(u, v) = h(F_1^{-1}(u), F_2^{-1}(v))$$
$$+ \Big\langle \big(\partial H/\partial x, \partial H/\partial y\big) \circ (F_1^{-1}, F_2^{-1})(u, v), \Big(- \frac{h(F_1^{-1}(u), \infty)}{f_1(F_1^{-1}(u))}, - \frac{h(F_2^{-1}(v), \infty)}{f_2(F_2^{-1}(v))} \Big) \Big\rangle,$$

so that we may apply the functional δ-method, Theorem 1.17, to Theorem 1.22 to obtain the desired result. □

Numeric example 1.2 (Empirical copula approximation) In Figure 1.5 we illustrate how sample size improves both the empirical copula and the associated empirical copula process. The copula considered is the Gaussian copula, constructed as per $C(u, v) = H(F_1^{-1}(u), F_2^{-1}(v))$. More precisely, the right panels display the centred and scaled field

$$(u, v) \mapsto \sqrt{n}\big(\mathbb{C}^{(n)}(u, v) - C(u, v)\big),$$

that is, the process appearing in Theorem 1.24. For $n = 100$, the empirical copula surface still exhibits visible sampling roughness, and the empirical-process surface does not yet resemble a typical realisation of the Gaussian limit particularly well. For $n = 10{,}000$, both visual approximations are substantially more stable. The true copula is not depicted, since it is virtually indistinguishable from the $n = 10{,}000$ approximation to the eye. ◇

1.6 Stochastic simulation

We have already encountered some simulation studies, without providing details on *how* to simulate. Although stochastic simulation is not, strictly speaking, a topic of statistics for insurance, it is an indispensable tool throughout modern actuarial work, and throughout this book. It is the foundation for parametric bootstrap methods, enables scenario generation for both random variables and stochastic processes, and makes it possible to approximate integrals and probabilities when closed-form expressions are unavailable or intractable. Below, we give a very basic treatment, and leave more sophisticated methods to the chapters which require them.

At the root of essentially every simulation method lies the ability to generate values that behave like independent draws from the standard uniform distribution. In practice these numbers are produced by deterministic pseudorandom

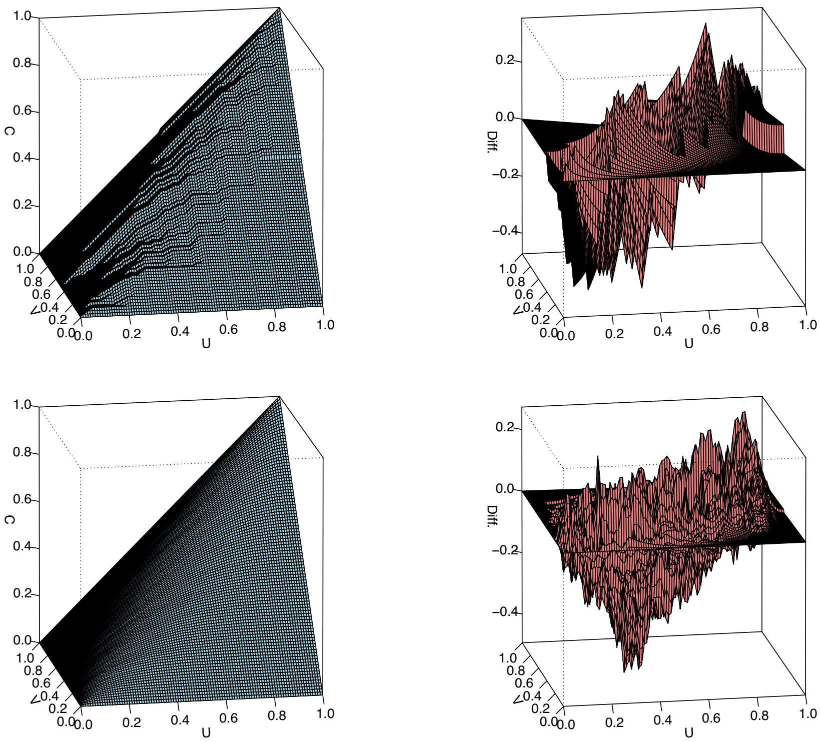

Figure 1.5: Empirical copulas (left) and their associated empirical processes (right) for simulated Gaussian vectors (correlation $\rho = 0.8$) of sample size $n = 100$ (top) and $n = 10,000$ (bottom).

number generators, designed so that their output has good statistical properties and appears random for computational purposes. Here we adopt the standard working assumption that the underlying random number generator provides independent and identically distributed samples from $\mathrm{Unif}(0,1)$, and we build subsequent simulation algorithms from that starting point.

Discrete random variables are then immediate to simulate. If Z is discrete and takes values $x_1,\dots,x_N$ with probabilities $p_1,\dots,p_N$, define cumulative sums $c_j = \sum_{i=1}^{j} p_i$ with $c_0 = 0$ and $c_N = 1$. Draw $U \sim \mathrm{Unif}(0,1)$ and return $Z = x_j$ whenever $c_{j-1} < U \le c_j$. In other words, we split $(0,1)$ into intervals of lengths $p_1,\dots,p_N$ and map each interval to its corresponding value.

For continuous random variables Z with density f, this direct interval partition is no longer available, and three general-purpose methods are common in practice: inversion, acceptance-rejection, and Fourier inversion. Beyond these, many specialised simulation routines exploit specific structural properties of the distribution of Z. We present theory for univariate laws. For multivariate laws, the most general direct construction is to factor the joint law into successive conditional laws and simulate componentwise (known as the Rosenblatt construction); when these conditional laws are not tractable, one typically turns to more specialised case-by-case techniques.

1.6.1 Inversion method

The inversion method is the most simple and robust and general-purpose simulation technique. Its caveat is that computing the inverse of a function might be numerically costly.

Definition 1.18 (Generalised inverse) Let F be a distribution function on $\mathbb{R}$. Its generalised inverse is

$$F^{-1}(u) = \inf\{y \in \mathbb{R} : F(y) \ge u\}, \quad u \in (0,1).$$

Proposition 1.3 (Generalised inverse and inversion) *Let F be a distribution function and let F^{-1} be its generalised inverse.*

1. *For $u \in (0,1)$ and $z \in \mathbb{R}$,*

$$u \le F(z) \quad \Leftrightarrow \quad F^{-1}(u) \le z.$$

2. *If $U \sim \mathrm{Unif}(0,1)$, then $F^{-1}(U)$ has distribution function F.*

3. *If Z has distribution function F and F is continuous, then $F(Z) \sim \mathrm{Unif}(0,1)$.*

Proof. The equivalence in the first statement follows directly from the definition

$$F^{-1}(u) = \inf\{y \in \mathbb{R} : F(y) \ge u\},$$

since $\{y : F(y) \ge u\} = [F^{-1}(u), \infty)$. Therefore, for any $z \in \mathbb{R}$,

$$\mathbb{P}(F^{-1}(U) \le z) = \mathbb{P}(U \le F(z)) = F(z),$$

which proves that $F^{-1}(U)$ has cdf F. Now assume $Z \sim F$ and that F is continuous. For $u \in (0,1)$,

$$\mathbb{P}(F(Z) \geq u) = \mathbb{P}(Z \geq F^{-1}(u)).$$

By continuity of F, $\mathbb{P}(Z \geq a) = \mathbb{P}(Z > a) = 1 - F(a)$ for every a, hence

$$\mathbb{P}(F(Z) \geq u) = 1 - F(F^{-1}(u)).$$

Now $F(F^{-1}(u)) \geq u$ by definition of F^{-1}. If $y_n < F^{-1}(u)$ and $y_n \uparrow F^{-1}(u)$, then $F(y_n) < u$ for all n, and continuity gives $F(F^{-1}(u)) \leq u$. Therefore $F(F^{-1}(u)) = u$, so

$$\mathbb{P}(F(Z) \geq u) = 1 - u, \quad u \in (0,1),$$

which is equivalent to $F(Z) \sim \mathrm{Unif}(0,1)$. □

In simulation, the key statement is the second one: a random variable with cdf F is obtained by generating $U \sim \mathrm{Unif}(0,1)$ and setting $Z = F^{-1}(U)$.

Example 1.15 (Exponential building blocks and inversion) Let $F(z) = 1 - \exp(-\lambda z)$, $z \geq 0$, be the exponential cdf with rate $\lambda > 0$. Its inverse is

$$F^{-1}(u) = -\frac{\log(1-u)}{\lambda}, \quad u \in (0,1),$$

so if $U \sim \mathrm{Unif}(0,1)$ then

$$Z = -\frac{\log(1-U)}{\lambda} \sim \mathrm{Exp}(\lambda).$$

Since $1 - U \overset{d}{=} U$, one may equivalently use $Z = -\log(U)/\lambda$. This basic construction is often exploited to generate richer laws. For instance, if $E_1, \dots, E_p$ are independent $\mathrm{Exp}(\lambda)$ random variables and $p \in \mathbb{N}$, then

$$G_p = \sum_{i=1}^{p} E_i \sim \mathrm{Gamma}(p, \lambda).$$

Likewise, if (E_n) are iid $\mathrm{Exp}(\lambda)$ and

$$N = \max\Big\{n \geq 0 : \sum_{i=1}^{n} E_i \leq 1\Big\},$$

then $N \sim \mathrm{Poisson}(\lambda)$. Indeed, this N is the unit-time count of the process with iid exponential inter-arrival times, hence of a Poisson process with rate λ. Writing $E_i = -\log(U_i)/\lambda$ with iid $U_i \sim \mathrm{Unif}(0,1)$ gives the equivalent product representation

$$N = \max\Big\{n \geq 0 : \prod_{i=1}^{n} U_i > e^{-\lambda}\Big\}.$$

At the same time, direct inversion is available for both families. One may generate

$$G_p = F_\Gamma^{-1}(U), \quad N = F_{\mathrm{Pois}}^{-1}(U),$$

where F_Γ and F_{Pois} denote the Gamma and Poisson cdf, respectively. For Gamma this means numerical inversion of the Gamma cdf, while for Poisson it is the smallest $m \in \mathbb{N}_0$ such that

$$\sum_{k=0}^{m} e^{-\lambda} \frac{\lambda^k}{k!} \geq U.$$

The broader message is that simulation can proceed either by direct inversion of the target cdf or by exploiting probabilistic identities that transform simpler random inputs into the target law. $\diamond$

1.6.2 Acceptance-rejection

We now turn to the second general-purpose method. Let f be a target density and let g be a proposal density that is easy to simulate, with

$$f(z) \leq Cg(z), \quad z \in \mathbb{R},$$

for some finite constant $C \geq 1$. The method repeatedly draws $Y \sim g$ and $U \sim \mathrm{Unif}(0,1)$, independently, and accepts the proposal whenever

$$U \leq \frac{f(Y)}{Cg(Y)}.$$

If the proposal is rejected, one draws a new pair and repeats until acceptance. The next result guarantees that such a method correctly samples from f.

Proposition 1.4 (Acceptance-rejection) *Let (Y_n, U_n) be iid with $Y_n \sim g$, $U_n \sim \mathrm{Unif}(0,1)$, and $Y_n \perp\!\!\!\perp U_n$. Define*

$$A_n = \Big\{U_n \leq \frac{f(Y_n)}{Cg(Y_n)}\Big\}, \quad T = \inf\{n \geq 1 : A_n \text{ occurs}\}, \quad Z = Y_T.$$

Then Z has density f.

Proof. Set $p = \mathbb{P}(A_1)$. Since U_1 is independent of Y_1,

$$p = \mathbb{E}\Big[\mathbb{P}\Big(U_1 \leq \frac{f(Y_1)}{Cg(Y_1)} \,\Big|\, Y_1\Big)\Big] = \mathbb{E}\Big[\frac{f(Y_1)}{Cg(Y_1)}\Big] = \frac{1}{C}\int f(z)\,\mathrm{d}z = \frac{1}{C}.$$

Now let $B \subseteq \mathbb{R}$ be measurable. Then

$$\begin{aligned}\mathbb{P}(Z \in B) = \sum_{n=1}^{\infty} \mathbb{P}(T = n,\ Y_n \in B) &= \sum_{n=1}^{\infty} \mathbb{P}(A_1^c, \ldots, A_{n-1}^c,\ A_n,\ Y_n \in B) \\ &= \sum_{n=1}^{\infty} (1-p)^{n-1} \mathbb{P}(A_1,\ Y_1 \in B),\end{aligned}$$

by independence of the iid trials. Furthermore,

$$\mathbb{P}(A_1,\ Y_1 \in B) = \int_B g(z)\frac{f(z)}{Cg(z)}\ \mathrm{d}z = \frac{1}{C}\int_B f(z)\ \mathrm{d}z = p\int_B f(z)\ \mathrm{d}z.$$

Hence

$$\mathbb{P}(Z \in B) = \sum_{n=1}^{\infty}(1-p)^{n-1}p\int_B f(z)\ \mathrm{d}z = \int_B f(z)\ \mathrm{d}z,$$

which proves that Z has density f. □

Remark 1.9 (Efficiency) The acceptance probability is

$$\mathbb{P}(A_1) = \mathbb{E}\Big[\mathbb{P}\Big(U_1 \le \frac{f(Y_1)}{Cg(Y_1)}\,|\,Y_1\Big)\Big] = \frac{1}{C},$$

so the expected number of proposals per accepted draw equals C. Therefore the method is efficient when C is close to 1, which in turn requires a proposal g that resembles f closely on both body and tail. ○

This following simple construction illustrates the bounded-support case.

Example 1.16 (Beta simulation) Suppose the target is Beta(α,β) on $(0,1)$ with $\alpha,\beta \ge 1$, so its density

$$f(z) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}z^{\alpha-1}(1-z)^{\beta-1}, \quad 0<z<1,$$

is bounded. Let

$$M = \sup_{0<z<1} f(z) < \infty, \quad g(z) = 1_{(0,1)}(z),$$

so g is the uniform density on $(0,1)$ and $f(z) \le Mg(z)$ on $\mathbb{R}$. Taking $C = M$ gives a valid acceptance-rejection scheme with acceptance probability $1/M$. ◇

On unbounded domains, or when densities are sharply peaked, good proposals are typically chosen by matching concavity and tail behaviour so that C remains moderate. When f itself is also costly to evaluate, the following technique is useful.

Remark 1.10 (Squeezing) When f is expensive to evaluate, one can reduce calls to f by adding simple bounds. Suppose there are simple functions l and u such that

$$0 \le l(z) \le \frac{f(z)}{Cg(z)} \le u(z) \le 1.$$

Given a proposal Y and a uniform U, accept immediately if $U \le l(Y)$ and reject immediately if $U > u(Y)$. Only on the intermediate event $l(Y) < U \le u(Y)$ is the exact ratio $f(Y)/(Cg(Y))$ evaluated. This leaves the target law unchanged while reducing expensive evaluations. ○

1.6.3 Fourier inversion

In some models the distribution function F or density f is not available in explicit form, while a transform is. Let $Z \sim F$ and define the characteristic function

$$\psi(s) = \widehat{F}[\mathrm{i}s] = \mathbb{E}\big[e^{\mathrm{i}sZ}\big], \quad s \in \mathbb{R}.$$

This permits simulation constructions based directly on Fourier inversion. The first result gives the density representation directly from ψ.

Proposition 1.5 (Fourier inversion) *If $\psi \in L^1(\mathbb{R})$, then Z has a bounded continuous density*

$$f(z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \psi(s) e^{-\mathrm{i}sz}\, \mathrm{d}s, \quad z \in \mathbb{R}.$$

Proof. Define f by the displayed integral. Since $\psi \in L^1(\mathbb{R})$, dominated convergence implies that f is bounded and continuous. Moreover, for $a < b$,

$$\int_a^b f(z)\, \mathrm{d}z = \lim_{M\to\infty} \frac{1}{2\pi}\int_{-M}^{M} \psi(s)\frac{e^{-\mathrm{i}sa} - e^{-\mathrm{i}sb}}{\mathrm{i}s}\, \mathrm{d}s.$$

By the Lévy inversion formula, the right-hand side equals $F(b) - F(a)$ at continuity points a, b of F. Hence the finite measure with density f agrees with F on intervals $(a, b]$ and therefore coincides with F. □

The next example gives the practical approximation route for simulation, and shows how the equidistant-grid case leads to an FFT implementation.

Example 1.17 (Grid approximation and FFT implementation) Assume the density f of Z is continuous and supported on $[a, b]$. For any fixed integer $n \geq 2$, we define an equidistant grid of step size $\epsilon = (b - a)/(n - 1)$ with points

$$z_m = a + (m-1)\epsilon, \quad m = 1, \dots, n.$$

We construct a discrete random variable $\widetilde{Z}_n$ taking values in $\{z_1, \dots, z_n\}$ with probabilities proportional to the density:

$$p_m = \mathbb{P}(\widetilde{Z}_n = z_m) = \frac{f(z_m)}{\sum_{j=1}^n f(z_j)}.$$

This discrete variable $\widetilde{Z}_n$ approximates Z as $n \to \infty$. Indeed, for every bounded continuous function h on $[a, b]$, Riemann-sum convergence yields

$$\mathbb{E}[h(\widetilde{Z}_n)] = \frac{\sum_{m=1}^n h(z_m) f(z_m)\epsilon}{\sum_{m=1}^n f(z_m)\epsilon} \to \int_a^b h(z) f(z)\, \mathrm{d}z = \mathbb{E}[h(Z)],$$

hence $\widetilde{Z}_n \xrightarrow{d} Z$. Typically, such an approximation requires the point values $f(z_m)$ to be evaluated via numerical quadrature (e.g. using Proposition 1.5).

However, because the grid is explicitly equidistant, we can significantly accelerate this computation using the Fast Fourier Transform (FFT) applied directly to the characteristic function $\psi(s) = \mathbb{E}[e^{\mathrm{i}sZ}]$. Let $w = e^{2\pi\mathrm{i}/n}$ and define the discrete frequencies $s_r = 2\pi(r-1)/(n\epsilon)$ for $r = 1, \dots, n$. The Discrete Fourier Transform (DFT) of the probability vector (p_m) is given by

$$\widehat{p}_r = \frac{1}{n}\sum_{m=1}^{n} p_m w^{(r-1)(m-1)} = \frac{e^{-\mathrm{i}s_r z_1}}{n}\sum_{m=1}^{n} p_m e^{\mathrm{i}s_r z_m}.$$

Recognising the sum as a discrete approximation to the characteristic function $\psi(s_r)$, we can bypass the direct computation of $f(z_m)$ by defining proxy Fourier coefficients:

$$\widehat{q}_r = \frac{e^{-\mathrm{i}s_r z_1}}{n}\psi(s_r).$$

Applying the inverse transform to $(\widehat{q}_r)$ yields candidate probability masses $\widetilde{p}_m$:

$$\widetilde{p}_m = \sum_{r=1}^{n} \widehat{q}_r w^{-(r-1)(m-1)}.$$

To correct for minor numerical artifacts and ensure a valid distribution, we take the absolute values and re-normalise:

$$p_m^\star = |\widetilde{p}_m|, \quad p_m = \frac{p_m^\star}{\sum_{j=1}^{n} p_j^\star}.$$

The resulting probability vector (p_m) over the grid (z_m) allows the approximate simulation of Z to be reduced to standard discrete sampling. $\diamond$

The final result gives an exact transform-based acceptance-rejection route, avoiding grid discretisation.

Proposition 1.6 (Transform bounds for acceptance-rejection) *Assume $\psi, \psi', \psi'' \in L^1(\mathbb{R})$. Let f be the density of Z, and define*

$$c = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\psi(s)|\,\mathrm{d}s, \quad k = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\psi''(s)|\,\mathrm{d}s.$$

Then

$$f(z) \le c \wedge \frac{k}{z^2}, \quad z \neq 0.$$

Consequently, with $b = \sqrt{k/c}$ and $A = 4\sqrt{kc}$, the density

$$g(z) = \frac{1}{A}\left(c\,1_{(|z|\le b)} + kz^{-2}1_{(|z|>b)}\right)$$

yields the valid acceptance-rejection envelope $f(z) \le Ag(z)$.

Proof. From Proposition 1.5,

$$|f(z)| \le \frac{1}{2\pi}\int_{-\infty}^{\infty} |\psi(s)|\,\mathrm{d}s = c.$$

Because $\psi, \psi', \psi'' \in L^1$ we have $\psi(s) \to 0$ and $\psi'(s) \to 0$ as $|s| \to \infty$. Hence for $z \neq 0$, integration by parts gives

$$f(z) = \frac{1}{2\pi \mathrm{i} z} \int_{-\infty}^{\infty} \psi'(s) e^{-\mathrm{i}sz} \, \mathrm{d}s,$$

and integrating by parts again gives

$$f(z) = -\frac{1}{2\pi z^2} \int_{-\infty}^{\infty} \psi''(s) e^{-\mathrm{i}sz} \, \mathrm{d}s.$$

Hence

$$|f(z)| \leq \frac{1}{2\pi z^2} \int_{-\infty}^{\infty} |\psi''(s)| \, \mathrm{d}s = \frac{k}{z^2}.$$

Therefore $f(z) \leq c \wedge kz^{-2}$ for $z \neq 0$, while $f(0) \leq c$. Now,

$$A = \int_{-\infty}^{\infty} (c \wedge kz^{-2}) \, \mathrm{d}z = 2\int_0^b c \, \mathrm{d}z + 2\int_b^{\infty} kz^{-2} \, \mathrm{d}z = 2cb + \frac{2k}{b} = 4\sqrt{kc}.$$

So g is a density. Also, on $|z| \leq b$, one has $Ag(z) = c \geq f(z)$; on $|z| > b$, one has $Ag(z) = kz^{-2} \geq f(z)$. Hence $f \leq Ag$. □

Simulation from g is straightforward. With $b = \sqrt{k/c}$, the density g is a mixture of a uniform law on $(-b, b)$ and a symmetric shifted Pareto component with density $(b/2)z^{-2}1_{(|z|>b)}$. A convenient representation for the tail component is $S\,b/U$, where $S \in \{-1, 1\}$ is symmetric and $U \sim \mathrm{Unif}(0, 1)$. The acceptance-rejection step is then to accept Z with probability

$$\frac{f(Z)}{Ag(Z)},$$

where $A = 4\sqrt{kc}$ and $f(Y)$ is evaluated from Proposition 1.5.

1.7 Notes and Comments

The classical limit results and tools on stochastic convergence have long histories that can be traced back to the development of measure-theoretic probability in the early twentieth century. In Kolmogoroff [1933] the modern axiomatic framework was laid out, within which the principal convergence modes are formulated. The Law of Large Numbers and the Central Limit Theorem both have eighteenth-century origins; for the CLT under broad moment conditions, key proofs are due to Lyapunov [1901] and Lindeberg [1922]. The technical tools of asymptotic statistics were developed in parallel, with Slutsky's lemma [Slutsky, 1925], the Continuous Mapping Theorem [Mann and Wald, 1943], and the Cramér–Wold device [Cramér and Wold, 1936] all entering the literature in this period, together with the Glivenko–Cantelli theorem on uniform convergence of the empirical distribution function [Glivenko, 1933, Cantelli, 1933]. Stochastic order notation was formalised in Mann and Wald [1943], and Cramér [1946] gave a systematic asymptotic treatment that helped make this notation and delta-method arguments standard tools. The classical monograph Feller [1971] offers a thorough probabilistic treatment of these results, while modern references that present these results systematically include van der Vaart [1998], Billingsley [1999], Kallenberg [2002]. At the finite-dimensional level, standard references for estimator examples and finite-sample behaviour include Serfling [1980], together with sharp nonasymptotic inequalities for empirical distributions such as Dvoretzky et al. [1956], Massart [1990].

Convergence in law for stochastic processes required further structural tools. Tightness and relative compactness criteria were given by Prokhorov [1956], and almost-sure representations were introduced by Skorokhod [1956], forming the early foundations of weak convergence theory in metric spaces. A complementary development was martingale theory, as provided in Doob [1953], where maximal inequalities, optional-sampling arguments, and martingale convergence theorems became standard tools in stochastic-process limit proofs. The invariance principle of Chung and Fuchs [1951], establishing convergence of rescaled partial-sum processes to Brownian motion, marks the early beginning of functional central limit theorems. Various of these results were unified by Billingsley [1999], whose work largely systematised weak convergence of probability measures on function spaces.

Building on the foundational work, the development of empirical process theory is treated systematically in Dudley [1984], Pollard [1984], Shorack and Wellner [2009], van der Vaart and Wellner [1996]. The functional delta method, formulated in terms of Hadamard differentiability, connects these weak convergence results to asymptotic expansions for estimators and test statistics [van der Vaart, 1998, van der Vaart and Wellner, 1996]. The formulation of weak convergence using outer probability and outer expectation, which elegantly deals with measurability isues, follows Hoffmann–Jørgensen's work [Hoffmann-Jørgensen, 1974, Hoffmann-Jorgensen and Pisier, 1976] and has become standard in the empirical-process literature. The k-nearest neighbours empirical process treatment is taken from Portier [2025] and builds on the same empirical-process framework as van der Vaart and Wellner [1996]. The entropy-theorem block, based on bracketing arguments, fits directly into this same empirical-process line of work, as does the section on empirical distributions and quantiles follows standard weak-convergence treatments from Shorack and Wellner [2009], van der Vaart and Wellner [1996].

Finally, the simulation subsections are classically covered in random-variates monographs such as Devroye [1986], Hörmann et al. [2004], Ripley [1987], Robert and Casella [2004], with the material being inspired by the treatment of Asmussen and Glynn [2007]. Inversion and acceptance-rejection are treated there in depth, while acceptance-rejection traces back to von Neumann [1951], and transform-based inversion dates back to the inversion identity of Gil-Pelaez [1951] and practical numerical inversion methods such as Abate and Whitt [1995].

Beyond the provided material, these theoretical results have found systematic application in statistical methodology, as expanded in subsequent chapters; here we only mention a few directions. Survival analysis and event-history modelling rely heavily on martingale limit theorems and counting-process techniques, as developed in Andersen et al. [1993], Fleming and Harrington [2005]. Extreme value theory rests on weak convergence arguments for maxima, with standard references in Resnick [1987], de Haan and Ferreira [2006]. Dependence modelling via copulas has led to the study of empirical copula processes; foundational copula references include Sklar [1959], Nelsen [2006], Joe [2014], Durante and Sempi [2015]. Other specialized developments include treatment of stochastic-process limits in Whitt [2002], on Markov processes in Ethier and Kurtz [1986], and a semimartingale framework in Jacod and Shiryaev [2003]. The rigorous treatment of stochastic simulation methods of Asmussen and Glynn [2007] is used throughout the book.

1.8 Exercises

1. By applying Theorem 1.3 to $-\gamma$, show that we also obtain

$$\mathbb{P}\Big(\Big|\frac{1}{n}\sum_{i=1}^{n}\gamma(Z_i)\Big| \geq \kappa t + \sqrt{2t}\Big) \leq 2e^{-nt}.$$

Equivalently, show that there exists a constant $C > 0$ such that for all $\varepsilon > 0$,

$$\mathbb{P}\Big(\Big|\frac{1}{n}\sum_{i=1}^{n}\gamma(Z_i)\Big| \geq \varepsilon\Big) \leq 2\exp\Big(-\frac{Cn\varepsilon^2}{1+\kappa\varepsilon}\Big).$$

2. Let (Z_n) be iid with $\mathbb{P}(Z_1 > z) = z^{-\alpha}$ for $z \geq 1$ and $\alpha \in (0,1)$.

   (a) Show that $\mathbb{E}[|Z_1|] = \infty$.

   (b) Show that $\limsup_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} Z_i = \infty$, and hence Theorem 1.2 fails without the moment assumption.

   Hint: Use Borel–Cantelli with the events $\{Z_n > Mn\}$ for any $M > 0$.

3. Let (Z_n) be iid symmetric with $\mathbb{P}(|Z_1| > z) = z^{-\alpha}$ for $z \geq 1$ and $\alpha \in (1,2)$. Show that the Lindeberg condition for the normalisation $\sqrt{n}$ fails, by proving that for any fixed $\varepsilon > 0$,

$$n\,\mathbb{E}\big[Z_1^2\,;|Z_1| > \varepsilon\sqrt{n}\big] \to \infty.$$

Show that $n^{-1/2}\sum_{i=1}^{n} Z_i$ cannot converge in distribution to a non-degenerate normal law. Conclude that Theorem 1.6 fails when $\mathbb{E}[Z_1^2] = \infty$.

4. Let $Z_n = Z_1$ almost surely for all n, where Z_1 has a continuous distribution function F. Show that $\mathbb{F}^{(n)}(t) = 1_{(Z_1 \leq t)}$ for all n and hence

$$\sup_{t\in\mathbb{R}} |\mathbb{F}^{(n)}(t) - F(t)| = \max\{F(Z_1),\, 1 - F(Z_1)\} \geq \tfrac{1}{2} \quad \text{a.s.}$$

Conclude that the Glivenko–Cantelli theorem fails without independence.

5. Explain why the extended Glivenko–Cantelli Theorem 1.9 cannot be proven using a combination of Theorem 1.7 and Lemma 1.5.

6. Let (Z_n) and (W_n) be $\mathbb{R}^p$-valued random variables. Show that if $\|Z_n - W_n\| = o_{\mathbb{P}}(1)$ and $Z_n \xrightarrow{d} Z$, then also $W_n \xrightarrow{d} Z$.

   Hint: Use the definition of convergence in distribution.

7. Go through the details of Example 1.5, and discuss the practical implications of this result.

8. Prove a multidimensional version of (1.4). That is, for $-\infty < t_1 < t_2 < \cdots < t_M < \infty$, $M \in \mathbb{N}$, show that

$$\sqrt{n}\big(\mathbb{F}^{(n)}(t_1) - F(t_1), \ldots, \mathbb{F}^{(n)}(t_M) - F(t_M)\big) \xrightarrow{d} N\left(0, \Sigma(t_1, \ldots, t_M)\right),$$

where

$$(\Sigma(t_1, \ldots, t_M))_{i,j} = \sigma(t_i, t_j) = F(t_i \wedge t_j) - F(t_i)F(t_j).$$

9. Derive the mean excess function for each of the following distributions:

   (a) If Z is a Pareto-distributed random variable with scale parameter $\kappa > 0$ and shape parameter $\alpha > 0$, then $\bar{F}_Z(z) = (\kappa/z)^\alpha$, $z \geq \kappa$. Show that if $\alpha > 1$ then $e_F(u) = u/(\alpha - 1)$, $u \geq \kappa$, where $e_F(u) = \mathbb{E}[Z - u \,|\, Z > u]$ is the mean excess function.

   (b) If Z is log-normally distributed with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, then the distribution function of Z is given by

$$G_Z(z) = \Phi\Big(\frac{\log z - \mu}{\sigma}\Big), \quad z > 0,$$

   where Φ denotes the distribution function for a standard normal random variable. Straightforward calculations yield that

$$e_G(u) = e^{\mu + \frac{\sigma^2}{2}} \frac{\Phi\left(\frac{(\mu+\sigma^2)-\log u}{\sigma}\right)}{\Phi\left(\frac{\mu - \log u}{\sigma}\right)} - u.$$

   Show that as $u \to \infty$,

$$e_G(u) = \frac{\sigma^2 u}{\log u - (\mu + \sigma^2)} \left(1 + o(1)\right),$$

   where $o(1)$ is understood as $u \to \infty$.

   Hint: Recall Mills' ratio, which states that for $z \to -\infty$ we have that $\Phi(z) \sim -\phi(z)/z$, where ϕ denotes the density of a standard normal distribution.

10. Generalise Proposition 1.1 to independent but not identically distributed risks, using a uniformly bounded condition as in Example 1.6.

11. In Proposition 1.1 we saw that the empirical variance is asymptotically Gaussian. While empirical variance is commonly used for measuring the spread of a distribution, it is also common to use the empirical skewness to measure the asymmetry of a distribution.

$$\hat{\xi}_n = \frac{\frac{1}{n}\sum_{i=1}^n (Z_i - \bar{Z}_n)^3}{\left(\frac{1}{n}\sum_{i=1}^n (Z_i - \bar{Z}_n)^2\right)^{\frac{3}{2}}}, \quad \bar{Z}_n = \frac{1}{n}\sum_{i=1}^n Z_i,$$

    which targets the standardised central third moment

$$\xi := \frac{\mathbb{E}[(Z - \mathbb{E}[Z])^3]}{(\mathbb{E}[(Z - \mathbb{E}[Z])^2])^{\frac{3}{2}}}.$$

    Let (Z_n) be iid $\mathbb{R}$-valued random variables with finite sixth moment.

   (a) Show that $\hat{\xi}_n \xrightarrow{\mathbb{P}} \xi$.

   (b) State and prove a central limit theorem for $\sqrt{n}\big(\hat{\xi}_n - \xi\big)$.

12. Show that $\int_{(0,1]} \sqrt{\log\left(2/\varepsilon^2 + 1\right)}\, \mathrm{d}\varepsilon < \infty$.

13. Prove Lemma 1.9 by applying Lemma 1.8.

14. Go through the details of the proof of Theorem 1.22 in the univariate case, and provide the details in the bivariate case.

15. In Chapter 1 we proved that the empirical process (scaled difference) of the cumulative hazard is asymptotically Gaussian for each t with $F(t) < 1$. The goal of this exercise is to prove a functional (pathwise) CLT for the empirical process of the cumulative hazard. A useful route is to first identify the functional map and its derivative, and then apply the functional δ-method.

    (a) Consider the map ϕ, given by

    $$\phi\left(h\right)\left(t\right) = -\log(1 - h(t)),$$

    such that $h(t) < 1$ for all t. Show that the Hadamard derivative of ϕ at a function f in the direction of h is given by

    $$\phi'_f(h)(t) = \frac{h(t)}{1 - f(t)}.$$

    Hint: For small x, $-\log(1-x)$ is approximated by the identity function.

    (b) State and prove a functional limit theorem of $\sqrt{n}(\mathbb{A}^{(n)} - \Lambda)$ and compute the corresponding covariance function.

16. State and prove a functional CLT for the mean excess function.

17. Go through the details for obtaining the derivatives of Theorem 1.24.

18. Prove the following:

    Lemma 1.14 (Duhamel equation) *Let $B_1 = \phi(A_1)$ and $B_2 = \phi(A_2)$ be two product integrals. They satisfy the Duhamel equation*

    $$B_2(s,t] - B_1(s,t] = \int_{(s,t]} B_1(s,u)B_2(u,t](A_2 - A_1)\,\mathrm{d}u.$$

19. Prove the following:

    Lemma 1.15 (Integration by parts for the Duhamel equation) *Suppose that $A, B \in D(0,\infty)$ and that $h \in D(0,\infty)$ is of bounded variation. Then it holds that*

    $$\begin{aligned}
    &\int_{(0,t]} \prod\left(1 + \mathrm{d}B\right)(0,u) \prod\left(1 + \mathrm{d}A\right)(u,t]\,\mathrm{d}h(u) \\
    &= h(t) + \int_{(0,t]} \prod\left(1 + \mathrm{d}B\right)(0,s)\left[h(t) - h(s)\right]\mathrm{d}B(s) \\
    &\quad + \int_{(0,t]} h(r-) \prod\left(1 + \mathrm{d}A\right)(r,t]\,\mathrm{d}A(r) \\
    &\quad + \int_{(0,t]}\int_{(s,t]} \prod\left(1 + \mathrm{d}B\right)(0,s)\left[h(r-) - h(s)\right] \prod\left(1 + \mathrm{d}A\right)(r,t]\,\mathrm{d}A(r)\,\mathrm{d}B(s).
    \end{aligned}$$

20. Let (Z_n) be iid with characteristic function $\varphi(t) = \mathbb{E}[e^{itZ_1}]$ and define the empirical characteristic function

$$\varphi_n(t) = \frac{1}{n}\sum_{i=1}^{n} e^{itZ_i}, \quad t \in [-T, T].$$

   (a) Show that the class $\{e^{it\cdot} : t \in [-T, T]\}$ is Donsker and conclude that $\sqrt{n}(\varphi_n - \varphi) \xrightarrow{d} \mathbb{G}$ in $\ell^\infty([-T, T])$ for a centred Gaussian process with covariance

$$\mathrm{Cov}(\mathbb{G}(s), \mathbb{G}(t)) = \varphi(s - t) - \varphi(s)\varphi(-t).$$

   (b) For fixed t_0, use the δ-method (viewing $\varphi_n(t_0)$ as a complex-valued random element) to derive the asymptotic distribution of $|\varphi_n(t_0)|$.

21. Let $w\colon [0, 1] \to \mathbb{R}$ be Lipschitz and define the L-statistic functional

$$T(F) = \int_0^1 w(u)F^{\leftarrow}(u)\, \mathrm{d}u.$$

Assume F is continuous with strictly positive density on an interval containing the support of w. Show that T is Hadamard differentiable at F and derive the asymptotic distribution of $T(\mathbb{F}^{(n)})$ by the functional δ-method.

22. Let H be a bivariate distribution function with continuous marginals F and G, and copula C.

   (a) Show that if a and b are strictly increasing, then the copula of $(a(X), b(Y))$ equals C.

   (b) Assume that H has a density h with marginal densities f and g. Show that the copula density exists and is given by

$$c(u, v) = \frac{h(F^{\leftarrow}(u), G^{\leftarrow}(v))}{f(F^{\leftarrow}(u))\, g(G^{\leftarrow}(v))}, \quad (u, v) \in (0, 1)^2.$$

23. Establish consistency of the Lorenz curve and Gini coefficient. Comment on the truncated versus untruncated case.

24. Let Z be a discrete random variable with values $x_1, \ldots, x_N$ and probabilities $p_1, \ldots, p_N$, and define $c_j = \sum_{i=1}^{j} p_i$ with $c_0 = 0$ and $c_N = 1$. Let F be the distribution function of Z and F^{-1} its generalised inverse.

Use Proposition 1.3 to show that if $U \sim \mathrm{Unif}(0, 1)$ and $\widetilde{Z} = F^{-1}(U)$, then

$$\widetilde{Z} = x_j \quad \Leftrightarrow \quad c_{j-1} < U \le c_j,$$

and therefore

$$\mathbb{P}(\widetilde{Z} = x_j) = c_j - c_{j-1} = p_j, \quad j = 1, \ldots, N.$$

Conclude that the standard discrete simulation algorithm (partitioning $(0, 1)$ into intervals of lengths p_j and mapping each interval to x_j) is exactly the inversion method applied to a discrete target law.

25. Prove that the squeezing variant of acceptance-rejection method of Remark 1.10 produces a random variable with density f.

Practical exercises

1. Verify through simulation the variance of the approximations in Theorems 1.1 and 1.11.

   (a) Let (Z_n) be a sequence of iid random variables where $Z_n \sim N(0,1)$ for all n. Verify through simulation the asymptotic distribution $\sqrt{n}(\hat{\sigma}_n^2 - \mu^{(2)})$, where $\mu^{(2)} = \mathbb{E}[(Z_1 - \mathbb{E}[Z_1])^2]$.

   (b) Let (Z_n) be a sequence of iid random variables where Z_n is Pareto distributed with shape parameter $\alpha = 3$ and scale parameter $\kappa = 100$ for all n. Verify through simulation the asymptotic distribution for $u = 200$ of $\sqrt{n}\,(\mathbb{e}_n(u) - e_F(u))$.

   Hint: In `R` you can simulate Pareto distributed variables using the function `rPareto()` from the `R` package `Pareto`.

2. In the `R` package `CASdatasets` Dutang and Charpentier [2024] one may find the dataset `frecomfire`, within which there is a column called `ClaimCost2007`, a vector of claim costs in thousand euros from commercial French fire insurance, normalised to 2007. In this exercise we assume that the data in `ClaimCost2007` are iid and examine the variance, mean excess, and cumulative hazard.

   (a) Plot a histogram of the column `ClaimCost2007`,

   (b) Construct a 95% confidence interval for the variance using Theorem 1.1,

   (c) Construct a 95% confidence interval for the mean excess function in $u = 5000$, using Theorem 1.11,

   (d) Construct a 95% confidence interval for the cumulative hazard in $t = 2000$, using Theorem 1.12.

3. Consider the following simulation studies:

   (a) For $Z \sim \text{Exp}(\lambda)$, $\lambda > 0$ we have $F_Z(z) = 1 - e^{-\lambda z}$. In `R` simulate $n \in \{10, 100, 1000\}$ iid Exp(1/2)-variables, construct and plot $\mathbb{F}^{(n)}(z)$ for $z \in (0, 10]$ with 95% confidence bounds using Theorem 1.22.

   (b) Denote $D_n = \sup_x |\mathbb{F}^{(n)}(x) - F(x)|$. State the limit distribution for $\sqrt{n}D_n$. With the same setup as in (a), simulate 100 realisations of $\sqrt{n}D_n$ for each n. Plot the histograms and compare with the limit distribution of $\sqrt{n}D_n$ *either* through the exact limit density or through simulations; for the latter, the function `rbridge()` from the `R` package `e1071` can be useful.

   Hint: The supremum of a continuously time-changed Brownian bridge is the same as a non-time-changed one.

4. If $Y \sim N(\mu, \Sigma)$ is an r-dimensional multivariate random vector then its corresponding copula is the so-called *Gaussian copula*. The marginal distributions can be standardised through strictly increasing transformations to obtain a new vector $X \sim N(0, \rho_\Sigma)$, where $\rho_\Sigma = \rho$ is the correlation

matrix of Y. Since copulas are invariant under strictly increasing transformations it follows that Y and X have the same copula. It is given by

$$\begin{aligned} C_\rho^{\mathrm{Ga}} &= \mathbb{P}\left(\Phi(X_1) \le u_1, \ldots, \Phi(X_r) \le u_r\right) \\ &= \Phi_\rho\big(\Phi^{-1}(u_1), \ldots, \Phi^{-1}(u_r)\big), \end{aligned}$$

where Φ_ρ is the cdf of an r-dimensional zero-mean Gaussian vector with covariance matrix ρ. We consider $r = 2$ and X a centred Gaussian vector with $\sigma_1^2 = \sigma_2^2 = 1$ and $\sigma_{12} = 0.7$.

(a) Simulate $n = 1000$ observations from $N(0, \rho)$ and set $U_i = \Phi(X_{i1})$, $V_i = \Phi(X_{i2})$. Construct the empirical copula

$$\mathbb{C}^{(n)}(u, v) = \frac{1}{n} \sum_{i=1}^{n} 1_{(U_i \le u,\, V_i \le v)}$$

on a grid in $(0,1)^2$ and compare it visually with the true Gaussian copula $C(u, v) = \Phi_\rho(\Phi^{-1}(u), \Phi^{-1}(v))$. (Either numerically evaluate or approximate C by Monte Carlo simulation using a very large sample).

(b) To investigate the scaled differences $\sqrt{n}\big(\mathbb{C}^{(n)} - C\big)$, simulate 1000 realisations of sample size $n = 1000$ and then compute $\sup_{(u,v)\in[0,1]^2} |\sqrt{n}\big(\mathbb{C}^{(n)}(u, v) - C(u, v)\big)|$ for each simulation. Plot a histogram of these values. Compare with $n = 100,000$.

5. Implement simulation of Gamma(p, λ) (with integer p) and Poisson(λ) by two conceptually different methods. Use the distributional identities from the chapter to construct Gamma(p, λ) as a sum of p iid exponentials with rate λ, and to construct Poisson(λ) from exponential inter-arrival times (or equivalently from the product representation involving uniforms). Then implement direct inversion, i.e. $F_\Gamma^{-1}(U)$ and $F_{\mathrm{Pois}}^{-1}(U)$ with $U \sim \mathrm{Unif}(0, 1)$. Using a large sample, compare the two routes through empirical means and variances against the theoretical values, compare computation times, and discuss briefly which method is preferable in your implementation and why.

6. Consider the three distribution functions

$$F_P(z) = 1 - z^{-1/\gamma}, \quad z \ge 1, \quad F_{Fr}(z) = \exp\big(-z^{-1/\gamma}\big), \quad z > 0,$$

and

$$F_B(z) = 1 - \left(1 + (z/\theta)^\eta\right)^{-\beta}, \quad z \ge 0, \quad \theta, \eta, \beta > 0,$$

which are the Pareto, Fréchet and Burr parameterisations used in Chapter **??**. Derive their inverse distribution functions and implement inversion-based simulation for all three laws from a uniform sample.

7. Consider the standard normal target law, for which $\psi(s) = \exp(-s^2/2)$. Use Example 1.17 and Proposition 1.6 to compare Fourier simulation by FFT approximation and by exact acceptance-rejection.

(a) Compute explicitly

$$c = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi(s)| \, \mathrm{d}s, \quad k = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi''(s)| \, \mathrm{d}s,$$

and deduce $b = \sqrt{k/c}$ and $A = 4\sqrt{kc}$ for the acceptance-rejection envelope.

(b) Implement the FFT-based method of Example 1.17 on a fixed equidistant grid.

(c) Implement the exact transform-based acceptance-rejection method from Proposition 1.6.

(d) Using large samples, compare approximation quality (moments, selected quantiles, cdf values), computational cost, and, for acceptance-rejection, empirical acceptance probability.

(e) For the FFT method, study sensitivity with respect to truncation window and mesh size.

Conclude which method is preferable in your implementation for a given accuracy target.

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